GOODNESS-OF-FIT FOR DISTRIBUTIONS ON METRIC SPACES

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We propose a general goodness-of-fit framework for distributions on separable metric spaces. Under suitable identifiability conditions, probability distributions are characterized by distance profiles, which motivates their use in goodness-of-fit testing, for simple and composite null hypotheses. For composite null hypotheses, parameter estimation is incorporated via a Bahadur-type expansion, and the asymptotic distribution of the empirical process for distance profiles is obtained under the null. We define test statistics based on this empirical process and derive their asymptotic null distributions. We further study the behavior of the proposed tests under fixed and local alternatives, establishing consistency results. The methodology is illustrated with simulation studies and real-data experiments.

1. Basic concepts and notation. Consider a separable metric space (Ω, d). Let (S, Σ, P) be a probability space, where S is a sample space, Σ is a sigma algebra of subsets of S, and P is a probability measure. An Ω-valued random variable is called a random object, which is a measurable map $X : S \to \Omega$. Suppose that X is distributed according to a Borel probability measure μ, that is, $\mu(A) = \mathsf{P}\left(\{s \in S : X(s) \in A\}\right) =: \mathsf{P}\left(X \in A\right) = \mathsf{P}\left(X^{-1}(A)\right)$, for any Borel measurable set $A \subseteq \Omega$. For any $t \geq 0$, we define the *distance profile* at $\omega \in \Omega$ of X as

$$F_\omega^\mu(t) = \mathsf{P}\left(d(\omega, X) \leq t\right).$$

To estimate the distance profiles F_ω^μ from a sample $X_1, \ldots, X_n$ of independent realizations of X, $n \geq 1$, we use empirical estimates

$$F_{n,\omega}^\mu(t) := \frac{1}{n}\sum_{i=1}^n 1_{\{d(\omega, X_i) \leq t\}}, \quad t \geq 0.$$

Distance profiles characterize the underlying probability measures under suitable conditions. Consider two Borel probability measures μ_1 and μ_2 on (Ω, d). (Dubey, Chen and Müller, 2024, Proposition 1) provide a sufficient condition for $\mu_1 = \mu_2$ in terms of distance profiles: if there exists $\theta > 0$ such that (Ω, d^θ) is of strong negative type (Lyons, 2013), where $d^\theta(\omega, \omega') := d(\omega, \omega')^\theta$, then

$$\mu_1 = \mu_2 \quad \Longleftrightarrow \quad F_\omega^{\mu_1}(t) = F_\omega^{\mu_2}(t) \text{ for all } \omega \in \Omega,\ t \geq 0. \tag{1}$$

Chen and Dubey (2025) obtain an alternative characterization under a condition imposed on the probability measures. Let μ be a Borel probability measure on (Ω, d). If there exists $L_\mu < \infty$ such that, for every $\omega \in \operatorname{supp}(\mu)$,

$$\limsup_{r \downarrow 0} \frac{\mu\big(B(\omega, 2r)\big)}{\mu\big(B(\omega, r)\big)} \leq L_\mu, \tag{2}$$

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then μ is said to satisfy the doubling condition. If μ_1, μ_2 satisfy this condition, then Theorem 1 in Chen and Dubey (2025) shows that

$$\mu_1 = \mu_2 \quad \Longleftrightarrow \quad \mu_1\big(\overline{B}(\omega,t)\big) = \mu_2\big(\overline{B}(\omega,t)\big) \text{ for all } \omega \in \operatorname{supp}(\mu_1),\ t > 0, \tag{3}$$

i.e., $\mu_1 = \mu_2$ whenever $F_\omega^{\mu_1}(t) = F_\omega^{\mu_2}(t)$ for all $\omega \in \operatorname{supp}(\mu_1)$ and $t \geq 0$. The same result holds with $\operatorname{supp}(\mu_1)$ replaced by $\operatorname{supp}(\mu_2)$. Here, $\operatorname{supp}(\mu)$ denotes the support of μ, which is the closure of the set of all points to which μ assigns positive mass in every open neighborhood, i.e.,

$$\operatorname{supp}(\mu) = \operatorname{cl}(\{x \in \Omega : \mu(U) > 0 \text{ for all open sets } U \ni x\}).$$

Chen and Dubey (2025, p. 6) further discuss the set of values of ω for which (1) and (3) hold. In ibid., it is emphasized that, under the doubling condition (2), μ_1 and μ_2 coincide as long as they agree on all closed balls centered in *either* of their supports (3). Under the strong negative type condition, they claim that the characterization in (1) holds if μ_1 and μ_2 agree on all closed balls centered in *both* of their supports, i.e., $\omega \in \operatorname{supp}(\mu_1) \cup \operatorname{supp}(\mu_2)$. However, in Proposition 1 of Dubey, Chen and Müller (2024), condition (1) is written in terms of $\omega \in \Omega$ (the whole metric space), so the claim is false unless $\Omega \subset \operatorname{supp}(\mu_1) \cup \operatorname{supp}(\mu_2)$.

In this manuscript, we assume that at least one of the following two conditions holds:

(*a.1*) (Ω, d) is of strong negative type.
(*a.2*) The probability measures satisfy the doubling condition.

Assumption *(a.1)* is a condition on the metric space, whereas *(a.2)* is an assumption on the probability measures.

Under certain conditions, (Dubey, Chen and Müller, 2024, Theorem 5.1) and Chen and Dubey (2025, Theorem 3) establish that the distance profiles satisfy the Donsker condition. Let

$$\mathcal{Y} := \{y_{\omega,t} : (\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}\}, \quad \text{where } y_{\omega,t}(x) := 1_{\{d(\omega,x) \leq t\}}. \tag{4}$$

Then, $\mathcal{Y}$ is μ-Donsker. The conditions under which this property is established for distance profiles differ between Dubey, Chen and Müller (2024) and Chen and Dubey (2025).

Dubey, Chen and Müller (2024) require the following two conditions:

(*b.1*) Let $N(\varepsilon, \Omega, d)$ be the covering number of (Ω, d), for $\varepsilon > 0$. Then

$$\varepsilon \log N(\varepsilon, \Omega, d) \ \to \ 0 \text{ as } \varepsilon \to 0^+.$$

(*b.2*) For every $\omega \in \Omega$, F_ω^μ is absolutely continuous with continuous density f_ω. Defining

$$\underline{\Delta}_\omega := \inf_{t \in \operatorname{supp}(F_\omega)} f_\omega(t), \qquad \bar{\Delta}_\omega := \sup_{t \in \mathbb{R}} f_\omega(t),$$

it holds that $\underline{\Delta}_\omega > 0$ for each $\omega \in \Omega$ and there exists $\bar{\Delta} < \infty$ such that $\sup_{\omega \in \Omega} \bar{\Delta}_\omega \leq \bar{\Delta}$.

REMARK 1.1. Condition *(b.1)* implies that (Ω, d) is totally bounded. The assumption that $d(\omega, X)$ has a continuous distribution for every $\omega \in \Omega$, which is implied by *(b.2)*, dates back to Szabados (1987) (page 383), who replaced the supremum over all $\omega \in \Omega$ in the Kolmogorov–Smirnov test statistic with maximization over a finite sample, as in our Proposition 3.4. To achieve this, Szabados (1987) restricted the analysis to *continuous measures*, i.e., measures P such that $P(B(\omega,t))$ depends continuously on $t > 0$ for any $\omega \in \Omega$.

(Chen and Dubey, 2025, Assumption 1) assume the bracketing condition

$$\varepsilon \log N_{[]}\big(\varepsilon, \mathcal{Y}, L^1(P)\big) \ \to \ 0 \quad \text{as } \varepsilon \to 0^+, \tag{5}$$

where $N_{[]}(\varepsilon, \mathcal{Y}, L^1(P))$ denotes the $L^1(P)$ bracketing number of $\mathcal{Y}$. The condition given in (5) holds if both *(b.1)* and a modification of *(b.2)* are satisfied (Chen and Dubey, 2025, Proposition 4). The modification is:

(*b.2'*) There exists a constant $L_G > 0$ such that

$$\sup_{\omega \in \Omega} \left| F_\omega^\mu(r) - F_\omega^\mu(s) \right| \leq L_G |r - s|,$$

for every $r, s \in \mathbb{R}_{\geq 0}$.

Assumption 1 in Chen and Dubey (2025) does not require the ambient spaces to be totally bounded. A sufficient condition for this assumption to hold is the following. Assume that for any $\varepsilon > 0$ there exists a closed ball $\overline{B}(r_\varepsilon) \subset \Omega$ of radius $r_\varepsilon > 0$, such that $\mathsf{P}\left(X \notin \overline{B}(r_\varepsilon)\right) \leq \varepsilon$. If (*b.2'*) holds and

$$\varepsilon \log N\left(\varepsilon, \overline{B}(r_\varepsilon), d\right) \to 0 \quad \text{as } \varepsilon \to 0^+, \tag{6}$$

then Assumption 1 in Chen and Dubey (2025) is satisfied. Examples of distributions on Euclidean spaces satisfying (6) and (*b.2'*) include sub-Gaussian and sub-exponential distributions.

In this manuscript, we will assume (*b.1*) and (*b.2*) hold.

2. Distance profiles on particular metric spaces. This section aims to provide closed formulas for the distance profile of a random variable X following a specific distribution on a given metric space, with probability measure denoted by μ.

EXAMPLE 2.1 (multivariate normal distribution on Euclidean space). Consider a random variable $\boldsymbol{X} \sim \mathcal{N}_q(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ taking values in $(\mathbb{R}^q, d)$, with $\boldsymbol{\Sigma}$ positive definite and d the Euclidean distance in $\mathbb{R}^q$. The distribution of $d(\boldsymbol{\omega}, \boldsymbol{X})^2 = (\boldsymbol{\omega} - \boldsymbol{X})^\top (\boldsymbol{\omega} - \boldsymbol{X})$ characterizes $F_{\boldsymbol{\omega}}$ at $\boldsymbol{\omega} \in \mathbb{R}^q$. It can be shown that

$$(\boldsymbol{\omega} - \boldsymbol{X})^\top (\boldsymbol{\omega} - \boldsymbol{X}) = \boldsymbol{\lambda}^\top \boldsymbol{P},$$

where $\boldsymbol{\Sigma} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$ is the spectral decomposition of $\boldsymbol{\Sigma}$, with $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_q)$, $\boldsymbol{\eta} := \boldsymbol{U}^\top(\boldsymbol{X} - \boldsymbol{\omega})$, $\boldsymbol{\nu} := \boldsymbol{U}^\top(\boldsymbol{\mu} - \boldsymbol{\omega})$, and $P_i := \eta_i^2/\lambda_i$. It holds that P_i are independent, and $P_i \sim \chi_1^2\left(\nu_i^2/\lambda_i\right)$, $i = 1, \ldots, q$. In the special case when $\boldsymbol{\omega} = \boldsymbol{\mu}$, then $P_i \sim \chi_1^2$.

EXAMPLE 2.2 (the Brownian motion in $L^2[0,1]$). We study the distance profile of the Brownian motion $B(t)$, $t \in [0,1]$, as an element in the metric space $(L^2[0,1], d_{L^2})$, where d_{L^2} denotes the distance induced by the L_2 norm $\|\cdot\|_{L^2}$. The distribution of $\|B - \omega\|_{L^2}^2$ characterizes F_ω at $\omega \in L^2[0,1]$. It can be shown that

$$\|B - \omega\|_{L^2}^2 = \sum_{i=1}^{\infty} X_i^2,$$

where X_i are independent, with $X_i \sim \mathcal{N}(c_i, \lambda_i)$, $c_i := \langle \omega, e_i \rangle$, $\lambda_i = (i - \frac{1}{2})^{-2}\pi^{-2}$. Here, $\{e_i\}_{i=1}^\infty$ are the eigenfunctions of the covariance kernel of B. In particular, when $\omega(t) = 0$ for all $t \in [0,1]$, then $X_i^2 = \lambda_i Q_i$, where $(Q_i)_{i \geq 1}$ are iid and $Q_i \sim \chi_1^2$.

EXAMPLE 2.3 (von Mises–Fisher (vMF) distribution on the sphere). Consider the unit sphere $\mathbb{S}^q := \{\boldsymbol{x} \in \mathbb{R}^{q+1} : \|\boldsymbol{x}\|_2 = 1\}$. Assume $\boldsymbol{X} \sim \operatorname{vMF}(\boldsymbol{\mu}, \kappa)$, for some given $\boldsymbol{\mu} \in \mathbb{S}^q$, $\kappa > 0$. For $\boldsymbol{x} \in \mathbb{S}^q$, the density of $\boldsymbol{X}$ is given by $f_{\mathrm{vMF}}(\boldsymbol{x}; \boldsymbol{\mu}, \kappa) \propto e^{\kappa \boldsymbol{\mu}^\top \boldsymbol{x}}$. We denote the normalization constant by $c_q^{\mathrm{vMF}}(\kappa)$.

Proposition 2.1 characterizes the distance profile of $\boldsymbol{X}$ at $\boldsymbol{\omega} \in \mathbb{S}^q$ under two common distances on $\mathbb{S}^q$: the chordal distance, given by $d(\boldsymbol{X}, \boldsymbol{\omega})^2 = 2(1 - \boldsymbol{\omega}^\top \boldsymbol{X})$; and the geodesic distance, given by $d(\boldsymbol{X}, \boldsymbol{\omega}) = \cos^{-1}(\boldsymbol{\omega}^\top \boldsymbol{X})$. To characterize $F_{\boldsymbol{\omega}}$, the distribution of $T = \boldsymbol{\omega}^\top \boldsymbol{X}$ plays a central role. Its density is given by

$$f_T(t) = \frac{c_q^{\mathrm{vMF}}(\kappa) e^{\kappa t \boldsymbol{\mu}^\top \boldsymbol{\omega}} (1 - t^2)^{\frac{q-2}{2}}}{c_{q-1}^{\mathrm{vMF}}\left(\kappa \sqrt{(1 - t^2)(1 - (\boldsymbol{\mu}^\top \boldsymbol{\omega})^2)}\right)}, \quad -1 < t < 1. \tag{7}$$

EXAMPLE 2.4 (hyperboloid von Mises–Fisher distribution (HvMF) on the hyperboloid). Consider the Minkowski pseudo-inner product $(\boldsymbol{x},\boldsymbol{y}) := -x_1y_1 + \sum_{i=2}^{q+1} x_iy_i$ on the unit hyperboloid $\mathbb{H}^q := \{\boldsymbol{x}\in\mathbb{R}^{q+1} : (\boldsymbol{x},\boldsymbol{x}) = -1,\, x_1 > 0\}$. In this surface, distances are calculated as $d_{\mathbb{H}^q}(\boldsymbol{x},\boldsymbol{y}) = \operatorname{arccosh}(-(\boldsymbol{x},\boldsymbol{y}))$.

The analogous distribution to the vMF on $\mathbb{H}^q$ has density $f_{\mathrm{HvMF}}(\boldsymbol{y};\boldsymbol{\mu},\kappa) \propto \exp\{\kappa(\boldsymbol{y},\boldsymbol{\mu})\}$, for $\boldsymbol{y},\boldsymbol{\mu}\in\mathbb{H}^q$, and $\kappa>0$, with respect to the Lebesgue measure on $\mathbb{H}^q$ (Barndorff-Nielsen, 1978; Jensen, 1981). We denote the normalization constant by c_q^{HvMF}. For $\boldsymbol{X}\sim\mathrm{HvMF}(\boldsymbol{\mu},\kappa)$ and a fixed $\boldsymbol{\omega}\in\mathbb{H}^q$, set $\rho := d_{\mathbb{H}^q}(\boldsymbol{\omega},\boldsymbol{\mu})$. An important quantity in the characterization of $F_{\boldsymbol{\omega}}$ is $Z_r := d_{\mathbb{H}^q}(\boldsymbol{X},\boldsymbol{\omega})$, with density given by

$$f_{Z_r}(z_r) = \frac{c_q^{\mathrm{HvMF}}(\kappa)}{c_{q-1}^{\mathrm{vMF}}(\kappa\sinh(z_r)\sinh(\rho))}(\sinh(z_r))^{q-1}e^{\kappa(\boldsymbol{\mu},\boldsymbol{\omega})\cosh(z_r)},\quad z_r\geq 0. \tag{8}$$

EXAMPLE 2.5 (logistic Gaussian distribution on the simplex). Consider the simplex $\Delta^q = \{\boldsymbol{x}\in\mathbb{R}^{q+1} : x_i>0,\ \sum_{i=1}^{q+1}x_i = 1\}$, endowed with the Aitchison distance $d_A(\boldsymbol{x},\boldsymbol{\omega}) := \|\mathrm{clr}(\boldsymbol{x}) - \mathrm{clr}(\boldsymbol{\omega})\|_2$. The centered log-ratio transform is defined by $\mathrm{clr}(\boldsymbol{x}) = \big(\log x_1 - \bar{\ell},\ldots,\log x_{q+1} - \bar{\ell}\big)$, $\bar{\ell} := \frac{1}{q+1}\sum_{k=1}^{q+1}\log x_k$. Let $\mathbf{Y}\sim\mathcal{N}_{q+1}(\boldsymbol{\mu},\boldsymbol{\Sigma})$, with $\boldsymbol{\Sigma}$ positive definite, and consider its softmax transformation

$$\mathbf{X} := \boldsymbol{\sigma}(\mathbf{Y}) = \left(\frac{e^{Y_1}}{\sum_{j=1}^{q+1}e^{Y_j}},\ldots,\frac{e^{Y_{q+1}}}{\sum_{j=1}^{q+1}e^{Y_j}}\right).$$

Then, $\mathbf{X}$ follows a logistic Gaussian distribution on Δ^q, with parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

Denote by $\boldsymbol{H} = \boldsymbol{I}_{q+1} - \frac{1}{q+1}\mathbf{1}_{q+1}\mathbf{1}_{q+1}^\top$ the centering matrix projecting onto $\{\boldsymbol{z}\in\mathbb{R}^{q+1} : \sum_{i=1}^{q+1} z_i = 0\}$. Let $\boldsymbol{m} := \boldsymbol{H}\boldsymbol{\mu} - \mathrm{clr}(\boldsymbol{\omega})$ and $\boldsymbol{C} := \boldsymbol{H}\boldsymbol{\Sigma}\boldsymbol{H}^\top$, and let $\boldsymbol{C} = \boldsymbol{U}\boldsymbol{A}\boldsymbol{U}^\top$ be a spectral decomposition with $\boldsymbol{A} = \mathrm{diag}(\alpha_1,\ldots,\alpha_q,0)$, $\alpha_1,\ldots,\alpha_q>0$. Define $\delta_i := (\boldsymbol{u}_i^\top\boldsymbol{m})^2\alpha_i^{-1}$, $i=1,\ldots,q$, where $\boldsymbol{u}_i$ denotes the i-th column of $\boldsymbol{U}$. It can be shown that $d_A(\boldsymbol{X},\boldsymbol{\omega})^2 = \boldsymbol{\alpha}^\top\boldsymbol{Q}$, where $\boldsymbol{\alpha} = (\alpha_1,\ldots,\alpha_q)^\top$, and $(Q_i)_{i=1}^q$ are independent with $Q_i\sim\chi_1^2(\delta_i)$.

REMARK 2.1 (connection between isometric and central log-ratio transformations). The Aitchison distance can be defined using the isometric log-ratio (ilr) transform (see, e.g., Egozcue et al., 2003) instead of clr, that is, $d_A(\boldsymbol{x},\boldsymbol{\omega}) = \|\mathrm{ilr}(\boldsymbol{x}) - \mathrm{ilr}(\boldsymbol{\omega})\|_2$. See the proof of Proposition 2.1 for this equivalence.

PROPOSITION 2.1. *We provide distributional characterizations of the distance profiles, under the notation of the examples in Section 2:*

- *If $\boldsymbol{X}\sim\mathcal{N}_q(\boldsymbol{\mu},\boldsymbol{\Sigma})$, with $\boldsymbol{\Sigma}$ positive definite, then $F_{\boldsymbol{\omega}}(t) = \mathsf{P}\left(\boldsymbol{\lambda}^\top\boldsymbol{P}\leq t^2\right)$.*
- *If B is a Brownian motion in $L^2[0,1]$, then $F_\omega(t) = \mathsf{P}\left(\sum_{i=1}^\infty X_i^2\leq t^2\right)$.*
- *If $\boldsymbol{X}\sim\mathrm{vMF}(\boldsymbol{\mu},\kappa)$, for the chordal distance,*

$$F_{\boldsymbol{\omega}}(t) = \begin{cases} 1 - F_T\big(1-\frac{t^2}{2}\big), & 0\leq t\leq 2,\\ 1, & t>2.\end{cases}$$

Under the geodesic distance,

$$F_{\boldsymbol{\omega}}(t) = \begin{cases} 1 - F_T(\cos(t)), & 0\leq t\leq \pi,\\ 1, & t>\pi.\end{cases}$$

The density of T is given in (7).

- *If $\boldsymbol{X}\sim\mathrm{HvMF}(\boldsymbol{\mu},\kappa)$, then $F_{\boldsymbol{\omega}}(t) = \mathsf{P}(Z_r\leq t)$. The density of Z_r is given in (8).*

- *If $\boldsymbol{X}$ follows a logistic Gaussian distribution on Δ^q with positive definite covariance matrix $\boldsymbol{\Sigma}$, then* $F_{\boldsymbol{\omega}}(t) = \mathsf{P}\left(\boldsymbol{\alpha}^\top \boldsymbol{Q} \leq t^2\right)$.

The proof of this result is given in Appendix A. See Section 6 for numerical illustrations of the distance profiles covered by Proposition 2.1.

3. Null asymptotics. Consider a random object X taking values in a separable metric space (Ω, d) and denote by μ the underlying probability measure of X. In this section, we study the asymptotic distribution of $\sqrt{n}\,(F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t))$ under the null hypothesis H_0, for simple and composite null hypotheses, and use it to prove the consistency of the Kolmogorov–Smirnov and Cramér–von Mises type tests based on $\sqrt{n}\,(F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t))$.

3.1. *Simple null hypothesis.* The goal is to test $\mu = \mu_0$, where μ_0 is a given probability measure on (Ω, d). Without loss of generality, we assume condition *(a.2)*, so that distance profiles characterize distributions on (Ω, d). Hence, testing $\mu = \mu_0$ is equivalent to:

$$H_0 : F^{\mu}_{\omega} = F^{\mu_0}_{\omega}, \text{ for every } \omega \in \operatorname{supp}(\mu_0) \subset \Omega.$$

The alternative hypothesis is thus

$$H_1 : \text{ there exists } \omega \in \operatorname{supp}(\mu_0) \text{ such that } F^{\mu}_{\omega} \neq F^{\mu_0}_{\omega}.$$

Alternatively, one could assume condition *(a.1)* and replace $\operatorname{supp}(\mu_0)$ by Ω in the above hypotheses. Unless otherwise specified, all the results hold under either condition.

Consider a sample $X_1, \ldots, X_n$ of iid random objects taking values in (Ω, d), and following a common distribution μ. Denote by $F^{\mu}_{n,\omega}$ the empirical distance profile associated to this sample. Define the random field

$$\mathbb{G}^{\mu_0}_{n,\mu}(\omega, t) := \sqrt{n}\left(F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t)\right), \quad (\omega, t) \in (\Omega \times \mathbb{R}_{\geq 0}).$$

Observe that $\mathbb{G}^{\mu_0}_{n,\mu}(\omega, t)$ is the empirical process indexed by the class of functions $\mathcal{Y}$ given in (4). In other words, we can write $\mathbb{G}^{\mu_0}_{n,\mu}(\omega, t) = \sqrt{n}(P_n y_{\omega,t} - \mu_0 y_{\omega,t})$ where P_n is the empirical measure associated to the sample $X_1, \ldots, X_n$, i.e., $P_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$. Thus, for notational convenience, we will also write $\mathbb{G}^{\mu_0}_{n,\mu}(y_{\omega,t})$ to denote $\mathbb{G}^{\mu_0}_{n,\mu}(\omega, t)$ in the proofs. Theorem 5.1 in Dubey, Chen and Müller (2024) and Theorem 3 in Chen and Dubey (2025) prove that, under H_0, the class $\mathcal{Y}$ is μ_0-Donsker, i.e.,

$$\mathbb{G}^{\mu_0}_{n,\mu} \xrightarrow{\mathcal{L}} \mathbb{G}_0 \text{ in } \ell^{\infty}(\mathcal{Y}), \tag{9}$$

as $n \to \infty$, where $\mathbb{G}_0$ is a zero mean Gaussian process with covariance given by

(10)
$$\mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} := \mathbb{C}\mathrm{ov}\left[y_{\omega_1,t_1}, y_{\omega_2,t_2}\right] = \mathsf{P}\left(d(\omega_1, X) \leq t_1, d(\omega_2, X) \leq t_2\right) - F^{\mu_0}_{\omega_1}(t_1) F^{\mu_0}_{\omega_2}(t_2),$$

for every $(\omega_1, t_1), (\omega_2, t_2) \in \Omega \times \mathbb{R}_{\geq 0}$.

3.2. *Composite null hypothesis.* Assume that μ belongs to a parametric family of probability measures on (Ω, d), indexed by a parameter $\boldsymbol{\theta}$ in an open subset Θ of $\mathbb{R}^p$. One may be interested in testing $\mu \in \{\mu_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \Theta\}$. Denote by $\boldsymbol{\theta}_0 \in \Theta$ the true parameter value, i.e., $\mu = \mu_{\boldsymbol{\theta}_0}$. In this case, the (composite) null hypothesis can be expressed as

$$H_0 : F^{\mu}_{\omega} = F^{\mu_{\boldsymbol{\theta}_0}}_{\omega}, \text{ for every } \omega \in \operatorname{supp}(\mu_{\boldsymbol{\theta}_0}) \subset \Omega,\ \boldsymbol{\theta}_0 \in \Theta \subset \mathbb{R}^p \text{ unknown}.$$

The construction of test statistics for the composite null hypothesis requires estimating the unknown parameter $\boldsymbol{\theta}_0$. Shorack and Wellner (2009, Chapter 5, Section 5) provides an overview of goodness-of-fit tests with estimated parameters. Let $\hat{\boldsymbol{\theta}}$ be a measurable estimator of $\boldsymbol{\theta}_0$ based on the sample $X_1, \ldots, X_n$. Let $\mu_{\hat{\boldsymbol{\theta}}}$ be the probability measure corresponding to $\hat{\boldsymbol{\theta}}$, and define $F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}$ as the distance profile of $\mu_{\hat{\boldsymbol{\theta}}}$ at ω. We can then define the random field

$$\mathbb{G}_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}(\omega, t) := \sqrt{n}\left(F_{n,\omega}^{\mu}(t) - F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t)\right), \quad (\omega, t) \in (\Omega \times \mathbb{R}_{\geq 0}).$$

We study the asymptotic behavior of $\mathbb{G}_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}$. The quantity $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ will play a central role in deriving this asymptotic distribution. For estimators based on n replications of an experiment, the order at which $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ converges to zero is often $\sqrt{n}$ (van der Vaart, 1998, p. 51), and its asymptotic distribution is usually normal. Under certain regularity conditions, every consistent estimator $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}_0$ satisfies the so-called Bahadur's representation:

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\Big(\mathsf{E}\big(\dot{\boldsymbol{\psi}}_{\boldsymbol{\theta}_0}(X)\big)\Big)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X_i) + o_P(1). \tag{11}$$

In particular, the sequence $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$ is asymptotically normal with mean zero and covariance matrix $\Big(\mathsf{E}\big(\dot{\boldsymbol{\psi}}_{\boldsymbol{\theta}_0}(X)\big)\Big)^{-1} \mathsf{E}\big(\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}^T(X)\big)\bigg(\Big(\mathsf{E}\big(\dot{\boldsymbol{\psi}}_{\boldsymbol{\theta}_0}(X)\big)\Big)^{-1}\bigg)^\top$. See Theorem 5.41 in van der Vaart (1998). See also, e.g., Corollary 10.16 in Henze (2024) for this representation in the context of maximum likelihood.

We now list the regularity conditions needed for (11) to hold. The so-called "classical conditions" are the following:

($c.1$) For each $\boldsymbol{\theta}$ in an open subset of Euclidean space, the mapping $\boldsymbol{\theta} \mapsto \boldsymbol{\psi}_{\boldsymbol{\theta}}(x)$ is twice continuously differentiable for every x.
($c.2$) $\mathsf{E}\big(\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\big) = 0$
($c.3$) $\mathsf{E}\big(\|\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\|^2\big) < \infty$.
($c.4$) The matrix $\mathsf{E}\big(\dot{\boldsymbol{\psi}}_{\boldsymbol{\theta}_0}(X)\big)$ exists and is non-singular.
($c.5$) The second-order partial derivatives with respect to $\boldsymbol{\theta}$ exist for every x and are dominated by a fixed integrable function $M(x)$ for every $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}_0$; that is,

$$\left\| \frac{\partial^2 \boldsymbol{\psi}_{\boldsymbol{\theta}}(x)}{\boldsymbol{\theta}^2} \right\| \leq M(x)$$

for some integrable function $M(x)$.
($c.6$) $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_0$.
($c.7$) For every n, $\frac{1}{n}\sum_{i=1}^n \boldsymbol{\psi}_{\hat{\boldsymbol{\theta}}}(X_i) = 0$.

The classical conditions are too stringent and require the existence of third derivatives (van der Vaart, 1998). This is why it is sometimes convenient to replace these assumptions using a Lipschitz condition. The alternative regularity conditions are:

($\tilde{c}.1$) For each $\boldsymbol{\theta}$ in an open subset of Euclidean space, the mapping $x \mapsto \boldsymbol{\psi}_{\boldsymbol{\theta}}(x)$ is a vector-valued function such that, for every $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ in a neighborhood of $\boldsymbol{\theta}_0$ and a measurable function with $\mathsf{E}(L(X)^2) < \infty$,

$$\|\boldsymbol{\psi}_{\boldsymbol{\theta}_1}(x) - \boldsymbol{\psi}_{\boldsymbol{\theta}_2}(x)\| \leq L(x)\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|.$$

($\tilde{c}.2$) $\mathsf{E}\big(\|\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\|^2\big) < \infty$.
($\tilde{c}.3$) The map $\boldsymbol{\theta} \mapsto \mathsf{E}\big(\boldsymbol{\psi}_{\boldsymbol{\theta}}(X)\big)$ is differentiable at a zero $\boldsymbol{\theta}_0$, with non-singular derivative matrix $\boldsymbol{V}_{\boldsymbol{\theta}_0}$.
($\tilde{c}.4$) $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_0$.
($\tilde{c}.5$) $\frac{1}{n}\sum_{i=1}^n \boldsymbol{\psi}_{\hat{\boldsymbol{\theta}}}(X_i) = o_P(n^{-1/2})$.

Under conditions ($\tilde{c}.1$)–($\tilde{c}.5$), Bahadur's representation becomes

$$\sqrt{n}\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\right)=-\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X_i)+o_P(1). \tag{12}$$

In particular, the sequence $\sqrt{n}(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0)$ is asymptotically normal with mean zero and covariance matrix $\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathsf{E}\big(\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)^T\big)\left(\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\right)^T$. See Theorem 5.21 in van der Vaart (1998) for this result.

Diego: As of now, I include both sets of conditions for completeness. But I think we should keep only ($\tilde{c}.1$)–($\tilde{c}.5$) in the future.

Proposition 3.1 provides Bahadur's representation for the von Mises–Fisher distribution on the sphere.

PROPOSITION 3.1 (a Bahadur representation, vMF$(\boldsymbol{\mu}_0,\kappa_0)$). *Let $\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n$ be iid with common distribution* vMF$(\boldsymbol{\mu}_0,\kappa_0)$ *on $\mathbb{S}^q$, where $\kappa_0>0$, and let $\boldsymbol{\xi}_0=\kappa_0\boldsymbol{\mu}_0$ be the canonical parameter. Let $\hat{\boldsymbol{\xi}}$ be the Maximum Likelihood Estimator (MLE) of $\boldsymbol{\xi}_0$. Then*

$$\sqrt{n}\big(\hat{\boldsymbol{\xi}}-\boldsymbol{\xi}_0\big)=-\big(\dot{\psi}_{\boldsymbol{\xi}_0}\big)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X}_i)+o_{\mathsf{P}}(1), \tag{13}$$

where

$$\psi_{\boldsymbol{\xi}}(\boldsymbol{x}):=\boldsymbol{x}-A_q(\|\boldsymbol{\xi}\|)\frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|} \tag{14}$$

with $A_q(r):=\mathcal{I}_{\frac{q+1}{2}}(r)/\mathcal{I}_{\frac{q-1}{2}}(r)$, $r>0$, and, writing $\boldsymbol{\xi}=\kappa\boldsymbol{\mu}$,

$$\dot{\psi}_{\boldsymbol{\xi}}(\boldsymbol{x})=-\left(A_q'(\kappa)\boldsymbol{\mu}\boldsymbol{\mu}^\top+\frac{A_q(\kappa)}{\kappa}\left(\boldsymbol{I}_{q+1}-\boldsymbol{\mu}\boldsymbol{\mu}^\top\right)\right). \tag{15}$$

The proof of Proposition 3.1 is given in Section E.1. See Section 6 for a numerical validation of the Bahadur representation (13).

We will use the notation $\dot{F}_\omega^{\boldsymbol{\theta}_0}(t):=\frac{\partial}{\partial\boldsymbol{\theta}}F_\omega^{\mu_{\boldsymbol{\theta}}}(t)\big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}$ for the partial derivative of $F_\omega^{\mu_{\boldsymbol{\theta}}}(t)$ with respect to $\boldsymbol{\theta}$ at $\boldsymbol{\theta}_0$. The following differentiability condition is required in the derivation of the asymptotic distribution of $\mathbb{G}_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}$:

(d) The map $\boldsymbol{\theta}\mapsto F^{\mu_{\boldsymbol{\theta}}}$ is differentiable at $\boldsymbol{\theta}_0$ as a map from Θ into $\ell^\infty(\Omega\times\mathbb{R}_{\geq 0})$; that is,

$$\sup_{\omega\in\Omega,t\geq 0}\left|F_\omega^{\mu_{\boldsymbol{\theta}}}(t)-F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)-\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top(\boldsymbol{\theta}-\boldsymbol{\theta}_0)\right|=o(\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|)\quad\text{as }\boldsymbol{\theta}\to\boldsymbol{\theta}_0.$$

In particular, for every $(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}$, the map $\boldsymbol{\theta}\mapsto F_\omega^{\mu_{\boldsymbol{\theta}}}(t)$ is differentiable at $\boldsymbol{\theta}_0$, and $\sup_{\omega\in\Omega,t\geq 0}\|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)\|<\infty$.

PROPOSITION 3.2. *Assumption (d) holds for the following parametric models: the normal distribution $\mathcal{N}(\mu,\sigma^2)$ on $\mathbb{R}$, when either μ, σ, or both parameters are unknown; and the* vMF *distribution on $\mathbb{S}^q$ with canonical parameter $\boldsymbol{\xi}=\kappa\boldsymbol{\mu}\in\mathbb{R}^{q+1}\setminus\{\mathbf{0}\}$.*

The proof of this result is given in Section E.2.

The following result characterizes the asymptotic behavior of $\mathbb{G}_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}$. The result follows by proving that the class of functions

$$\mathcal{F}:=\{f_{\omega,t}:\omega\in\Omega,t\in\mathbb{R}_{\geq 0}\},\ \text{where }f_{\omega,t}(x):=y_{\omega,t}(x)-F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)+\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(x), \tag{16}$$

is Donsker.

Diego: Something key here, and that has motivated how I have written the Donsker results, is that there is an extra term $o_P(1)$ that makes $\mathbb{G}_{n,\mu}^{\mu_{\hat{\theta}}}$ not be exactly the empirical process indexed by the class $\mathcal{F}$, but only asymptotically. This is why the Donsker result implies the weak convergence of $\mathbb{G}_{n,\mu}^{\mu_{\hat{\theta}}}$, but it is not an equivalence, as usually.

THEOREM 3.1 (asymptotic distribution of $\mathbb{G}_{n,\mu}^{\mu_{\hat{\theta}}}$ under the null hypothesis). *Consider an estimator $\hat{\boldsymbol{\theta}}$ satisfying the conditions (c̃.1)–(c̃.5), and assume that (d) holds. Then, under H_0, the class of functions $\mathcal{F}$ defined in (16) is Donsker. Therefore,*

$$\mathbb{G}_{n,\mu}^{\mu_{\hat{\theta}}} \xrightarrow{\mathcal{L}} \tilde{\mathbb{G}}_0 \text{ in } \ell^\infty(\mathcal{F}),$$

where $\tilde{\mathbb{G}}_0$ is a zero mean Gaussian process with covariance given by

$$\begin{aligned}\tilde{\mathcal{C}}_{(\omega_1,t_1),(\omega_2,t_2)} = {}& \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} + \dot{F}_{\omega_2}^{\boldsymbol{\theta}_0}(t_2)^\top \left(\boldsymbol{V}_{\boldsymbol{\theta}_0}\right)^{-1} \mathsf{E}\left(y_{\omega_1,t_1}(X)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\right)\\ &+ \dot{F}_{\omega_1}^{\boldsymbol{\theta}_0}(t_1)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} \mathsf{E}\left(y_{\omega_2,t_2}(X)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\right)\\ &+ \dot{F}_{\omega_1}^{\boldsymbol{\theta}_0}(t_1)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} \mathsf{E}\left(\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)^\top\right) \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} \dot{F}_{\omega_2}^{\boldsymbol{\theta}_0}(t_2),\end{aligned} \tag{17}$$

with $\mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)}$ the covariance process under the simple null, given in (10).

See Section B.1 in the Appendix for the proof of this result.

PROPOSITION 3.3 (covariance process for specific parametric models). *The following expressions hold when the unknown parameters are estimated by maximum likelihood:*

(*i*) *If $X \sim \mathcal{N}(\mu_0, \sigma^2)$, with $\sigma > 0$ fixed and μ_0 unknown, then*

$$\tilde{\mathcal{C}}_{(\omega_1,t_1),(\omega_2,t_2)} = \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} - \sigma^2 \dot{F}_{\omega_1}^{\mu_0}(t_1)\dot{F}_{\omega_2}^{\mu_0}(t_2),$$

where

$$\dot{F}_{\omega}^{\mu_0}(t) = -\frac{1}{\sigma}\phi\left(\frac{\omega - \mu_0 + t}{\sigma}\right) + \frac{1}{\sigma}\phi\left(\frac{\omega - \mu_0 - t}{\sigma}\right). \tag{18}$$

(*ii*) *Take $X \sim \mathcal{N}(\mu, \sigma_0^2)$, with $\mu \in \mathbb{R}$ fixed and σ_0 unknown, then*

$$\tilde{\mathcal{C}}_{(\omega_1,t_1),(\omega_2,t_2)} = \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} - \frac{\sigma_0^2}{2}\dot{F}_{\omega_1}^{\sigma_0}(t_1)\dot{F}_{\omega_2}^{\sigma_0}(t_2),$$

where

$$\dot{F}_{\omega}^{\sigma_0}(t) = -\frac{\omega - \mu + t}{\sigma_0^2}\phi\left(\frac{\omega - \mu + t}{\sigma_0}\right) + \frac{\omega - \mu - t}{\sigma_0^2}\phi\left(\frac{\omega - \mu - t}{\sigma_0}\right). \tag{19}$$

(*iii*) *If $X \sim \mathcal{N}(\mu_0, \sigma_0^2)$, with both μ_0 and σ_0 unknown, then*

$$\tilde{\mathcal{C}}_{(\omega_1,t_1),(\omega_2,t_2)} = \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} - \left(\dot{F}_{\omega_1}^{\mu_0,\sigma_0}(t_1)\right)^\top \mathcal{I}_{\mu_0,\sigma_0}^{-1}\dot{F}_{\omega_2}^{\mu_0,\sigma_0}(t_2),$$

where $\mathcal{I}_{\mu_0,\sigma_0} = \operatorname{diag}\left(\frac{1}{\sigma_0^2}, \frac{2}{\sigma_0^2}\right)$ and $\dot{F}_{\omega}^{\mu_0,\sigma_0}(t) := \left(\dot{F}_{\omega}^{\mu_0}(t), \dot{F}_{\omega}^{\sigma_0}(t)\right)^\top$, with $\dot{F}_{\omega}^{\mu_0}(t)$ and $\dot{F}_{\omega}^{\sigma_0}(t)$ given in (18) and (19), respectively, with σ replaced by σ_0 and μ replaced by μ_0.

(*iv*) *Take $\boldsymbol{X} \sim \mathrm{vMF}(\boldsymbol{\mu}_0, \kappa_0)$ on $\mathbb{S}^q$, and the canonical parameter $\boldsymbol{\xi}_0 = \kappa_0\boldsymbol{\mu}_0$ is considered, then*

$$\tilde{\mathcal{C}}_{(\boldsymbol{\omega}_1,t_1),(\boldsymbol{\omega}_2,t_2)} = \mathsf{P}\left(d(\boldsymbol{\omega}_1,\boldsymbol{X}) \le t_1, d(\boldsymbol{\omega}_2,\boldsymbol{X}) \le t_2\right) - F_{\boldsymbol{\omega}_1}^{\boldsymbol{\xi}_0}(t_1)F_{\boldsymbol{\omega}_2}^{\boldsymbol{\xi}_0}(t_2)\left(1 + \boldsymbol{m}_1^\top \mathsf{Var}(\boldsymbol{X})^{-1}\boldsymbol{m}_2\right),$$

where $\boldsymbol{m}_i = \mathsf{E}(\boldsymbol{X} \mid d(\boldsymbol{X},\boldsymbol{\omega}_i) \le t_i) - A_q(\kappa_0)\boldsymbol{\mu}_0$, $i = 1, 2$.

The proof of Proposition 3.3 is given in Section E.3 of the Appendix.

3.3. *Test statistics.* First, consider the Kolmogorov–Smirnov test statistic, which we denote by either T_n^{KS} (simple null hypothesis), or $\tilde{T}_n^{\mathrm{KS}}$ (composite). These are defined as

$$T_n^{\mathrm{KS}} := \sup_{\omega\in\Omega, t\geq 0} \left|\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t)\right|, \qquad \tilde{T}_n^{\mathrm{KS}} := \sup_{\omega\in\Omega, t\geq 0} \left|\mathbb{G}_{n,\mu}^{\mu_{\hat{\theta}}}(\omega,t)\right|. \tag{20}$$

Given a sample $X_1,\dots,X_n$, define the empirical probability distribution P_n as the discrete measure that assigns mass $\frac{1}{n}$ to each of the points $X_1,\dots,X_n$. Consider the Cramér–von Mises test statistic, which we denote by either T_n^{CM} (simple null hypothesis), and $\tilde{T}_n^{\mathrm{CM}}$ (composite). These are defined, respectively, as

(21)
$$T_n^{\mathrm{CM}} := \int_{\Omega\times\mathbb{R}_{\geq 0}} (\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2 \,\mathrm{d}F_{n,\omega}^{\mu}(t)\,\mathrm{d}P_n(\omega), \quad \tilde{T}_n^{\mathrm{CM}} := \int_{\Omega\times\mathbb{R}_{\geq 0}} (\mathbb{G}_{n,\mu}^{\mu_{\hat{\theta}}}(\omega,t))^2 \,\mathrm{d}F_{n,\omega}^{\mu}(t)\,\mathrm{d}P_n(\omega).$$

Observe that this is an empirical version of the Cramér–von Mises test statistic, since the integration is with respect to the empirical measure $\mathrm{d}F_{n,\omega}^{\mu}(t)\,\mathrm{d}P_n(\omega)$ instead of the population version $\mathrm{d}F_{\omega}^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega)$. The use of the empirical measure is motivated by the fact that $\mathrm{d}F_{\omega}^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega)$ depends on the distribution under the null μ_0 (and, under a composite null, on unknown parameters). Replacing it by its empirical counterpart yields a statistic whose integration weight is model-free and fully data-driven.

Theorem 3.2 proves the asymptotic distributions under the simple null hypothesis of the Kolmogorov–Smirnov and Cramér–von Mises test statistics given in (20) and (21), respectively.

THEOREM 3.2 (asymptotic distributions of T_n^{KS} and T_n^{CM} under H_0). *The following statements are true under the simple null hypothesis:*

(*i*) *For the Kolmogorov–Smirnov test statistic, as n diverges to infinity, it holds that*

$$T_n^{\mathrm{KS}} \xrightarrow{\mathcal{L}} \sup_{\omega\in\Omega, t\geq 0} \left|\mathbb{G}_0(\omega,t)\right|,$$

(*ii*) *For the Cramér–von Mises test statistic, one has*

$$T_n^{\mathrm{CM}} \xrightarrow{\mathcal{L}} \int_{\Omega\times\mathbb{R}_{\geq 0}} (\mathbb{G}_0(\omega,t))^2 \mathrm{d}F_{\omega}^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega),$$

as $n\to\infty$.

See Section B.2 in the Appendix for the proof of Theorem 3.2. See Section 6 for numerical illustrations of the convergence of T_n^{KS}.

REMARK 3.1. We know that the class $\mathcal{Y}$ is Donsker, thus $\mathbb{G}_0(\omega,t)$ is tight in $\ell^\infty(\mathcal{Y})$. Tightness is equivalent to there being a σ-compact set that has probability one (see, e.g., page 15 in Van der Vaart and Wellner, 2023, and page 105 in Kosorok, 2008). Every σ-compact set is Lindelöf, and a metric space is Lindelöf if and only if it is separable (see, e.g., Theorem 16.11 of Willard, 2004). Since there is a measurable and separable set with probability one, then $\mathbb{G}_0(\omega,t)$ is separable.

REMARK 3.2. The random variable $\sup_{\omega,t}\mathbb{G}_0(\omega,t)$ may fail to be measurable, since $\Omega\times\mathbb{R}_{\geq 0}$ is uncountable (see, e.g., page 15 in Adler and Taylor, 2007). Separability solves the measurability issues, since it guarantees the existence of a countable subset $D\subset\Omega\times\mathbb{R}_{\geq 0}$ such that $\sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}}\mathbb{G}_0(\omega,t) = \sup_{(\omega,t)\in D}\mathbb{G}_0(\omega,t)$ almost surely, and the supremum of a countable set of measurable random variables is measurable. Thus, it is sensible to speak of weak convergence of $\sup_{\omega,t} \mathbb{G}_{n,\mu}^{\mu_0}(\omega,t)$ to $\sup_{\omega,t} \mathbb{G}_0(\omega,t)$, since the limiting process is required to be separable. Observe that this also implies that $\sup_{\omega,t} \mathbb{G}_{n,\mu}^{\mu_0}(\omega,t)$ is asymptotically measurable by Lemma 1.3.8 in Van der Vaart and Wellner (2023).

Theorem 3.3 is the analog of Theorem 3.2 for the composite null hypothesis.

THEOREM 3.3 (asymptotic distributions of $\tilde{T}_n^{\mathrm{KS}}$ and $\tilde{T}_n^{\mathrm{CM}}$ under H_0). *Assume the conditions (c̃.1)–(c̃.5) and (d). The following statements are true under the composite null hypothesis:*

(*i*) *For the Kolmogorov–Smirnov test statistic, as n diverges to infinity, it holds that*

$$\tilde{T}_n^{\mathrm{KS}} \xrightarrow{\mathcal{L}} \sup_{\omega\in\Omega, t\geq 0} \big|\tilde{\mathbb{G}}_0(\omega,t)\big|.$$

(*ii*) *If, in addition, the map $(\omega,t)\mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$ is uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big) := d(\omega,\omega') + |t-t'|$, then*

$$\tilde{T}_n^{\mathrm{CM}} \xrightarrow{\mathcal{L}} \int_{\Omega\times\mathbb{R}_{\geq 0}} \tilde{\mathbb{G}}_0(\omega,t)^2 \,\mathrm{d}F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\,\mathrm{d}\mu_{\boldsymbol{\theta}_0}(\omega)$$

as $n\to\infty$.

See Section B.3 in the Appendix for the proof of Theorem 3.3. See Section 6 for numerical illustrations of the convergence of $\tilde{T}_n^{\mathrm{KS}}$.

REMARK 3.3. The uniform continuity assumption on $(\omega,t)\mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$ is automatically satisfied in many parametric models. In particular, consider maximum likelihood, so that $\mu_{\boldsymbol{\theta}}$ admits a density $p_{\boldsymbol{\theta}}$ with respect to a dominating measure and the score function is given by $\psi_{\boldsymbol{\theta}}(x) = \frac{\partial}{\partial\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(x)$. Assume moreover that the regularity conditions allowing differentiation under the integral sign are satisfied. Fix $(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}$. Then

$$\begin{aligned}\dot{F}_\omega^{\boldsymbol{\theta}_0}(t) &= \frac{\partial}{\partial\boldsymbol{\theta}} \int y_{\omega,t}(x)\, p_{\boldsymbol{\theta}}(x)\,\mathrm{d}x\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \\ &= \int y_{\omega,t}(x)\, p_{\boldsymbol{\theta}_0}(x)\, \frac{\partial}{\partial\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(x)\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \mathrm{d}x \\ &= \mathsf{E}\left[y_{\omega,t}(X)\,\psi_{\boldsymbol{\theta}_0}(X)\right].\end{aligned} \tag{22}$$

Then, for any pair $(\omega,t),(\omega',t')\in\Omega\times\mathbb{R}_{\geq 0}$, one has by Cauchy–Schwarz and Lemma B.5 that

$$\begin{aligned}\big\|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t) - \dot{F}_{\omega'}^{\boldsymbol{\theta}_0}(t')\big\| &\leq \Big(\mathsf{E}\|\psi_{\boldsymbol{\theta}_0}(X)\|^2\Big)^{1/2} \Big(\mathsf{E}\big(y_{\omega,t}(X) - y_{\omega',t'}(X)\big)^2\Big)^{1/2} \\ &\leq \Big(\mathsf{E}\|\psi_{\boldsymbol{\theta}_0}(X)\|^2\Big)^{1/2} C_1 \sqrt{d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big)}\end{aligned}$$

for some constant $C_1>0$. Hence, $(\omega,t)\mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$ is uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}$.

Taking the supremum over all values of $\omega\in\Omega$ can be infeasible in practice, especially in high dimensions. For this reason, we provide conditions under which the supremum over the whole metric space can be replaced with maximization over the sample elements. Proposition 3.4 deals precisely with this.

PROPOSITION 3.4. *The following results hold:*

(i) *Assume that* $\Omega \subset \operatorname{supp}(\mu_0)$. *Under the simple null hypothesis, it holds that*

$$\sup_{\omega\in\Omega, t\geq 0} \left|\mathbb{G}^{\mu_0}_{n,\mu}(\omega,t)\right| = \max_{1\leq i,j\leq n} \left|\mathbb{G}^{\mu_0}_{n,\mu}(X_i, d(X_i,X_j))\right| + o_{\mathsf{P}^*}(1). \tag{23}$$

(ii) *Assume also that the conditions of Theorem 3.1 hold,* $\Omega \subset \operatorname{supp}(\mu_{\boldsymbol{\theta}_0})$, *and that the map* $(\omega,t) \mapsto \dot{F}^{\boldsymbol{\theta}_0}_{\omega}(t)$ *is uniformly continuous with respect to* $d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big) := d(\omega,\omega') + |t-t'|$. *Then, under the composite null hypothesis, it holds that*

$$\sup_{\omega\in\Omega, t\geq 0} \left|\mathbb{G}^{\mu_{\hat{\boldsymbol{\theta}}}}_{n,\mu}(\omega,t)\right| = \max_{1\leq i,j\leq n} \left|\mathbb{G}^{\mu_{\hat{\boldsymbol{\theta}}}}_{n,\mu}(X_i, d(X_i,X_j))\right| + o_{\mathsf{P}^*}(1). \tag{24}$$

See Section B.4 in the Appendix for the proof of Proposition 3.4.

4. Non-null asymptotics.

4.1. *Fixed alternatives.* Suppose that $X_1,\ldots,X_n$ are generated from a distribution μ that is different from μ_0, the distribution under the null hypothesis. In other words, we are interested in the behavior of the test statistics under alternatives of the form

$$H_1 : \mu \neq \mu_0 \text{ (simple null)},$$

$$H_1 : \mu \notin \{\mu_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \boldsymbol{\Theta}\} \text{ (composite null)}.$$

4.1.1. *Simple null hypothesis.* In terms of distance profiles, the alternative $H_1 : \mu \neq \mu_0$ is equivalent to the existence of a pair $(\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}$ such that $F^{\mu}_{\omega}(t) \neq F^{\mu_0}_{\omega}(t)$. Theorem 4.1 shows that the test based on T_n^{KS} is consistent against fixed alternatives.

THEOREM 4.1 (consistency of T_n^{KS} under a fixed alternative). *Under H_1, the Kolmogorov–Smirnov statistic T_n^{KS} diverges in outer probability.*

A similar result holds for the Cramér–von Mises statistic T_n^{CM}. Observe that under the alternative hypothesis, we only know that $F^{\mu}_{\omega}(t) \neq F^{\mu_0}_{\omega}(t)$ at a single point $(\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}$. Theorem 4.2 is proved by extending (ω,t) to a neighborhood where the separation is uniform, in Lemma C.2. Then, Lemma C.3 shows that the probability measure of this neighborhood is positive under the alternative, so that the Cramér von-Mises statistic can detect the separation between the distance profiles under the null and the alternative. Finally, in Theorem 4.2, a Glivenko–Cantelli argument allows us to replace the empirical ingredients in T_n^{CM} by their population counterparts, and the law of large numbers for U-statistics yields the desired result.

THEOREM 4.2 (divergence of T_n^{CM} under a fixed alternative). *Assume that either (a.2) holds, or (a.1) holds and* $\operatorname{supp}(\mu) = \Omega$. *Then, the test statistic T_n^{CM} diverges in outer probability under H_1.*

REMARK 4.1. Assumption *(a.2)* guarantees the existence of a point (ω,t) at which $F^{\mu}_{\omega}(t) \neq F^{\mu_0}_{\omega}(t)$, and such that $\omega \in \operatorname{supp}(\mu)$. In contrast, Assumption *(a.1)* alone only ensures the existence of such a point with $\omega \in \Omega$, which need not belong to $\operatorname{supp}(\mu)$. The extra assumption $\operatorname{supp}(\mu) = \Omega$ guarantees $\omega \in \operatorname{supp}(\mu)$. This ensures that the differences between $F^{\mu}_{\omega}(t)$ and $F^{\mu_0}_{\omega}(t)$ are detected by the Cramér–von Mises statistic.

See Section C.1.1 for the proof of Theorems 4.1 and 4.2.

4.1.2. *Composite null hypothesis.* Focus now on the composite null hypothesis. In terms of distance profiles, the alternative hypothesis means that for every $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ there exists at least one pair $(\omega, t) \in \Omega \times \mathbb{R}_{\geq 0}$ such that $F^{\mu}_{\omega}(t) \neq F^{\mu_{\boldsymbol{\theta}}}_{\omega}(t)$. Under H_1, the estimated parameter $\hat{\boldsymbol{\theta}}$ does not converge to a "true" parameter value, since the parametric model under which $\hat{\boldsymbol{\theta}}$ is computed is misspecified. Hence, we work with a value $\boldsymbol{\theta}_1$, which is the limit to which $\hat{\boldsymbol{\theta}}$ converges in probability under H_1.

Diego: The term "true" is written in quotation marks because, under the alternative, the data-generating distribution may lie outside the parametric family under the composite null, so a true parameter value may actually not exist.

THEOREM 4.3 (divergence of $\tilde{T}^{\mathrm{KS}}_n$ under a fixed alternative). *Consider an estimator $\hat{\boldsymbol{\theta}}$, and assume that there exists $\boldsymbol{\theta}_1 \in \boldsymbol{\Theta}$ such that $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_1$. Assume also that the map $\boldsymbol{\theta} \mapsto F^{\mu_{\boldsymbol{\theta}}}$ is continuous at $\boldsymbol{\theta}_1$ as a map from $\boldsymbol{\Theta}$ into $\ell^\infty(\Omega \times \mathbb{R}_{\geq 0})$; that is,*

$$\sup_{\omega \in \Omega, t \geq 0} \left| F^{\mu_{\boldsymbol{\theta}}}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t) \right| \to 0 \quad \text{as } \boldsymbol{\theta} \to \boldsymbol{\theta}_1. \tag{25}$$

Then, under H_1, the statistic $\tilde{T}^{\mathrm{KS}}_n$ diverges in outer probability, i.e., $\tilde{T}^{\mathrm{KS}}_n \xrightarrow{\mathsf{P}^*} \infty$.

Condition (25) is a weaker analogue of Assumption *(d)* at $\boldsymbol{\theta}_1$, requiring continuity rather than differentiability.

REMARK 4.2 (sufficient conditions for (25)). A sufficient condition for (25) is that there exist $\rho > 0$ and $L < \infty$ such that

$$\sup_{\boldsymbol{\theta} \in B(\boldsymbol{\theta}_1, \rho)} \sup_{\omega \in \Omega, t \geq 0} \left\| \dot{F}^{\boldsymbol{\theta}}_{\omega}(t) \right\| \leq L \tag{26}$$

where the differentiability of $\boldsymbol{\theta} \mapsto F^{\mu_{\boldsymbol{\theta}}}_{\omega}(t)$ in $B(\boldsymbol{\theta}_1, \rho)$, for every $(\omega, t) \in \Omega \times \mathbb{R}_{\geq 0}$, is implicitly assumed. Indeed, if (26) holds, then the mean value theorem gives, for every $\boldsymbol{\theta} \in B(\boldsymbol{\theta}_1, \rho)$, that $\sup_{\omega, t} \left| F^{\mu_{\boldsymbol{\theta}}}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t) \right| \leq L \|\boldsymbol{\theta} - \boldsymbol{\theta}_1\|$, which implies (25).

Recall the score representation in (22) (in the setting of Remark 3.3). It follows that $\left\| \dot{F}^{\boldsymbol{\theta}}_{\omega}(t) \right\| = \left\| \mathsf{E}_{\boldsymbol{\theta}}\big(y_{\omega,t}(X)\, \psi_{\boldsymbol{\theta}}(X)\big) \right\| \leq \mathsf{E}_{\boldsymbol{\theta}} \left\| \psi_{\boldsymbol{\theta}}(X) \right\|$. Consequently, (26) follows if there exists $\rho > 0$ such that

$$\sup_{\boldsymbol{\theta} \in B(\boldsymbol{\theta}_1, \rho)} \mathsf{E}_{\boldsymbol{\theta}} \left\| \psi_{\boldsymbol{\theta}}(X) \right\| < \infty, \tag{27}$$

which is independent of (ω, t).

Condition (27) is satisfied, for instance, by the vMF distribution on $\mathbb{S}^q$ with canonical parameter $\boldsymbol{\xi} = \kappa \boldsymbol{\mu}$, since by (14), $\|\psi_{\boldsymbol{\xi}}(x)\| \leq \|x\| + A_q(\|\boldsymbol{\xi}\|) \leq 2$ for all $x \in \mathbb{S}^q$. For $X \sim \mathcal{N}(\mu_0, \sigma^2)$ with μ_0 fixed and $\sigma > 0$ unknown, it holds by (120) that $\sup_{\omega,t} \left| \dot{F}^{\sigma}_{\omega}(t) \right| \leq \frac{2}{\sigma} \sup_{u \in \mathbb{R}} |u| \varphi(u) = \frac{2}{\sigma\sqrt{2\pi e}}$. Thus, (26) holds for any $\rho \in (0, \sigma_1)$. For $X \sim \mathcal{N}(\mu_0, \sigma^2)$ with μ unknown and $\sigma_0 > 0$ fixed, by (119) and the bound $\phi(u) \leq \frac{1}{\sqrt{2\pi}}$ for all $u \in \mathbb{R}$, we have $\sup_{\omega,t} \left| \dot{F}^{\mu}_{\omega}(t) \right| \leq \frac{\sqrt{2}}{\sigma_0 \sqrt{\pi}} < \infty$. Hence, (26) holds for every $\rho > 0$. The case when both μ and σ are unknown follows by combining the two previous cases.

Theorem 4.4 is the analog of Theorem 4.2 for the composite null hypothesis.

THEOREM 4.4 (divergence of $\tilde{T}^{\mathrm{CM}}_n$ under a fixed alternative, composite null). *Consider an estimator $\hat{\boldsymbol{\theta}}$, and assume that there exists $\boldsymbol{\theta}_1 \in \boldsymbol{\Theta}$ such that $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}_\mu} \boldsymbol{\theta}_1$. Assume also that the continuity condition* (25) *holds. Assume that either (a.2) holds, or (a.1) holds and* $\operatorname{supp}(\mu) = \Omega$*. It holds that $\tilde{T}^{\mathrm{CM}}_n \xrightarrow{\mathsf{P}^*_\mu} \infty$ under H_1.*

See Section C.1.2 for the proof of the results in this section.

4.2. *Local alternatives.* We study the asymptotic behavior of the Kolmogorov–Smirnov and Cramér–von Mises statistics under local alternatives, for simple and composite null hypotheses. Fix a probability measure G on $(\Omega, \mathcal{B}(\Omega))$, which we assume that satisfies (*b.2*), so that Theorem 5.1 in Dubey, Chen and Müller (2024) holds (Diego: maybe not necessary to say it, since we stated that (*b.2*) is assumed globally in the paper).

4.2.1. *Simple null hypothesis.* Define the sequence of local alternatives converging to H_0 at rate $n^{-1/2}$ by

$$Q_n := (1 - n^{-\frac{1}{2}})\,\mu_0 + n^{-\frac{1}{2}}\,G, \qquad n \in \mathbb{N}. \tag{28}$$

For $(\omega, t) \in \Omega \times \mathbb{R}_{\geq 0}$, this implies

$$F_\omega^{Q_n}(t) := (1 - n^{-\frac{1}{2}})F_\omega^{\mu_0}(t) + n^{-\frac{1}{2}} F_\omega^{G}(t) = F_\omega^{\mu_0}(t) + n^{-\frac{1}{2}} h(\omega, t),$$

where $h(\omega, t) := F_\omega^G(t) - F_\omega^{\mu_0}(t)$. Hence,

$$\sqrt{n}\Big(F_{n,\omega}^{Q_n}(t) - F_\omega^{\mu_0}(t)\Big) = \sqrt{n}\Big(F_{n,\omega}^{Q_n}(t) - F_\omega^{Q_n}(t)\Big) + h(\omega, t) = \mathbb{G}_{n,Q_n}^{Q_n}(\omega, t) + h(\omega, t), \tag{29}$$

where $\mathbb{G}_{n,Q_n}^{Q_n}(\omega, t) := \sqrt{n}\Big(F_{n,\omega}^{Q_n}(t) - F_\omega^{Q_n}(t)\Big) = \sqrt{n}(P_n - Q_n)y_{\omega,t}$. Observe that $\mathbb{G}_{n,Q_n}^{Q_n}$ is a triangular-array empirical process, since the underlying law Q_n varies with n.

The following result shows that $\mathbb{G}_{n,Q_n}^{Q_n}$ admits a coupling with $\mathbb{G}_{n,\mu_0}^{\mu_0}$ such that their difference converges in probability to zero in $\ell^\infty(\mathcal{F})$.

PROPOSITION 4.1. *Let $\mathcal{F}$ be a μ_0- and G-Donsker class of measurable real-valued functions. Assume that $\sup_{f \in \mathcal{F}} \big|(G - \mu_0)f\big| < \infty$. Then, under the local alternative (28), $\mathbb{G}_{n,Q_n}^{Q_n}$ admits a coupling with $\mathbb{G}_{n,\mu_0}^{\mu_0}$ such that*

$$\sup_{f \in \mathcal{F}} \Big|\mathbb{G}_{n,Q_n}^{Q_n}(f) - \mathbb{G}_{n,\mu_0}^{\mu_0}(f)\Big| \xrightarrow{\mathsf{P}} 0. \tag{30}$$

COROLLARY 4.1 (asymptotic distribution of $\mathbb{G}_{n,Q_n}^{Q_n}$ under the local alternative). *Under the assumptions of Proposition 4.1, one has*

$$\mathbb{G}_{n,Q_n}^{Q_n} \xrightarrow{\mathcal{L}} \mathbb{G}_0 \quad \text{in } \ell^\infty(\mathcal{F}), \tag{31}$$

under the local alternative (28).

PROOF OF COROLLARY 4.1. Using the coupled versions provided by Proposition 4.1, one has $\sup_{f \in \mathcal{F}} \big|\mathbb{G}_{n,Q_n}^{Q_n}(f) - \mathbb{G}_{n,\mu_0}^{\mu_0}(f)\big| \xrightarrow{\mathsf{P}} 0$. Since $\mathbb{G}_{n,\mu_0}^{\mu_0} \xrightarrow{\mathcal{L}} \mathbb{G}_0$ in $\ell^\infty(\mathcal{F})$ (Donsker), Slutsky's theorem (recall that $\mathbb{G}_0$ is separable) yields (31). □

The following theorem gives the weak limit of the Kolmogorov–Smirnov statistic under the local alternative.

THEOREM 4.5 (asymptotic distribution of T_n^{KS} under the local alternative). *Under the local alternative (28), it holds that*

$$T_n^{\mathrm{KS}} \xrightarrow{\mathcal{L}} \sup_{(\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}} |\mathbb{G}_0(\omega, t) + h(\omega, t)|.$$

Theorem 4.6 is the analog of Theorem 4.5 for the Cramér–von Mises statistic.

THEOREM 4.6 (asymptotic distribution of T_n^{CM} under the local alternative). *Under the local alternative* (28)*, it holds that*

$$T_n^{\mathrm{CM}} \xrightarrow{\mathcal{L}} \int_{\Omega\times\mathbb{R}_{\geq 0}} \big(\mathbb{G}_0(\omega,t)+h(\omega,t)\big)^2 \,\mathrm{d}F_\omega^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega).$$

See Section C.2 in the Appendix for the proofs of Proposition 4.1 and Theorems 4.5 and 4.6.

4.2.2. *Composite null hypothesis.* For the composite case, define the sequence of local alternatives converging to H_0 at rate $n^{-1/2}$ by

$$Q_n := (1-n^{-\frac{1}{2}})\,\mu_{\boldsymbol{\theta}_0} + n^{-\frac{1}{2}}\,G, \qquad n\in\mathbb{N}. \tag{32}$$

Accordingly, $h(\omega,t) := F_\omega^G(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)$.

Under the local alternative, there are two relevant parameters: the estimator $\hat{\boldsymbol{\theta}}$ based on P_n, and the true population parameter under the local alternative, $\tilde{\boldsymbol{\theta}}_n$, defined as any solution to $\mathsf{E}_{Q_n}\big(\psi_{\boldsymbol{\theta}}(X)\big)=0$. We first give sufficient local conditions under which $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ both converge ($\hat{\boldsymbol{\theta}}$ in probability) to $\boldsymbol{\theta}_0$ under the local alternative (32).

Diego: The minimal assumptions I am able to come up with are below. It all boils down to finding a good neighborhood of $\boldsymbol{\theta}_0$ in which the score functions are "sufficiently well-behaved". The only complaint I have about this is that we assume that the estimated parameters $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ belong to that neighborhood, which is more than just saying that there is **some** neighborhood that contains them. Still, this is less restrictive in general than assuming the consistency of $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ directly, and all the conditions are local around $\boldsymbol{\theta}_0$. It would be interesting to check how restrictive our conditions are in examples, but right now I prioritize making progress.

PROPOSITION 4.2 (sufficient conditions for the convergence of $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ to $\boldsymbol{\theta}_0$). *Assume conditions (c̃.2) and (c̃.3) and* $\mathsf{E}_G\|\psi_{\boldsymbol{\theta}_0}(X)\|^2<\infty$*. Consider a sufficiently small bounded neighborhood U of $\boldsymbol{\theta}_0$. Assume that there exists a measurable function L such that* $\mathsf{E}_{\mu_{\boldsymbol{\theta}_0}}\big(L(X)^2\big)<\infty$, $\mathsf{E}_G\big(L(X)^2\big)<\infty$*, and*

$$\|\psi_{\boldsymbol{\theta}_1}(x)-\psi_{\boldsymbol{\theta}_2}(x)\| \leq L(x)\|\boldsymbol{\theta}_1-\boldsymbol{\theta}_2\| \quad \textit{for all } \boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in U \textit{ and all } x. \tag{33}$$

The following results hold under the local alternative (32)*:*

(*i*) *Let* $\tilde{\boldsymbol{\theta}}_n\in\Theta$ *be such that* $\mathsf{E}_{Q_n}\big(\psi_{\tilde{\boldsymbol{\theta}}_n}(X)\big)=0$ *and* $\tilde{\boldsymbol{\theta}}_n\in U$ *for all n sufficiently large. Then,* $\tilde{\boldsymbol{\theta}}_n\to\boldsymbol{\theta}_0$ *as* $n\to\infty$.

(*ii*) *Let* $\hat{\boldsymbol{\theta}}\in\Theta$ *be such that* $\mathsf{E}_{P_n}\big(\psi_{\hat{\boldsymbol{\theta}}}(X)\big)=0$ *and* $\mathsf{P}(\hat{\boldsymbol{\theta}}\in U)\to 1$*. Then,* $\hat{\boldsymbol{\theta}}\xrightarrow{\mathsf{P}}\boldsymbol{\theta}_0$ *as* $n\to\infty$.

Observe that condition (33) is a local version of (*c̃.1*).

Obtaining the asymptotic distribution of the test statistics under the local alternative requires a Bahadur-type asymptotic linear expansion of $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ as in (12), under the local alternative. Since the parametric model is misspecified under Q_n, it is important to provide sufficient conditions under which such expansions hold. This is given in Theorem 4.7. To make the following result more general, we only require that $\tilde{\boldsymbol{\theta}}_n$ and $\hat{\boldsymbol{\theta}}$ converge to $\boldsymbol{\theta}_0$, instead of assuming the conditions of Proposition 4.2 directly.

THEOREM 4.7 (a Bahadur expansion of $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ under the local alternative). *Assume conditions (c̃.1)–(c̃.3),* $\mathsf{E}_G\big(L(X)^2\big)<\infty$*, and* $\mathsf{E}_G\|\psi_{\boldsymbol{\theta}_0}(X)\|^2<\infty$*. Under the local alternative, the following results hold:*

(*i*) *Let* $\tilde{\boldsymbol{\theta}}_n \in \Theta$ *be such that* $\mathsf{E}_{Q_n}\big(\psi_{\tilde{\boldsymbol{\theta}}_n}(X)\big) = 0$ *and* $\tilde{\boldsymbol{\theta}}_n \to \boldsymbol{\theta}_0$. *Then,*

$$\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0 = -\frac{1}{\sqrt{n}}\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathsf{E}_G\big(\psi_{\boldsymbol{\theta}_0}(X)\big) + o(n^{-\frac{1}{2}}), \tag{34}$$

(*ii*) *Let* $\hat{\boldsymbol{\theta}} \in \Theta$ *be such that* $\mathsf{E}_{P_n}\big(\psi_{\hat{\boldsymbol{\theta}}}(X)\big) = o_P(n^{-1/2})$ *and* $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_0$. *Then,*

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 = -\frac{1}{n}\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\sum_{i=1}^{n}\psi_{\tilde{\boldsymbol{\theta}}_n}(X_i) - \frac{1}{\sqrt{n}}\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathsf{E}_G\big(\psi_{\boldsymbol{\theta}_0}(X)\big) + o_P(n^{-\frac{1}{2}}), \tag{35}$$

The following result gives the asymptotic distribution of $\mathbb{G}_{n,Q_n}^{\mu_{\hat{\boldsymbol{\theta}}}} := \sqrt{n}\big(F_{n,\omega}^{Q_n}(t) - F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t)\big)$ under the local alternative (32) in the composite null case.

THEOREM 4.8 (asymptotic distribution of $\mathbb{G}_{n,Q_n}^{\mu_{\hat{\boldsymbol{\theta}}}}$ under the local alternative). *Assume conditions (c̃.1)–(c̃.3) and (d),* $\mathsf{E}_G\big(L(X)^2\big) < \infty$, *and* $\mathsf{E}_G\|\psi_{\boldsymbol{\theta}_0}(X)\|^2 < \infty$. *Let* $\tilde{\boldsymbol{\theta}}_n \in \Theta$ *be such that* $\mathsf{E}_{Q_n}\big(\psi_{\tilde{\boldsymbol{\theta}}_n}(X)\big) = 0$ *and* $\tilde{\boldsymbol{\theta}}_n \to \boldsymbol{\theta}_0$. *Let also* $\hat{\boldsymbol{\theta}} \in \Theta$ *be such that* $\mathsf{E}_{P_n}\big(\psi_{\hat{\boldsymbol{\theta}}}(X)\big) = o_P(n^{-1/2})$ *and* $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_0$ *under the local alternative* (32). *Then, it holds that*

$$\mathbb{G}_{n,Q_n}^{\mu_{\hat{\boldsymbol{\theta}}}} \xrightarrow{\mathcal{L}} \tilde{\mathbb{G}}_0(\omega,t) + h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathsf{E}_G\big(\psi_{\boldsymbol{\theta}_0}(X)\big) \quad \text{in } \ell^\infty(\mathcal{F}) \tag{36}$$

under the local alternative.

REMARK 4.3. In Theorem 4.8, we have not assumed directly the asymptotic linear expansions for $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}_n$ given in (34) and (35). This is because the conditions of Theorem 4.7 are also used in the proof of Theorem 4.8 beyond the derivation of these expansions themselves.

The next theorem uses this convergence result to derive the asymptotic behavior of the Kolmogorov–Smirnov and Cramér–von Mises statistics under local alternatives for the composite null hypothesis.

THEOREM 4.9 (asymptotic distribution of $\tilde{T}_n^{\mathrm{KS}}$ and $\tilde{T}_n^{\mathrm{CM}}$ under the local alternative). *Assuming the conditions of Theorem 4.8, it holds under the local alternative that*

$$\tilde{T}_n^{\mathrm{KS}} = \sup_{\omega\in\Omega, t\geq 0}\left|\mathbb{G}_{n,Q_n}^{\mu_{\hat{\boldsymbol{\theta}}}}(\omega,t)\right| \xrightarrow{\mathcal{L}} \sup_{\omega\in\Omega, t\geq 0}\left|\tilde{\mathbb{G}}_0(\omega,t) + h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathsf{E}_G(\psi_{\boldsymbol{\theta}_0}(X))\right|.$$

Assume also that $(\omega,t) \mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$ *is uniformly continuous with respect to* $d_{\Omega\times\mathbb{R}_{\geq 0}}$. *Then, under the local alternative, it holds that*

$$\tilde{T}_n^{\mathrm{CM}} \xrightarrow{\mathcal{L}} \int_{\Omega\times\mathbb{R}_{\geq 0}}\Big(\tilde{\mathbb{G}}_0(\omega,t) + h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathsf{E}_G(\psi_{\boldsymbol{\theta}_0}(X))\Big)^2\, dF_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\, d\mu_{\boldsymbol{\theta}_0}(\omega).$$

See Section C.2.2 for the proof of the results from this section.

5. Multiplier bootstrap. Calculations with the theoretical distribution of the test statistics, such as critical values or p-values, can be complicated in practice. For this reason, we propose a multiplier bootstrap approach and prove its consistency under some general conditions.

Some basic concepts and notation are required before introducing the multiplier bootstrap. Let $\mathcal{H} = \{h_{\omega,t} : \omega \in \Omega,\ t \geq 0\}$ denote the indexing class, with $\mathcal{H} = \mathcal{Y}$ (simple null) or $\mathcal{H} = \mathcal{F}$ (composite null), with $\mathcal{Y}$ and $\mathcal{F}$ as defined in (4) and (16), respectively. Consider an iid sequence $X_1, X_2, \ldots$ of random objects in (Ω, d) with common law P, where $P = \mu_0$ (simple null), or $P = \mu_{\boldsymbol{\theta}_0}$ (composite null). Let $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots)$ be a sequence of nonnegative iid random variables, independent of $X_1, X_2, \ldots$, with mean $0 < m < \infty$ and variance $0 < \tau^2 < \infty$. Assume that $\|\xi\|_{2,1} < \infty$ where $\|\xi\|_{2,1} = \int_0^\infty \sqrt{\mathsf{P}(|\xi| > x)}\, dx$. Let E_* and P_* denote conditional expectation and conditional probability with respect to the multipliers $\boldsymbol{\xi}$, given $X_1, \ldots, X_n$, i.e., we condition on the sample, and the remaining randomness is in the multipliers.

A key result (Kosorok, 2008) is that, for a sequence of random elements $(\mathbb{Z}_n)_{n\geq 1}$ and a random element $\mathbb{Z}$ in $\ell^\infty(\mathcal{H})$, endowed with the supremum norm, $\mathbb{Z}_n \xrightarrow{\mathcal{L}} \mathbb{Z}$ if and only if

$$\sup_{\varphi \in \mathrm{BL}_1(\ell^\infty(\mathcal{H}))} |\mathsf{E}^* \varphi(\mathbb{Z}_n) - \mathsf{E}\varphi(\mathbb{Z})| \to 0, \tag{37}$$

where E^* denotes outer expectation. Here, $\mathrm{BL}_1(\ell^\infty(\mathcal{H}))$ is the class of bounded Lipschitz functions $\varphi : \ell^\infty(\mathcal{H}) \mapsto \mathbb{R}$ satisfying $\|\varphi\|_\infty \leq 1$ and $|\varphi(z) - \varphi(z')| \leq \|z - z'\|_{\mathcal{H}}$ for all $z, z' \in \ell^\infty(\mathcal{H})$. This result is used to define the convergence of the bootstrap distributions.

For a tight random element $\mathbb{Z}$ in $\ell^\infty(\mathcal{H})$, the convergence $\mathbb{Z}_n^* \underset{*}{\overset{P}{\to}} \mathbb{Z}$ (Kosorok, 2008) means that, conditionally on $X_1, \ldots, X_n$,

$$\sup_{\varphi \in \mathrm{BL}_1(\ell^\infty(\mathcal{H}))} |\mathsf{E}_* \varphi(\mathbb{Z}_n^*) - \mathsf{E}(\varphi)(\mathbb{Z})| \xrightarrow{\mathsf{P}} 0, \tag{38}$$

and for all $\varphi \in \mathrm{BL}_1(\ell^\infty(\mathcal{H}))$, $\mathsf{E}_* \overline{\varphi(\mathbb{Z}_n^*)} - \mathsf{E}_* \underline{\varphi(\mathbb{Z}_n^*)} \xrightarrow{\mathsf{P}} 0$. The quantities $\overline{\varphi(\mathbb{Z}_n^*)}$ and $\underline{\varphi(\mathbb{Z}_n^*)}$ are measurable majorants and minorants, respectively, with respect to the data, including the weights $\boldsymbol{\xi}$. Contrary to (37), the convergence in (38) is in probability because the conditional expectations depend on the sample.

Let P_n^* denote the (multiplier) bootstrapped empirical measure, given by

$$P_n^* f = \frac{1}{n} \sum_{i=1}^n \frac{\xi_i}{\bar{\xi}_n} f(X_i),$$

where $\bar{\xi}_n = \frac{1}{n}\sum_{i=1}^n \xi_i$ and $P_n^* \equiv 0$ if $\bar{\xi}_n = 0$. Define

$$F_{n,\omega}^*(t) := P_n^* y_{\omega,t} = \frac{1}{n} \sum_{i=1}^n \frac{\xi_i}{\bar{\xi}_n} y_{\omega,t}(X_i).$$

We set $F_{n,\omega}^*(t) = 0$ if $\bar{\xi}_n = 0$.

Theorem 2.6 in Kosorok (2008) gives the asymptotic distribution of $\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)$. This motivates the following definition of the multiplier bootstrap processes $\mathbb{G}_n^*$ and $\tilde{\mathbb{G}}_n^*$, which are the bootstrap analogues of $\mathbb{G}_{n,\mu}^{\mu_0}$ and $\mathbb{G}_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}$, respectively. Under a simple null hypothesis, set

$$\mathbb{G}_n^*(\omega, t) := \frac{m}{\tau}\sqrt{n}\left[\left\{F_{n,\omega}^*(t) - F_\omega^{\mu_0}(t)\right\} - \left\{F_{n,\omega}^{\mu}(t) - F_\omega^{\mu_0}(t)\right\}\right] = \frac{m}{\tau}\sqrt{n}\left(F_{n,\omega}^*(t) - F_{n,\omega}^{\mu}(t)\right).$$

This definition connects with the bootstrapped empirical process, since $\mathbb{G}_n^*(\omega, t) = \frac{m}{\tau}\sqrt{n}(P_n^* - P_n) y_{\omega,t}$.

For a composite null hypothesis, let $\hat{\boldsymbol{\theta}}^*$ be the (multiplier) bootstrap version of $\hat{\boldsymbol{\theta}}$, obtained by replacing P_n with P_n^* in the estimating procedure. We assume that $\hat{\boldsymbol{\theta}}^*$ is conditionally measurable given the sample. Define

$$\tilde{\mathbb{G}}_n^*(\omega, t) := \frac{m}{\tau}\sqrt{n}\left[\left\{F_{n,\omega}^*(t) - F_\omega^{\mu_{\hat{\boldsymbol{\theta}}^*}}(t)\right\} - \left\{F_{n,\omega}^{\mu}(t) - F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t)\right\}\right].$$

For the composite case, the connection between $\tilde{\mathbb{G}}_n^*$ and the bootstrapped empirical process is given in Theorem 5.1. This is the essential ingredient to obtain the asymptotic distribution of $\tilde{\mathbb{G}}_n^*$ (similarly for $\mathbb{G}_n^*$).

THEOREM 5.1 (asymptotic distribution of the multiplier bootstrap processes). *The following statements are true:*

(i) *Under the simple null hypothesis, it holds that*

$$\mathbb{G}_n^* \underset{*}{\overset{P}{\to}} \mathbb{G}_0 \quad \textit{in } \ell^\infty(\mathcal{Y}).$$

(ii) *Assume the conditions of Theorem 3.1 and the bootstrap analogue of Bahadur's representation* (12)*:*

$$\frac{m}{\tau}\sqrt{n}\left(\hat{\boldsymbol{\theta}}^* - \hat{\boldsymbol{\theta}}\right) = -\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\frac{m}{\tau}\frac{1}{\sqrt{n}}\sum_{i=1}^n\left(\frac{\xi_i}{\bar{\xi}_n} - 1\right)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X_i) + o_{P_*}(1). \tag{39}$$

Then, it holds under the composite null hypothesis that

$$\sup_{\omega\in\Omega, t\geq 0}\left|\tilde{\mathbb{G}}_n^*(\omega,t) - \frac{m}{\tau}\sqrt{n}(P_n^* - P_n)f_{\omega,t}\right| = o_{P_*}(1). \tag{40}$$

This yields

$$\tilde{\mathbb{G}}_n^* \underset{*}{\overset{P}{\to}} \tilde{\mathbb{G}}_0 \quad \textit{in } \ell^\infty(\mathcal{F}).$$

The bootstrap versions of the test statistics under the simple null hypothesis are then

$$T_n^{*\mathrm{KS}} := \sup_{\omega\in\Omega, t\geq 0}\left|\mathbb{G}_n^*(\omega,t)\right|, \quad T_n^{*\mathrm{CM}} := \int_{\Omega\times\mathbb{R}_{\geq 0}} \left(\mathbb{G}_n^*(\omega,t)\right)^2 \,\mathrm{d}F_{n,\omega}^{\mu}(t)\,\mathrm{d}P_n(\omega),$$

and for the composite null hypothesis,

$$\tilde{T}_n^{*\mathrm{KS}} := \sup_{\omega\in\Omega, t\geq 0}\left|\tilde{\mathbb{G}}_n^*(\omega,t)\right|, \quad \tilde{T}_n^{*\mathrm{CM}} := \int_{\Omega\times\mathbb{R}_{\geq 0}} \left(\tilde{\mathbb{G}}_n^*(\omega,t)\right)^2 \,\mathrm{d}F_{n,\omega}^{\mu}(t)\,\mathrm{d}P_n(\omega).$$

THEOREM 5.2 (asymptotic distribution of the bootstrap test statistics). *The following statements are true:*

(i) *Under the simple null hypothesis, it holds that*

$$T_n^{*\mathrm{KS}} \underset{*}{\overset{P}{\to}} \sup_{\omega\in\Omega, t\geq 0}|\mathbb{G}_0(\omega,t)|, \quad T_n^{*\mathrm{CM}} \underset{*}{\overset{P}{\to}} \int_{\Omega\times\mathbb{R}_{\geq 0}} \mathbb{G}_0(\omega,t)^2\,\mathrm{d}F_\omega^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega).$$

(ii) *Under the composite null hypothesis, assuming the conditions of Theorem 5.1, one has that*

$$\tilde{T}_n^{*\mathrm{KS}} \underset{*}{\overset{P}{\to}} \sup_{\omega\in\Omega, t\geq 0}|\tilde{\mathbb{G}}_0(\omega,t)|.$$

If, in addition, the map $(\omega,t)\mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$ is uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}$ and, for every $(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}$, the map $\boldsymbol{\theta}\mapsto F_\omega^{\mu_{\boldsymbol{\theta}}}(t)$ is Borel measurable, then

$$\tilde{T}_n^{*\mathrm{CM}} \underset{*}{\overset{P}{\to}} \int_{\Omega\times\mathbb{R}_{\geq 0}} \tilde{\mathbb{G}}_0(\omega,t)^2\,\mathrm{d}F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\,\mathrm{d}\mu_{\boldsymbol{\theta}_0}(\omega).$$

See Section D in the Appendix for the proof of Theorems 5.1 and 5.2.

6. Numerical experiments. We present numerical results to validate some theoretical results derived in the previous sections. Figures **??**–**??** illustrate the theoretical distance profiles obtained in Proposition 2.1 by comparing them with empirical distance profiles from simulated samples. In each plot, the theoretical distance profile is compared with 10 empirical distance profiles obtained from 10 different samples, and pointwise 95% confidence bands (based on the normal distribution) for the distance profiles are also included.

Figures **??**–**??** show the convergence of T_n^{KS} and $\tilde{T}_n^{\mathrm{KS}}$ to their respective weak limits under the null hypothesis, for the multivariate normal and vMF distributions. The suprema defining the Kolmogorov–Smirnov statistics are approximated on finite 10×10 grids in $\Omega \times \mathbb{R}_{\geq 0}$. This grid varies depending on the example considered. For each setting, we simulate $M = 10000$ realizations of $\sup_{\omega,t} \left|\mathbb{G}_0(\omega,t)\right|$ (or $\sup_{\omega,t} \left|\tilde{\mathbb{G}}_0(\omega,t)\right|$) and of T_n^{KS} (or $\tilde{T}_n^{\mathrm{KS}}$) for each sample size $n = 50, 100, 500$. In composite settings, the unknown parameters were estimated by maximum likelihood in each simulated sample. The first row of each figure displays kernel density estimates of the simulated distributions, while the second row gives the corresponding quantile-quantile plots of the approximation errors with respect to the weak limits. For the kernel density estimation, the Gaussian kernel was employed, and the bandwidth was selected by biased cross-validation.

Figures **??** and **??** provide numerical evidence consistent with the Bahadur representation established in Proposition 3.1 for the MLE of the canonical parameter $\boldsymbol{\xi} = \kappa\boldsymbol{\mu}$ in the $\mathrm{vMF}(\boldsymbol{\mu}, 1)$ model on $\mathbb{S}^2$. For several mean directions $\boldsymbol{\mu}$, and for increasing sample sizes n, we examine the magnitude of the difference between $\sqrt{n}(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi})$ and the linear approximation given in (13). For each value of n, the figures display the pointwise mean over $M = 500$ simulated samples together with pointwise 95% confidence intervals (based on the normal distribution), and include the reference curve $\frac{1}{\sqrt{n}}$ for comparison. The results confirm that the convergence rate is of order $\frac{1}{\sqrt{n}}$, as expected.

7. Real data application. Some ideas:

i. Test the goodness-of-fit of sunspot positions with a mixture of two small circle distributions to model the positions in both hemispheres. Recall $f_{\mathrm{SC}}(\boldsymbol{x}; \boldsymbol{\mu}, \tau, \mu) := c(\tau, \mu) \exp(-\tau(\boldsymbol{x}^\top\boldsymbol{\mu} - \mu)^2)$. It is natural that the SCs for both hemispheres share the location parameter, have the same weights, and have symmetric effect of the parameters. So we can actually merge the mixtures under a rotationally symmetric model:

$$f(\boldsymbol{x}; \boldsymbol{\mu}, \tau, \mu) = \frac{c(\tau, \mu)}{2} \left\{ \exp(-\tau(\boldsymbol{x}^\top\boldsymbol{\mu} - \mu)^2) + \exp(-\tau(\boldsymbol{x}^\top\boldsymbol{\mu} + \mu)^2) \right\}.$$

We can estimate by MLE the axis $\boldsymbol{\mu}$ (very close to $(0,0,1)^\top$), concentration τ, and location μ.

ii. Jensen (1981) fitted von Mises distributions to wind direction data to test independence between speed and direction.

iii. Comets:

- Puedes replicar el test de los long-period comets con el abordaje de los distance profiles.
- El resultado del análisis de los short-period comets te lo deja a huevo (historia + justificación) para continuar el análisis de datos reales en tu artículo: el modelo adecuado para estos cometas es uno rotacionalmente simétrico y unimodal (más fácil) pero que se concentra rápidamente en su moda. Creo que una distribución esférica Cauchy (hay dos tipos) o la distribución de Jones and Pewsey podrían ser buenas candidatas. La vMF creo que no va a dar un buen ajuste porque no se concentra lo suficientemente rápido en la moda.

- Bonus: debido a la equivalencia entre la cardioide y Watson para concentraciones bajas, podría ser que los long-period comets también se pudiesen modelar adecuadamente con una distribución de Watson. Esto sería doblemente interesante, porque si consideras una distribución small circle (generaliza a la Watson) y calculas su maquinaria de proyecciones, esta podría ser doblemente útil: para modelar los long-period comets y para modelar las manchas solares (en este caso, con una mixtura de dos o más small circles con el mismo eje).
- Sobre las distribuciones:
  - Ver Jones and Pewsey (2005), Sección 5. Tendrás que ver cómo simularla. Inversión de la cdf proyectada es lo más directo: mira `r_sph_car()` en el código que te mando como plantilla.
  - Mira la prueba del Teorema 9, página 47, para ver cómo calcular distribuciones proyectadas de modelos rotacionalmente simétricos.
  - En el artículo de Alberto y mío en TEST puedes ver una variante de la spherical Cauchy, Sección B.1. Puedes simularla con `?sphunif::r_alt(alt = "C")`.

Para ti, el artículo es relevante por lo siguiente: Usa el abordaje projected-ecdf, muy relacionado con los distance profiles. La diferencia es que en las proyecciones el signo se tiene en cuenta. La distribución de las proyecciones en la cardioide es más sencilla que la de la vMF, por lo que será más fácil de usar con la filosofía de los distance profiles. En la prueba del Teorema 8 tienes la "representación de Bahadur" del estimador MLE. Se discuten varios pesos en el estadístico de CvM. No encontré grandes diferencias, por lo que podría estar justificado tomar el más sencillo (Pn). Si bien los estudios numéricos no son exhaustivos y por tanto tampoco concluyentes. Puedes replicar el test de los long-period comets con el abordaje de los distance profiles. El regalito: el resultado del análisis de los short-period comets te lo deja a huevo (historia + justificación) para continuar el análisis de datos reales en tu artículo: el modelo adecuado para estos cometas es uno rotacionalmente simétrico y unimodal (más fácil) pero que se concentra rápidamente en su moda. Creo que una distribución esférica Cauchy (hay dos tipos) o la distribución de Jones and Pewsey podrían ser buenas candidatas. La vMF creo que no va a dar un buen ajuste porque no se concentra lo suficientemente rápido en la moda. Bonus: debido a la equivalencia entre la cardioide y Watson para concentraciones bajas, podría ser que los long-period comets también se pudiesen modelar adecuadamente con una distribución de Watson. Esto sería doblemente interesante, porque si consideras una distribución small circle (generaliza a la Watson) y calculas su maquinaria de proyecciones, esta podría ser doblemente útil: para modelar los long-period comets y para modelar las manchas solares (en este caso, con una mixtura de dos o más small circles con el mismo eje). Sobre las distribuciones: - Ver Jones and Pewsey (2005), Sección 5. Tendrás que ver cómo simularla. Inversión de la cdf proyectada es lo más directo: mira $r_sph_car()$ en el código que te mando como plantilla. - Mira la prueba del Teorema 9, página 47, para ver cómo calcular distribuciones proyectadas de modelos rotacionalmente simétricos. - En el artículo de Alberto y mío en TEST puedes ver una variante de la spherical Cauchy, Sección B.1. Puedes simumarla con $?sphunif :: r_alt(alt = ”C”)$.

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APPENDIX A: PROOF OF PROPOSITION 2.1

PROOF OF PROPOSITION 2.1. The proof is done following the same order as Proposition 2.1.

- Example 2.1. The distance profile at $\boldsymbol{\omega} \in \mathbb{R}^q$ is given by

$$F^{\mu}_{\boldsymbol{\omega}}(t) = \mathsf{P}\left(d(\boldsymbol{\omega}, \boldsymbol{X}) \leq t\right) = \mathsf{P}\left(\|\boldsymbol{\omega} - \boldsymbol{X}\|_2 \leq t\right) = \mathsf{P}\left(\sqrt{\boldsymbol{Z}^\top \boldsymbol{Z}} \leq t\right),$$

where $\boldsymbol{Z} = \boldsymbol{X} - \boldsymbol{\omega}$ and $\|\cdot\|_2$ is the Euclidean norm in $\mathbb{R}^q$. Thus, $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{\mu} - \boldsymbol{\omega}, \boldsymbol{\Sigma})$. We are interested in the distribution of $\boldsymbol{Z}^\top \boldsymbol{Z}$. Consider $\boldsymbol{\eta} = \boldsymbol{U}^\top \boldsymbol{Z}$, where $\boldsymbol{U}$ is an orthogonal matrix such that $\boldsymbol{\Sigma} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$ (spectral decomposition). Thus, $\boldsymbol{\eta} \sim \mathcal{N}_q(\boldsymbol{U}^\top(\boldsymbol{\mu} - \boldsymbol{\omega}), \boldsymbol{\Lambda})$. We conclude that

$$\boldsymbol{Z}^\top \boldsymbol{Z} = \boldsymbol{\eta}^\top \boldsymbol{\eta} = \sum_{i=1}^{q} \eta_i^2 = \sum_{i=1}^{q} \lambda_i P_i,$$

with η_i independent, $\eta_i \sim \mathcal{N}(\nu_i, \lambda_i)$, and thus P_i are also independent and follow a non-central chi-squared distribution, $P_i \sim \chi^2_1\left(\frac{\nu_i^2}{\lambda_i}\right)$. Here, $\boldsymbol{\nu} = \boldsymbol{U}^\top(\boldsymbol{\mu} - \boldsymbol{\omega})$ and $\boldsymbol{\lambda}$ is the vector of eigenvalues of $\boldsymbol{\Sigma}$. In the special case when $\boldsymbol{\omega} = \boldsymbol{\mu}$, we have that

$$\boldsymbol{Z}^\top \boldsymbol{Z} = \sum_{i=1}^{q} \lambda_i P_i,$$

where P_i are independent, $P_i \sim \chi^2_1$. If additionally, $\boldsymbol{\Sigma} = \boldsymbol{I}_q$, where $\boldsymbol{I}_q$ is the $q \times q$ identity matrix, then $\sqrt{\boldsymbol{Z}^\top \boldsymbol{Z}}$ follows a χ_q distribution.

- Example 2.2. The distance profile of B at a point $\omega \in L^2[0,1]$ is given by

$$F_\omega(t) = \mathsf{P}\left(d_{L^2}(B, \omega) \leq t\right) = \mathsf{P}\left(\|B - \omega\|_{L^2} \leq t\right).$$

In order to analyze the distribution of $\|B - \omega\|_{L^2}$, express B and ω in the Karhunen-Loève expansion. Let $\{e_k\}_{k=1}^{\infty}$ denote the eigenfunctions of the covariance kernel of B, given by $K_B(s,t) = \mathbb{C}\mathrm{ov}\left[B(s), B(t)\right] = \min\{s,t\}$. Then, B and ω can be expressed as

$$B(t) = \sum_{i=1}^{\infty} Z_i e_i(t), \text{ and } \omega(t) = \sum_{i=1}^{\infty} c_i e_i(t),$$

where Z_i are independent random variables with $Z_i \sim \mathcal{N}(0, \lambda_i)$ and the coefficients c_i are given by $c_i = \langle \omega, e_i \rangle = \int_0^1 \omega(t) e_i(t)\,\mathrm{d}t$. Hence, $B(t) - \omega(t) = \sum_{i=1}^{\infty}(Z_i - c_i)e_i(t)$, where $\lambda_k = \frac{1}{\left(k - \frac{1}{2}\right)^2 \pi^2}$. Thus,

$$\|B - \omega\|^2_{L^2} = \sum_{i=1}^{\infty}(Z_i - c_i)^2 = \sum_{i=1}^{\infty} X_i^2,$$

where $X_i := c_i - Z_i$; with $X_1, \ldots$ a sequence of independent random variables, $X_i \sim \mathcal{N}(c_i, \lambda_i)$. As a special case, if $\omega(t) = 0$ for all $t \in [0,1]$, then $\|B - \omega\|^2_{L^2} = \sum_{i=1}^{\infty} \lambda_i Q_i$, where Q_i are iid and $Q_i \sim \chi^2_1$. Observe the similarity to the distance profile for the multivariate normal when $\boldsymbol{\omega} = \boldsymbol{\mu}$.

- Example 2.3. Assume first that $|\boldsymbol{\mu}^\top \boldsymbol{\omega}| < 1$. To characterize the distance profile, the distribution of $T = \boldsymbol{\omega}^\top \boldsymbol{X}$ plays a central role. To simplify notation, let $\boldsymbol{\mu}_\perp := \boldsymbol{\mu} - (\boldsymbol{\mu}^\top \boldsymbol{\omega})\boldsymbol{\omega}$, where $\|\boldsymbol{\mu}_\perp\| = \sqrt{1 - (\boldsymbol{\mu}^\top \boldsymbol{\omega})^2}$. Define also $\tilde{\boldsymbol{\mu}} = \frac{\boldsymbol{\mu}_\perp}{\|\boldsymbol{\mu}_\perp\|}$. The tangent-normal decomposition allows expressing any $\boldsymbol{x} \in \mathbb{S}^q$ as $\boldsymbol{x} = t\boldsymbol{\omega} + \sqrt{1-t^2}\boldsymbol{u}$, where $\boldsymbol{u} \in \mathbb{S}^{q-1}$ and $\boldsymbol{u}^\top \boldsymbol{\omega} = 0$. We have that

$$\boldsymbol{\mu}^\top \boldsymbol{x} = t\boldsymbol{\mu}^\top \boldsymbol{\omega} + \sqrt{1-t^2}\boldsymbol{\mu}^\top \boldsymbol{u} = t\boldsymbol{\mu}^\top \boldsymbol{\omega} + \sqrt{1-t^2}\boldsymbol{\mu}_\perp^\top \boldsymbol{u} = t\boldsymbol{\mu}^\top \boldsymbol{\omega} + \sqrt{(1-t^2)(1-(\boldsymbol{\mu}^\top\boldsymbol{\omega})^2)}\tilde{\boldsymbol{\mu}}^\top \boldsymbol{u}.$$

Hence,

$$f_{\mathrm{vMF}}(\boldsymbol{x}; \boldsymbol{\mu}, \kappa) = c_q^{\mathrm{vMF}}(\kappa) e^{\kappa \boldsymbol{\mu}^\top \boldsymbol{\omega} t} e^{\kappa \sqrt{(1-t^2)(1-(\boldsymbol{\mu}^\top\boldsymbol{\omega})^2)}\tilde{\boldsymbol{\mu}}^\top \boldsymbol{u}}.$$

We know that $d\sigma_{\mathbb{S}^q}(\boldsymbol{x}) = (1-t^2)^{\frac{q-2}{2}} d\sigma_{\mathbb{S}^{q-1}}(\boldsymbol{u})$, where $d\sigma_{\mathbb{S}^q}$ is the Lebesgue measure on the q-dimensional unit sphere. So

$$f_T(t) = c_q^{\mathrm{vMF}}(\kappa) e^{\kappa \boldsymbol{\mu}^\top \boldsymbol{\omega} t} (1-t^2)^{\frac{q-2}{2}} \int_{\boldsymbol{u} \in \mathbb{S}^{q-1}} e^{\kappa \sqrt{(1-t^2)(1-(\boldsymbol{\mu}^\top\boldsymbol{\omega})^2)}\tilde{\boldsymbol{\mu}}^\top \boldsymbol{u}} \, \mathrm{d}\sigma_{\mathbb{S}^{q-1}}(\boldsymbol{u}). \tag{41}$$

The integrand in (41) corresponds to the density of a $\mathrm{vMF}\big(\tilde{\boldsymbol{\mu}}, \kappa\sqrt{(1-t^2)(1-(\boldsymbol{\mu}^\top\boldsymbol{\omega})^2)}\big)$ (up to the normalization constant). We conclude that (7) holds. If $\boldsymbol{\omega} = \pm\boldsymbol{\mu}$, then the tangent-normal decomposition gives $\boldsymbol{\mu}^\top \boldsymbol{x} = t\,\boldsymbol{\mu}^\top \omega$. Integrating over $\mathbb{S}^{q-1}$ yields (7). Hence, (7) holds for all $\boldsymbol{\omega} \in \mathbb{S}^q$. Obtaining F_T, the CDF of T, from (7) is not direct.

For the chordal distance, if $0 \le t \le 2$, then

$$F_{\boldsymbol{\omega}}^{\mu}(t) = \mathsf{P}\left(\sqrt{2(1-\boldsymbol{X}^\top\boldsymbol{\omega})} \le t\right) = 1 - F_T\left(1 - \frac{t^2}{2}\right),$$

whereas $F_{\boldsymbol{\omega}}^{\mu}(t) = 1$ for $t > 2$. Under the geodesic distance, if $0 \le t \le \pi$, then

$$F_{\boldsymbol{\omega}}^{\mu}(t) = \mathsf{P}\left(\cos^{-1}(\boldsymbol{X}^\top\boldsymbol{\omega}) \le t\right) = 1 - F_T\left(\cos(t)\right),$$

whereas $F_{\boldsymbol{\omega}}^{\mu}(t) = 1$ for $t > \pi$.

- Example 2.4. Following Jensen (1981), for $\boldsymbol{x} \in \mathbb{H}^q$, define the hyperbolic-spherical coordinates:

$$\begin{cases} x_1 = \cosh(x_r) \\ x_2 = \sinh(x_r) \end{cases}$$

when $q = 1$, with $x_r \ge 0$, and

$$\begin{cases} x_1 = \cosh(x_r) \\ x_2 = \sinh(x_r)\cos(\theta_1) = \sinh(x_r)(x_s)_1 \\ x_3 = \sinh(x_r)\sin(\theta_1)\cos(\theta_2) = \sinh(x_r)(x_s)_2 \\ \quad\vdots \\ x_q = \sinh(x_r)\sin(\theta_1)\cdots\sin(\theta_{q-2})\cos(\theta_{q-1}) = \sinh(x_r)(x_s)_{q-1} \\ x_{q+1} = \sinh(x_r)\sin(\theta_1)\cdots\sin(\theta_{q-2})\sin(\theta_{q-1}) = \sinh(x_r)(x_s)_q \end{cases}$$

when $q > 1$, with $(\theta_1, \ldots, \theta_{q-1})^\top \in (0,\pi)^{q-2} \times [0, 2\pi)$ being the angular hyperspherical coordinates of $\boldsymbol{x}_s \in \mathbb{S}^{q-1}$. Denoting by μ_q the Lebesgue measure on $\mathbb{H}^q$ and σ_{q-1} the Lebesgue measure on $\mathbb{S}^{q-1}$, it holds that $\mu_q(\mathrm{d}\boldsymbol{x}) = \sinh^{q-1}(x_r)\,\mathrm{d}x_r\,\sigma_{q-1}(\mathrm{d}\boldsymbol{x}_s)$.

Define the group of hyperbolic transformations on $\mathbb{R}^{q+1}$ as:

$$\mathrm{SH}(q) := \{\boldsymbol{A} \in \mathbb{R}^{(q+1)\times(q+1)} : \boldsymbol{A}^\top \tilde{\boldsymbol{I}}_{q+1}\boldsymbol{A} = \tilde{\boldsymbol{I}}_{q+1}, \, \det(\boldsymbol{A}) = 1, \, A_{11} > 0\},$$

where

$$\tilde{\boldsymbol{I}}_{q+1} = \begin{bmatrix} -1 & \boldsymbol{0}_q \\ \boldsymbol{0}_q & \boldsymbol{I}_q \end{bmatrix}.$$

The group of hyperbolic transformations acts transitively on $\mathbb{H}^q$ (see Lemma 2(ii) of Jensen, 1981), i.e., for any pair of mean vectors $\boldsymbol{x},\boldsymbol{y}\in\mathbb{H}^q$, there exists an action $\boldsymbol{A}\in\mathrm{SH}(q)$ such that $\boldsymbol{x}=\boldsymbol{A}\boldsymbol{y}$. The Minkowski pseudo-inner product is invariant under hyperbolic transformations, i.e., $(\boldsymbol{x},\boldsymbol{y})=(\boldsymbol{A}\boldsymbol{x},\boldsymbol{A}\boldsymbol{y})$ for every $\boldsymbol{x},\boldsymbol{y}\in\mathbb{H}^q$ and every $\boldsymbol{A}\in\mathrm{SH}(q)$. The invariance follows from the identity $(\boldsymbol{x},\boldsymbol{y})=\boldsymbol{x}^\top\tilde{\boldsymbol{I}}_{q+1}\boldsymbol{y}$ for any $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^{q+1}$. Take $A\in\mathrm{SH}(q)$ such that $\boldsymbol{A}\boldsymbol{\omega}=\boldsymbol{e}_1=(1,0,\ldots,0)^\top$.

We are interested in the distance profile

$$\begin{aligned}F_{\boldsymbol{\omega}}^{\mu}(t)&=\mathsf{P}\left(d_{\mathbb{H}^q}(\boldsymbol{X},\boldsymbol{\omega})\le t\right)=\mathsf{P}\left(-(\boldsymbol{X},\boldsymbol{\omega})\le\cosh(t)\right)=\mathsf{P}\left(-(\boldsymbol{A}\boldsymbol{X},\boldsymbol{A}\boldsymbol{\omega})\le\cosh(t)\right)\\&=\mathsf{P}\left(-(\boldsymbol{A}\boldsymbol{X},\boldsymbol{e}_1)\le\cosh(t)\right)=\mathsf{P}\left((\boldsymbol{A}\boldsymbol{X})_1\le\cosh(t)\right)\end{aligned}$$

for $t\ge 0$. Therefore,

$$F_{\boldsymbol{\omega}}^{\mu}(t)=\int_{\mathbb{H}^q}1_{\{(\boldsymbol{A}\boldsymbol{x})_1\le\cosh(t)\}}f_{\mathrm{HvMF}}(\boldsymbol{x};\boldsymbol{\mu},\kappa)\,\mu_q(\mathrm{d}\boldsymbol{x}).$$

Write $\boldsymbol{z}:=\boldsymbol{A}\boldsymbol{x}=(\cosh(z_r),\sinh(z_r)\boldsymbol{z}_s)^\top$, where $(z_r,\boldsymbol{z}_s)^\top\in\mathbb{R}_{\ge0}\times\mathbb{S}^{q-1}$. We thus have that

$$\begin{aligned}F_{\boldsymbol{\omega}}^{\mu}(t)&=\int_{\mathbb{H}^q}1_{\{z_r\le t\}}f_{\mathrm{HvMF}}(\boldsymbol{A}^{-1}\boldsymbol{z};\boldsymbol{\mu},\kappa)\,\mu_q(\mathrm{d}(\boldsymbol{A}^{-1}\boldsymbol{z})) &(42)\\&=\int_{\mathbb{H}^q}1_{\{z_r\le t\}}f_{\mathrm{HvMF}}(\boldsymbol{z};\boldsymbol{A}\boldsymbol{\mu},\kappa)\,\mu_q(\mathrm{d}(\boldsymbol{A}^{-1}\boldsymbol{z})) &(43)\\&=\int_{\mathbb{H}^q}1_{\{z_r\le t\}}f_{\mathrm{HvMF}}(\boldsymbol{z};\boldsymbol{A}\boldsymbol{\mu},\kappa)\,\mu_q(\mathrm{d}\boldsymbol{z}) &(44)\\&=\int_0^t\int_{\mathbb{S}^{q-1}}f_{\mathrm{HvMF}}\left((\cosh(z_r),\sinh(z_r)\boldsymbol{z}_s);\boldsymbol{A}\boldsymbol{\mu},\kappa\right)\sinh^{q-1}(z_r)\sigma_{q-1}(\mathrm{d}\boldsymbol{z}_s)\mathrm{d}z_r\end{aligned}$$

Equality (42) follows from the substitution $\boldsymbol{x}=\boldsymbol{A}^{-1}\boldsymbol{z}$. Observe that the domain of integration remains invariant since the mapping $\boldsymbol{x}\mapsto\boldsymbol{A}^{-1}\boldsymbol{x}$ is a bijection from $\mathbb{H}^q$ to itself. Surjectivity holds because for any $\boldsymbol{z}\in\mathbb{H}^q$, $\boldsymbol{z}=\boldsymbol{A}^{-1}(\boldsymbol{A}\boldsymbol{x})$, and $\boldsymbol{A}\boldsymbol{x}\in\mathbb{H}^q$ because the unit hyperboloid is invariant under hyperbolic transformations (Lemma 1(ii) of Jensen, 1981). The equality (43), i.e., $f_{\mathrm{HvMF}}(\boldsymbol{A}^{-1}\boldsymbol{z};\boldsymbol{\mu},\kappa)=f_{\mathrm{HvMF}}(\boldsymbol{z};\boldsymbol{A}\boldsymbol{\mu},\kappa)$, is verified because $(\boldsymbol{A}^{-1}\boldsymbol{z},\boldsymbol{\mu})=(\boldsymbol{z},\boldsymbol{A}\boldsymbol{\mu})$, and the normalization constant of the HvMF distribution, $c_q^{\mathrm{HvMF}}(\kappa)$, does not depend on the mean vector (see the proof of Proposition 4.1 in Serrano and García-Portugués, 2025). Equality (44) holds since μ_q is invariant under hyperbolic transformations (see Section 2C of Jensen, 1981). Note that $\det(\boldsymbol{A}^{-1})=1$.

Assume first that $\rho>0$. Writing $\boldsymbol{A}\boldsymbol{\mu}=(\cosh(\rho),\sinh(\rho)\boldsymbol{\xi})^\top$, with $(\rho,\boldsymbol{\xi})\in\mathbb{R}_{\ge0}\times\mathbb{S}^{q-1}$, we have that

$$(\boldsymbol{\omega},\boldsymbol{\mu})=(\boldsymbol{A}\boldsymbol{\omega},\boldsymbol{A}\boldsymbol{\mu})=(\boldsymbol{e}_1,\boldsymbol{A}\boldsymbol{\mu})=-\cosh(\rho).$$

Hence,

$$\begin{aligned}(\boldsymbol{A}\boldsymbol{\mu},\boldsymbol{z})&=-\cosh(\rho)\cosh(z_r)+\sinh(\rho)\sinh(z_r)\langle\boldsymbol{z}_s,\boldsymbol{\xi}\rangle\\&=(\boldsymbol{\mu},\boldsymbol{\omega})\cosh(z_r)+\sinh(\rho)\sinh(z_r)\langle\boldsymbol{z}_s,\boldsymbol{\xi}\rangle.\end{aligned}\tag{45}$$

Therefore,

$$F_{\boldsymbol{\omega}}^{\mu}(t)=c_q^{\mathrm{HvMF}}(\kappa)\int_0^t e^{\kappa(\boldsymbol{\mu},\boldsymbol{\omega})\cosh(z_r)}\sinh^{q-1}(z_r)\int_{\mathbb{S}^{q-1}}e^{\kappa\sinh(z_r)\sinh(\rho)\langle\boldsymbol{\xi},\boldsymbol{z}_s\rangle}\sigma_{q-1}(\mathrm{d}\boldsymbol{z}_s)\mathrm{d}z_r$$

Thus, $F_{\boldsymbol{\omega}}^{\boldsymbol{\mu}}(t)$ is the CDF of the real random variable $Z_r := d_{\mathbb{H}^q}(\boldsymbol{X},\boldsymbol{\omega}) = \operatorname{arccosh}((\boldsymbol{A}\boldsymbol{X})_1)$, with density given by

$$f_{Z_r}(z_r) = c_q^{\mathrm{HvMF}}(\kappa)(\sinh(z_r))^{q-1} e^{\kappa(\boldsymbol{\mu},\boldsymbol{\omega})\cosh(z_r)} \times \int_{\mathbb{S}^{q-1}} e^{\kappa\sinh(z_r)\sinh(\rho)\langle\boldsymbol{\xi},\boldsymbol{z}_s\rangle}\,\mathrm{d}\sigma_{\mathbb{S}^{q-1}}(\boldsymbol{z}_s).$$

This integral is proportional to the normalization constant of a von Mises–Fisher distribution on $\mathbb{S}^{q-1}$:

$$\int_{\mathbb{S}^{q-1}} e^{\kappa\sinh(z_r)\sinh(\rho)\langle\boldsymbol{\xi},\boldsymbol{z}_s\rangle}\,\mathrm{d}\sigma_{\mathbb{S}^{q-1}}(\boldsymbol{z}_s) = \frac{1}{c_{q-1}^{\mathrm{vMF}}(\kappa\sinh(z_r)\sinh(\rho))}.$$

Thus, the density of Z_r is

$$f_{Z_r}(z_r) = \frac{c_q^{\mathrm{HvMF}}(\kappa)}{c_{q-1}^{\mathrm{vMF}}(\kappa\sinh(z_r)\sinh(\rho))}(\sinh(z_r))^{q-1} e^{\kappa(\boldsymbol{\mu},\boldsymbol{\omega})\cosh(z_r)}, \quad z_r \geq 0,$$

for $\rho > 0$.

If $\rho = 0$, then $\boldsymbol{\omega} = \boldsymbol{\mu}$, and hence $\boldsymbol{A}\boldsymbol{\mu} = \boldsymbol{e}_1$. Therefore, $(\boldsymbol{A}\boldsymbol{\mu},\boldsymbol{z}) = -\cosh(z_r)$. Integrating over $\mathbb{S}^{q-1}$ yields (8). Hence, (8) holds for all $\rho \geq 0$. The distance profile is then given by

$$F_{\boldsymbol{\omega}}^{\boldsymbol{\mu}}(t) = \mathsf{P}\left(Z_r \leq t\right).$$

- Example 2.5. With the notation of Example 2.5, let

$$\boldsymbol{Z} := \boldsymbol{H}\boldsymbol{Y} - \operatorname{clr}(\boldsymbol{\omega}), \qquad \boldsymbol{W} := \boldsymbol{U}^\top \boldsymbol{Z}, \qquad \boldsymbol{v} := \boldsymbol{U}^\top \boldsymbol{m}.$$

Since $\operatorname{clr}(\boldsymbol{X}) = \operatorname{clr}(\boldsymbol{\sigma}(\boldsymbol{Y})) = \boldsymbol{H}\boldsymbol{Y}$, one has $\operatorname{clr}(\boldsymbol{X}) \sim \mathcal{N}(\boldsymbol{H}\boldsymbol{\mu}, \boldsymbol{H}\boldsymbol{\Sigma}\boldsymbol{H}^\top) = \mathcal{N}(\boldsymbol{H}\boldsymbol{\mu},\boldsymbol{C})$. Hence, $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{m},\boldsymbol{C})$ and $\boldsymbol{W} \sim \mathcal{N}(\boldsymbol{v},\boldsymbol{A})$. Since $\boldsymbol{\Sigma}$ is positive definite, $\operatorname{rank}(\boldsymbol{C}) = q$, so $\boldsymbol{A} = \operatorname{diag}(\alpha_1,\ldots,\alpha_q,0)$. Choosing $\boldsymbol{u}_{q+1}$ proportional to $\mathbf{1}_{q+1}$, one has $W_{q+1} = 0$ almost surely because $\boldsymbol{Z} \in H$ almost surely. Since $\boldsymbol{U}$ is orthogonal,

$$d_A(\boldsymbol{X},\boldsymbol{\omega})^2 = \boldsymbol{Z}^\top\boldsymbol{Z} = \boldsymbol{W}^\top\boldsymbol{W} = \sum_{i=1}^{q+1} W_i^2 = \sum_{i=1}^{q} W_i^2,$$

where $W_i \sim \mathcal{N}(v_i,\alpha_i)$, for $i = 1,\ldots,q$. Hence, we have that $W_i = v_i + \sqrt{\alpha_i}\,\varepsilon_i$ where ε_i are independent and identically distributed, $\varepsilon_i \sim \mathcal{N}(0,1)$. Hence, $W_i^2 = \alpha_i\left(\varepsilon_i + \frac{v_i}{\sqrt{\alpha_i}}\right)^2$. By definition, $\left(\varepsilon_i + \frac{v_i}{\sqrt{\alpha_i}}\right)^2$ is a noncentral χ_1^2 random variable with one degree of freedom and noncentrality parameter $\delta_i = \frac{v_i^2}{\alpha_i} = \frac{(\boldsymbol{u}_i^\top\boldsymbol{m})^2}{\alpha_i}$. Therefore, setting $Q_i := W_i^2/\alpha_i$, one has $Q_i \sim \chi_1^2(\delta_i)$. This completes the characterization of the distance profile of the logistic Gaussian distribution.

Finally, we prove the equality $\|\operatorname{clr}(\boldsymbol{x}) - \operatorname{clr}(\boldsymbol{\omega})\|_2 = \|\operatorname{ilr}(\boldsymbol{x}) - \operatorname{ilr}(\boldsymbol{\omega})\|_2$, shown in Remark 2.1. Egozcue et al. (2003, Equation (28)) show that $\operatorname{ilr}(\boldsymbol{x}) = \mathbf{B}^\top\operatorname{clr}(\boldsymbol{x})$, where $\mathbf{B}$ is the $(q+1)\times q$ matrix from Equation (17) of Egozcue et al. (2003). The columns $\mathbf{b}_i \in \mathbb{R}^{q+1}$ of $\mathbf{B}$ form a basis of $H = \{\boldsymbol{z} \in \mathbb{R}^{q+1} : \sum_{i=1}^{q+1} z_i = 0\}$, and they are orthonormal with respect to the Euclidean inner product in $\mathbb{R}^{q+1}$. Hence,

$$\begin{aligned}
\|\operatorname{ilr}(\boldsymbol{x}) - \operatorname{ilr}(\boldsymbol{\omega})\|_2^2 &= \|\mathbf{B}^\top\operatorname{clr}(\boldsymbol{x}) - \mathbf{B}^\top\operatorname{clr}(\boldsymbol{\omega})\|_2^2 \\
&= (\operatorname{clr}(\boldsymbol{x}) - \operatorname{clr}(\boldsymbol{\omega}))^\top\,\mathbf{B}\mathbf{B}^\top\,(\operatorname{clr}(\boldsymbol{x}) - \operatorname{clr}(\boldsymbol{\omega})) \\
&= \|\operatorname{clr}(\boldsymbol{x}) - \operatorname{clr}(\boldsymbol{\omega})\|_2^2 = d_A(\boldsymbol{x},\boldsymbol{\omega})^2.
\end{aligned}$$

The third equality follows because $\mathbf{B}\mathbf{B}^\top$ is the projection matrix onto H and $(\operatorname{clr}(\boldsymbol{x}) - \operatorname{clr}(\boldsymbol{\omega})) \in H$.

□

APPENDIX B: PROOFS OF SECTION 3

B.1. Proof of Theorem 3.1.

PROOF OF THEOREM 3.1. First, observe that

$$\mathbb{G}_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}(\omega,t) = \mathbb{G}_{n,\mu}^{\mu_{\boldsymbol{\theta}_0}}(\omega,t) - \sqrt{n}\left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right).$$

We know that $\mathbb{G}_{n,\mu}^{\mu_{\boldsymbol{\theta}_0}}(\omega,t)$ converges weakly to a Gaussian process by Theorem 5.1 of Dubey, Chen and Müller (2024):

$$\mathbb{G}_{n,\mu}^{\mu_{\boldsymbol{\theta}_0}}(\omega,t) \xrightarrow{\mathcal{L}} \mathbb{G}_{\boldsymbol{\theta}_0}(\omega,t) \text{ in } \ell^\infty(\mathcal{Y}), \tag{46}$$

where $\mathbb{G}_{\boldsymbol{\theta}_0}(\omega,t)$ is a zero-mean Gaussian process with covariance given by

$$\mathbb{C}\mathrm{ov}\left[y_{(\omega_1,t_1)}, y_{(\omega_2,t_2)}\right] = \mathsf{P}\left(d(\omega_1,X)\le t_1, d(\omega_2,X)\le t_2\right) - F_{\omega_1}^{\mu_{\boldsymbol{\theta}_0}}(t_1)F_{\omega_2}^{\mu_{\boldsymbol{\theta}_0}}(t_2), \tag{47}$$

for $\omega_1,\omega_2\in\Omega$ and $t_1,t_2\in\mathbb{R}_{\ge 0}$.

Now, focus on $F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)$. By differentiability of $\boldsymbol{\theta}\mapsto F_\omega^{\boldsymbol{\theta}}(t)$ at $\boldsymbol{\theta}_0$, it holds that

$$F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) = \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top\left(\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\right) + o_P\left(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|\right). \tag{48}$$

From (48) and Bahadur's representation (12), one obtains that

$$\sqrt{n}\left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right) = -\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n \boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X_i) + o_P(1) + \sqrt{n}o_P\left(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|\right). \tag{49}$$

Observe that $\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top o_P(1) = o_P(1)$ by *(d)*. By Bahadur's representation (12), $\sqrt{n}\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\| = O_P(1)$, so $\sqrt{n}o_P\left(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|\right) = o_P(1)$. Hence, by *(d)*, the remainders in (49) are $o_P(1)$ uniformly over $(\omega,t)\in\Omega\times\mathbb{R}_{\ge 0}$. Therefore, (46) and (49) yield that

$$G_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}(\omega,t) = \frac{1}{\sqrt{n}}\sum_{i=1}^n\left(y_{\omega,t}(X_i) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) + g_{\omega,t}(X_i)\right) + o_P(1) = \frac{1}{\sqrt{n}}\sum_{i=1}^n f_{\omega,t}(X_i) + o_P(1),$$

where

$$g_{\omega,t}(x) := \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(x),$$

and

$$f_{\omega,t}(x) := y_{\omega,t}(x) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) + g_{\omega,t}(x).$$

Observe that $\mathsf{E}(f_{\omega,t}(X)) = 0$, so $G_{n,\mu}^{\mu_{\hat{\boldsymbol{\theta}}}}$ is, up to an $o_P(1)$ remainder, the empirical process indexed by the class

$$\mathcal{F} := \{f_{\omega,t} :\ \omega\in\Omega,\ t\ge 0\}.$$

Thus, proving that $\mathcal{F}$ is Donsker concludes the proof. To prove the Donsker property, write $\mathcal{F}\subset\mathcal{Y}+\mathcal{G}$, where $\mathcal{Y} := \{y_{\omega,t} - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) :\ \omega\in\Omega,\ t\ge 0\}$ and $\mathcal{G} := \{g_{\omega,t} :\ \omega\in\Omega,\ t\ge 0\}$. If $\mathcal{Y}$ and $\mathcal{G}$ are Donsker and $\sup_{h\in\mathcal{Y}\cup\mathcal{G}}\mathsf{E}|h(X)| < \infty$, then $\mathcal{Y}+\mathcal{G}$ is also Donsker (see, e.g., Example 2.10.9 in van der Vaart, 1998), and since $\mathcal{F}$ is a subset of a Donsker class, it is Donsker.

The class $\mathcal{Y}$ is Donsker by (9). Corollary 16.1 in DasGupta (2011) shows that a sufficient condition for $\mathcal{G}$ to be Donsker is that $\mathcal{G}$ is finite-dimensional, and its envelope function is in $L_2(\mathsf{P})$. The fact that $\mathcal{G}$ is finite-dimensional can be seen by writing

$$g_{\omega,t}(x) = \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(x) = \sum_{i=1}^p \alpha_i(\omega,t)\psi_{\boldsymbol{\theta}_0,i}(x),$$

where $\boldsymbol{\alpha}(\omega,t)^\top := \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} \in \mathbb{R}^p$ (coefficients), and $\psi_{\boldsymbol{\theta}_0,i}$ represents the i-th component of $\boldsymbol{\psi}_{\boldsymbol{\theta}_0}$, $i = 1,\ldots,p$. Therefore, one has $\mathcal{G} \subset \operatorname{span}\{\boldsymbol{\psi}_{\boldsymbol{\theta}_0,1},\ldots,\boldsymbol{\psi}_{\boldsymbol{\theta}_0,p}\}$, a linear subspace of dimension at most p. By Assumption *(d)*, it holds that $\sup_{\omega,t}\|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)\| < \infty$, hence Assumption *($\tilde{c}$.2)* guarantees that the envelope function of $\mathcal{G}$ is in $L_2(\mathsf{P})$. Therefore, $\mathcal{G}$ is Donsker.

It is only left to show that $\sup_{h\in\mathcal{Y}\cup\mathcal{G}} \mathsf{E}(h(X)) < \infty$. This is trivial for class $\mathcal{Y}$. For $\mathcal{G}$, it follows from Assumption *($\tilde{c}$.3)*, since $\mathsf{E}(g_{\omega,t}(X)) = 0$ for every $(\omega,t) \in \Omega\times\mathbb{R}_{\geq 0}$. Therefore, $\mathcal{F}$ is Donsker. This implies that $G_{n,\mu}^{\mu_{\hat{\theta}}}$ converges weakly to a Gaussian process $\tilde{\mathbb{G}}_0$, with mean zero and covariance $\tilde{\mathcal{C}}_{(\omega_1,t_1),(\omega_2,t_2)} := \mathsf{E}(f_{\omega_1,t_1}(X)f_{\omega_2,t_2}(X))$, which is given in (17). □

B.2. Proof of Theorem 3.2. This section contains the proof of Theorem 3.2, together with auxiliary results used in the proof. In what follows, we denote by $d_{\Omega\times\mathbb{R}_{\geq 0}}$ the product distance on $(\Omega,d)\times(\mathbb{R}_{\geq 0},|\cdot|)$, that is,

$$d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big) := d(\omega,\omega') + |t-t'|.$$

Lemma B.1 is an auxiliary result in the proof of Lemma B.2.

LEMMA B.1. *Fix $\varepsilon > 0$. Let $g : \Omega\times\mathbb{R}_{\geq 0} \to \mathbb{R}$ be a bounded, uniformly continuous function on $\Omega\times\mathbb{R}_{\geq 0}$. For each $\omega\in\Omega$, define the following regularization of g:*

$$g_\varepsilon(\omega,t) := \inf_{s\in[0,M]}\left\{g(\omega,s) + \frac{2\|g\|_\infty}{\delta}|t-s|\right\}. \tag{50}$$

Here, $M := \operatorname{diam}(\Omega) = \sup_{\omega,\omega'\in\Omega} d(\omega,\omega')$, and $\delta(\varepsilon) > 0$ is such that for every $(\omega,t),(\omega',t') \in \Omega\times[0,M]$ with $d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big) < \delta$, it holds that $|g(\omega,t) - g(\omega',t')| < \varepsilon$. For every fixed $\omega\in\Omega$, the map $t\mapsto g_\varepsilon(\omega,t)$ is Lipschitz on $[0,M]$, with Lipschitz constant $\operatorname{Lip}(g_\varepsilon) = \frac{2\|g\|_\infty}{\delta}$, and for every $\omega\in\Omega$ and $t\in[0,M]$, it holds a.s. that

$$\left|\int_0^M g_\varepsilon(\omega,t)\,\mathrm{d}(F_{n,\omega}^\mu - F_\omega^{\mu_0})(t)\right| \leq \frac{2M\|g\|_\infty}{\delta}\sup_{t\in[0,M]}|F_{n,\omega}^\mu(t) - F_\omega^{\mu_0}(t)|.$$

PROOF. Observe that $M < \infty$, since Ω is totally bounded, which is implied by *(b.1)*. Thus, $F_\omega([0,M]) = F_{n,\omega}^\mu([0,M]) = 1$ and for any measurable function ϕ, $\int_{\mathbb{R}_{\geq 0}}\phi(t)\,\mathrm{d}F_\omega(t) = \int_{[0,M]}\phi(t)\,\mathrm{d}F_\omega(t)$ (similarly for $F_{n,\omega}$). Since g is uniformly continuous on $\Omega\times\mathbb{R}_{\geq 0}$, it is also uniformly continuous on $\Omega\times[0,M]$.

We first show that $g_\varepsilon(\omega,\cdot)$ is Lipschitz with Lipschitz constant $\operatorname{Lip}(g_\varepsilon) = \frac{2\|g\|_\infty}{\delta}$. Indeed, let $t,t'\in[0,M]$ with $t<t'$. For every $s\in[0,M]$, one has

$$g(\omega,s) + \frac{2\|g\|_\infty}{\delta}|t-s| \leq g(\omega,s) + \frac{2\|g\|_\infty}{\delta}|t'-s| + \frac{2\|g\|_\infty}{\delta}|t-t'|,$$

and taking the infimum over s yields $g_\varepsilon(\omega,t) \leq g_\varepsilon(\omega,t') + \frac{2\|g\|_\infty}{\delta}|t-t'|$. By symmetry, one also has $g_\varepsilon(\omega,t') \leq g_\varepsilon(\omega,t) + \frac{2\|g\|_\infty}{\delta}|t-t'|$, so that $|g_\varepsilon(\omega,t) - g_\varepsilon(\omega,t')| \leq \frac{2\|g\|_\infty}{\delta}|t-t'|$.

Fix $\omega \in \Omega$. Since $g_\varepsilon(\omega, \cdot)$ is Lipschitz on $[0, M]$, it is absolutely continuous and differentiable almost everywhere, with $|g'_\varepsilon(\omega, t)| \leq \mathrm{Lip}(g_\varepsilon)$. An integration by parts argument gives

$$
\begin{aligned}
\left| \int_0^M g_\varepsilon(\omega, t)\, \mathrm{d}(F^{\mu}_{n,\omega} - F^{\mu_0}_{\omega})(t) \right| &= \left| - \int_0^M g'_\varepsilon(\omega, t) \left(F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t) \right) \mathrm{d}t \right| \\
&\leq \int_0^M |g'_\varepsilon(\omega, t)|\, |F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t)|\, \mathrm{d}t \\
&\leq \frac{2M\|g\|_\infty}{\delta} \sup_{t \in [0,M]} |F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t)|. \qquad (51)
\end{aligned}
$$

This holds almost surely and concludes the proof. □

Lemma B.2 is used to prove convergence to zero of a term in Lemma B.4.

LEMMA B.2. *Let $g : \Omega \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a bounded, uniformly continuous function on $\Omega \times \mathbb{R}_{\geq 0}$. For each $\omega \in \Omega$, define*

$$
r(\omega) := \int_{\mathbb{R}_{\geq 0}} g(\omega, t)\, \mathrm{d}F^{\mu_0}_{\omega}(t), \quad r_n(\omega) := \int_{\mathbb{R}_{\geq 0}} g(\omega, t)\, \mathrm{d}F^{\mu}_{n,\omega}(t). \qquad (52)
$$

It holds that $\sup_{\omega \in \Omega} |r_n(\omega) - r(\omega)| \xrightarrow{a.s.} 0$ *as* $n \to \infty$.

PROOF. Observe that

$$
r(\omega) = \int_0^M g(\omega, t)\, \mathrm{d}F^{\mu_0}_{\omega}(t), \quad r_n(\omega) = \int_0^M g(\omega, t)\, \mathrm{d}F^{\mu}_{n,\omega}(t).
$$

Also, one has

$$
|r_n(\omega) - r(\omega)| \leq \left| \int_0^M (g(\omega, t) - g_\varepsilon(\omega, t))\, \mathrm{d}(F^{\mu}_{n,\omega} - F^{\mu_0}_{\omega})(t) \right| + \left| \int_0^M g_\varepsilon(\omega, t)\, \mathrm{d}(F^{\mu}_{n,\omega} - F^{\mu_0}_{\omega})(t) \right|. \qquad (53)
$$

Begin with the first term on the RHS of (53). Fix $\varepsilon > 0$, and define g_ε as in (50). If one takes $s = t$ in the definition of $g_\varepsilon(\omega, t)$, one has $g_\varepsilon(\omega, t) \leq g(\omega, t)$ for every $(\omega, t) \in \Omega \times [0, M]$. To obtain a lower bound for $g_\varepsilon(\omega, t)$, we consider two cases. If $|t - s| < \delta$, by uniform continuity of g, one has $g(\omega, s) \geq g(\omega, t) - \varepsilon$, so,

$$
g(\omega, s) + \frac{2\|g\|_\infty}{\delta} |t - s| \geq g(\omega, t) - \varepsilon.
$$

If $|t - s| \geq \delta$, then

$$
g(\omega, s) + \frac{2\|g\|_\infty}{\delta} |t - s| \geq -\|g\|_\infty + 2\|g\|_\infty = \|g\|_\infty \geq g(\omega, t) \geq g(\omega, t) - \varepsilon.
$$

Taking infima over s yields $g_\varepsilon(\omega, t) \geq g(\omega, t) - \varepsilon$ for every $(\omega, t) \in \Omega \times [0, M]$. Therefore, one has $|g_\varepsilon(\omega, t) - g(\omega, t)| \leq \varepsilon$ for every $(\omega, t) \in \Omega \times [0, M]$. This provides an upper bound for the second term in the RHS of (53):

$$
\left| \int_0^M (g_\varepsilon(\omega, t) - g(\omega, t))\, \mathrm{d}(F^{\mu}_{n,\omega} - F^{\mu_0}_{\omega})(t) \right| \leq \varepsilon\, |\mu_{\omega,n} - \mu_{\omega,0}|\, ([0, M]) \leq \varepsilon\, |\gamma|\, (\mathbb{R}) \leq 2\varepsilon
$$

where $|\gamma|$ denotes the total variation of a finite signed measure $\gamma = \mu_{\omega,n} - \mu_{\omega,0}$. Here, $\mu_{\omega,n}$ and $\mu_{\omega,0}$ are the measures induced by $F_{n,\omega}$ and $F^{\mu_0}_{\omega}$, respectively. The total variation of a measurable subset E is, in particular,

$$
|\gamma|(E) = \sup_{\text{Partitions } E_1, \ldots, E_m \text{ of } E} \left\{ \sum_{i=1}^m |\gamma(E_j)| \right\}
$$

(Exercise 8 in Folland, 1999). That $|\gamma|(\mathbb{R}) \leq 2$ follows easily from the calculation

$$\begin{aligned}|\gamma|(\mathbb{R}) &= \sup\left\{\sum_{i=1}^{m} |\gamma(E_j)|\right\} = \sup\left\{\sum_{i=1}^{m} |\mu_{\omega,n}(E_j) - \mu_{\omega,0}(E_j)|\right\} \\ &\leq \sup\left\{\sum_{i=1}^{m} |\mu_{\omega,n}(E_j) + \mu_{\omega,0}(E_j)|\right\} = \sup\left\{\sum_{i=1}^{m} \mu_{\omega,n}(E_j) + \sum_{i=1}^{m} \mu_{\omega,0}(E_j)\right\} \\ &= \sup\{1+1\} = 2.\end{aligned}$$

Now, focus on the second term in the RHS of (53). Since the class $\mathcal{Y} = \{y_{\omega,t} : (\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}\}$ is Donsker and $\mathbb{G}_0$ is separable, we obtain the Glivenko–Cantelli property:

$$\sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \left|F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t)\right| = \sup_{y\in\mathcal{Y}} \left|(P_n - \mu_0)y\right| \xrightarrow{a.s.} 0,$$

see page 130 in Van der Vaart and Wellner (2023). Thus, applying Lemma B.1 and taking the supremum over $\omega \in \Omega$ in (51) shows that the second term in the RHS of (53) converges to zero as $n \to \infty$. Therefore, we can conclude that

$$\sup_{\omega\in\Omega} |r_n(\omega) - r(\omega)| \leq 3\varepsilon,$$

for n sufficiently large such that $\frac{2M\|g\|_\infty}{\delta} \sup_{\omega\in\Omega} \sup_{t\in[0,M]} |F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t)| \leq \varepsilon$. Since $\varepsilon > 0$ is arbitrary, this shows that $\sup_{\omega\in\Omega} |r_n(\omega) - r(\omega)| \to 0$ as $n \to \infty$. □

Lemma B.3 is used in the proof of Lemma B.4.

LEMMA B.3. *The function r defined in (52) is continuous on Ω.*

PROOF. By the definition of expectation (law of the unconscious statistician), one has

$$r(\omega) = \int_0^M g(\omega,t)\,\mathrm{d}F^{\mu_0}_{\omega}(t) = \mathsf{E}(g(\omega, d(\omega, X))).$$

Consider a sequence $(\omega_k)_{k\geq 1}$ with $d(\omega_k, \omega_0) \to 0$ as $k \to \infty$. We want to prove $\lim_{k\to\infty} \mathsf{E}(g(\omega_k, d(\omega_k, X))) = \mathsf{E}(g(\omega_0, d(\omega_0, X)))$. Since $d(\omega_k, x) \to d(\omega_0, x)$ for every $x \in \Omega$ and g is continuous on $\Omega \times [0, M]$, it holds that $g(\omega_k, d(\omega_k, X)) \xrightarrow{a.s.} g(\omega_0, d(\omega_0, X))$ as $k \to \infty$. Now, since g is bounded, an application of the DCT gives

$$\lim_{k\to\infty} r(\omega_k) = \lim_{k\to\infty} \mathsf{E}(g(\omega_k, d(\omega_k, X))) = \mathsf{E}(g(\omega_0, d(\omega_0, X))) = r(\omega_0).$$

Thus, r is continuous on Ω. □

Lemma B.4 is used to prove convergence of a term in the proof of Theorem 3.2.

LEMMA B.4. *Let $g : \Omega \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a bounded, uniformly continuous function on $\Omega \times \mathbb{R}_{\geq 0}$. It holds under H_0 that*

$$\int_{\Omega\times\mathbb{R}_{\geq 0}} g(\omega,t)\,\mathrm{d}F^{\mu}_{n,\omega}(t)\,\mathrm{d}P_n(\omega) \xrightarrow{a.s.} \int_{\Omega\times\mathbb{R}_{\geq 0}} g(\omega,t)\,\mathrm{d}F^{\mu_0}_{\omega}(t)\,\mathrm{d}\mu_0(\omega) \text{ as } n \to \infty.$$

PROOF. One has

$$\left|\int_\Omega r_n(\omega)\,\mathrm{d}P_n(\omega) - \int_\Omega r(\omega)\,\mathrm{d}\mu_0(\omega)\right| \le \left|\int_\Omega r_n(\omega) - r(\omega)\,\mathrm{d}P_n(\omega)\right| + \left|\int_\Omega r(\omega)\,\mathrm{d}\left(P_n(\omega) - \mu_0(\omega)\right)\right|$$
$$\le \sup_{\omega\in\Omega} |r_n(\omega) - r(\omega)| + \left|\int_\Omega r(\omega)\,\mathrm{d}\left(P_n(\omega) - \mu_0(\omega)\right)\right| \to 0$$

as $n \to \infty$. The convergence follows by Lemma B.2 for the first term. For the second term, convergence follows by the weak convergence of P_n to μ_0 and the continuity of r by Lemma B.3. For the weak convergence of P_n to μ_0, see Theorems 1 and 3 of Varadarajan (1958). □

Lemma B.5 connects $d_{\Omega\times\mathbb{R}_{\ge 0}}$ with the $L^2(\mu)$ pseudometric induced by $\mathbb{G}_0$, which we denote by $\rho_{2,\mathbb{G}_0}$. This allows obtaining uniform continuity with respect to $\rho_{2,\mathbb{G}_0}$ from uniform continuity with respect to $d_{\Omega\times\mathbb{R}_{\ge 0}}$, which is required in Lemma B.6.

LEMMA B.5. *Let $\rho_{2,y}$ and $\rho_{2,\mathbb{G}_0}$ be the $L^2(\mu)$ pseudometrics induced by the class of functions $\mathcal{Y}$ and by the Gaussian process $\mathbb{G}_0$, respectively. That is, for $p \ge 1$ and $(\omega,t),(\omega',t') \in \Omega\times\mathbb{R}_{\ge 0}$, we define*

$$\rho_{p,y}\big((\omega,t),(\omega',t')\big) := \Big(\mathsf{E}\big|y_{\omega,t}(X) - y_{\omega',t'}(X)\big|^p\Big)^{\frac{1}{p}},$$
$$\rho_{p,\mathbb{G}_0}\big((\omega,t),(\omega',t')\big) := \Big(\mathsf{E}\big|\mathbb{G}_0(\omega,t) - \mathbb{G}_0(\omega',t')\big|^p\Big)^{1/p}. \tag{54}$$

It holds that

$$\rho_{2,\mathbb{G}_0}\big((\omega,t),(\omega',t')\big) \le \rho_{2,y}\big((\omega,t),(\omega',t')\big) \le 2\sqrt{\bar{\Delta} d_{\Omega\times\mathbb{R}_{\ge 0}}\big((\omega,t),(\omega',t')\big)}, \tag{55}$$

In particular, any function that is uniformly continuous with respect to $\rho_{2,\mathbb{G}_0}$ is also uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\ge 0}}$.

PROOF. To prove the first inequality in (55), recall the covariance structure (47)

$$\mathbb{C}\mathrm{ov}\left[\mathbb{G}_0(\omega,t),\mathbb{G}_0(\omega',t')\right] = \mu(y_{\omega,t}y_{\omega',t'}) - \mu y_{\omega,t}\,\mu y_{\omega',t'}.$$

One has

$$\rho_{2,\mathbb{G}_0}\big((\omega,t),(\omega',t')\big)^2 = \mathsf{E}\big(\mathbb{G}_0(\omega,t) - \mathbb{G}_0(\omega',t')\big)^2$$
$$= \mu\Big(\big(y_{\omega,t} - y_{\omega',t'}\big)^2\Big) - \Big(\mu\big(y_{\omega,t} - y_{\omega',t'}\big)\Big)^2$$
$$\le \mu\Big(\big(y_{\omega,t} - y_{\omega',t'}\big)^2\Big) = \rho_{2,y}\big((\omega,t),(\omega',t')\big)^2,$$

which yields the desired inequality.

Now, prove the second inequality in (55). Observe that since $y_{\omega,t}$ and $y_{\omega',t'}$ are indicator functions, one has $(y_{\omega,t} - y_{\omega',t'})^2 = 1_{B(\omega,t)\triangle B(\omega',t')}$, where $A\triangle B := (A\setminus B)\cup(B\setminus A)$ denotes the symmetric difference between two sets A and B. Therefore, it is verified that

$$\rho_{2,y}\big((\omega,t),(\omega',t')\big)^2 = \mathsf{E}((y_{\omega,t} - y_{\omega',t'})^2) = \mathsf{P}\big(B(\omega,t)\triangle B(\omega',t')\big).$$

Define $r := d_{\Omega\times\mathbb{R}_{\ge 0}}((\omega,t),(\omega',t'))$. Then,

$$B(\omega,t)\triangle B(\omega',t') \subset \{x : |d(\omega,x) - t| \le r\} \cup \{x : |d(\omega',x) - t'| \le r\}. \tag{56}$$

To see this, consider any $x \in B(\omega,t)\triangle B(\omega',t')$. Without loss of generality, assume $x \in B(\omega,t)\setminus B(\omega',t')$ (the other case is symmetric). Then, we have $d(\omega,x) \le t$ and $d(\omega',x) > t'$

From the reverse triangle inequality, one obtains $d(\omega,x) \geq d(\omega',x) - d(\omega,\omega') > t' - d(\omega,\omega')$. This provides a bound for $|d(\omega,x) - t| = t - d(\omega,x)$:

$$t - d(\omega,x) < t - \big(t' - d(\omega,\omega')\big) = (t - t') + d(\omega,\omega') \leq |t - t'| + d(\omega,\omega') = r.$$

Thus, $|d(\omega,x) - t| \leq r$. Analogously, if $x \in B(\omega',t') \setminus B(\omega,t)$, one obtains $|d(\omega',x) - t'| \leq r$. Therefore,

$$x \in \big\{x : |d(\omega,x) - t| \leq r\big\} \cup \big\{x : |d(\omega',x) - t'| \leq r\big\},$$

which proves (56).

From (56), it readily follows that

$$\mathsf{P}\big(B(\omega,t) \triangle B(\omega',t')\big) \leq \mathsf{P}\big(|d(\omega,X) - t| \leq r\big) + \mathsf{P}\big(|d(\omega',X) - t'| \leq r\big).$$

By Assumption *(b.2)*, for every $(\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}$ it holds that $\sup_{\omega,t} f_\omega(t) \leq \bar{\Delta}$. Thus, we obtain:

$$\mathsf{P}\big(|d(\omega,X) - t| \leq r\big) = \int_{t-r}^{t+r} f_\omega(s)\,\mathrm{d}s \leq \int_{t-r}^{t+r} \bar{\Delta}\,\mathrm{d}s = 2\bar{\Delta}r.$$

The same estimate applies to $\mathsf{P}\big(|d(\omega',X) - t'| \leq r\big)$. Hence,

$$\rho_{2,y}\big((\omega,t),(\omega',t')\big)^2 \leq 4\bar{\Delta}\,r = 4\bar{\Delta}\,d_{\Omega\times\mathbb{R}_{\geq 0}}((\omega,t),(\omega',t')) \tag{57}$$

which gives the second inequality in (55). $\square$

Lemma B.6 shows that the Gaussian process $\mathbb{G}_0(\omega,t)$ has almost surely uniformly continuous sample paths on $(\Omega \times \mathbb{R}_{\geq 0}, d_{\Omega\times\mathbb{R}_{\geq 0}})$. This result is used, combined with Lemma B.4, in the proof of Theorem 3.2.

LEMMA B.6. *The process $\mathbb{G}_0(\omega,t)$ has almost surely uniformly continuous sample paths on the metric space $(\Omega \times \mathbb{R}_{\geq 0}, d_{\Omega\times\mathbb{R}_{\geq 0}})$, considering the sample paths as functions from $(\Omega \times \mathbb{R}_{\geq 0}, d_{\Omega\times\mathbb{R}_{\geq 0}})$ to $(\mathbb{R}, |\cdot|)$.*

PROOF. Under the null hypothesis, the class $\mathcal{Y}$ is μ-Donsker. Hence, the limiting process $\mathbb{G}_0$ is a tight centered Gaussian process in $\ell^\infty(\mathcal{Y})$.

By Lemma 1.5.9 in Van der Vaart and Wellner (2023), there exists a semimetric ρ on $\mathcal{Y}$ such that $\mathcal{Y}$ is totally bounded with respect to ρ, and almost all sample paths of $\mathbb{G}_0$ are uniformly ρ-continuous. If, in addition, the map $\mathcal{Y} \mapsto \mathbb{R}$ defined by $y \mapsto \mathbb{G}_0(y)$ is uniformly ρ-continuous in second mean, then Lemma 1.5.9 in Van der Vaart and Wellner (2023) (equivalence between (i) and (iii)) can be applied (with $p = 2$). Therefore, $\mathcal{Y}$ is totally bounded with respect to $\rho_{2,\mathbb{G}_0}$, and almost all sample paths of $\mathbb{G}_0$ are uniformly $\rho_{2,\mathbb{G}_0}$-continuous. Here, $\rho_{2,\mathbb{G}_0}(y,y') = \big(\mathsf{E}\big|\mathbb{G}_0(y) - \mathbb{G}_0(y')\big|^2\big)^{1/2}$, which is the same as (54), but as a function of $\mathcal{Y} \times \mathcal{Y}$, rather than $(\Omega \times \mathbb{R}_{\geq 0}) \times (\Omega \times \mathbb{R}_{\geq 0})$. However, for every $y, y' \in \mathcal{Y}$, there exist $(\omega,t),(\omega',t') \in \Omega \times \mathbb{R}_{\geq 0}$ such that $y = y_{\omega,t}$ and $y' = y_{\omega',t'}$. Hence, all sample paths of $\mathbb{G}_0(\omega,t)$ are uniformly $\rho_{2,\mathbb{G}_0}$-continuous as functions from $(\Omega \times \mathbb{R}_{\geq 0}, \rho_{2,\mathbb{G}_0})$ to $(\mathbb{R}, |\cdot|)$. It follows from Lemma B.5 that any function that is uniformly continuous with respect to $\rho_{2,\mathbb{G}_0}$ is also uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}$. Therefore, $\mathbb{G}_0$ has almost surely uniformly continuous sample paths as a function from $(\Omega \times \mathbb{R}_{\geq 0}, d_{\Omega\times\mathbb{R}_{\geq 0}})$ to $(\mathbb{R}, |\cdot|)$.

It remains to show that the map $y \mapsto \mathbb{G}_0(y)$ is uniformly ρ-continuous in second mean. Let $(y_n)_{n\geq 0}$ and $(y_n')_{n\geq 0}$ be sequences in $\mathcal{Y}$ such that $\rho(y_n, y_n') \to 0$, as $n \to \infty$. Since almost all sample paths of $\mathbb{G}_0$ are uniformly ρ-continuous, it follows that $\mathbb{G}_0(y_n) - \mathbb{G}_0(y_n') \xrightarrow{a.s.} 0$, and so $\mathbb{G}_0(y_n) - \mathbb{G}_0(y_n') \xrightarrow{\mathcal{L}} 0$.

For every n, define

$$Z_n := \mathbb{G}_0(y_n) - \mathbb{G}_0(y_n').$$

Since $\mathbb{G}_0$ is a centered Gaussian process, one has that $Z_n \sim N(0, \sigma_n^2)$, with $\sigma_n^2 = \mathsf{E}(Z_n^2)$. Since $Z_n \xrightarrow{\mathcal{L}} 0$, it holds that $\exp(-\sigma_n^2 u^2/2) \to 1$ for every $u \in \mathbb{R}$, which implies that $\sigma_n^2 \to 0$. Therefore,

$$\mathsf{E}\left|\mathbb{G}_0(y_n) - \mathbb{G}_0(y_n')\right|^2 \to 0.$$

This proves that $y \mapsto \mathbb{G}_0(y)$ is uniformly ρ-continuous in second mean. □

Using Lemmas B.1–B.6, Theorem 3.2 is proved.

PROOF OF THEOREM 3.2. We divide the proof into its two parts, *(i)* and *(ii)*.

Proof of (i). The continuous mapping theorem gives the asymptotics of T_n^{KS}. Since we can identify $\mathbb{G}_{n,\mu}^{\mu_0}(\omega, t) = \mathbb{G}_{n,\mu}^{\mu_0}(y_{\omega,t})$ (see the definition of $\mathbb{G}_{n,\mu}^{\mu_0}$ in Section 3.1), it holds that

$$\sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} |\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t)| = \sup_{y\in\mathcal{Y}} |\mathbb{G}_{n,\mu}^{\mu_0}(y)|.$$

The mapping $\sup_{\mathcal{Y}} : \ell^\infty(\mathcal{Y}) \to \mathbb{R}$ defined by $f \mapsto \sup_{\mathcal{Y}} |f(y)|$ for every $f \in \ell^\infty(\mathcal{Y})$ is in fact 1-Lipschitz, hence it is continuous (with respect to the sup-norm).

Proof of (ii). Focus now on T_n^{CM}. We know that $\mathbb{G}_0(\omega, t)$ is separable by Remark 3.1. Therefore, Skorohod's construction (see Theorem 1.10.3 in Van der Vaart and Wellner, 2023, page 58) applies, which ensures that $\mathbb{G}_{n,\mu}^{\mu_0}(\omega, t)$ converges almost uniformly to $\mathbb{G}_0(\omega, t)$. Egorov's theorem (Lemma 1.9.2 in Van der Vaart and Wellner, 2023) states the equivalence of almost uniform and outer almost sure convergence for sequences of Borel measurable random variables. Therefore, there exist random variables (stochastic processes) $(\mathbb{G}_{n,\mu}^{\mu_0})^\star(\omega, t)$ and $\mathbb{G}_0^\star(\omega, t)$ defined on the same probability space such that

$$\sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \left|(\mathbb{G}_{n,\mu}^{\mu_0})^\star(\omega,t) - \mathbb{G}_0^\star(\omega,t)\right| \xrightarrow{a.s.} 0. \tag{58}$$

For simplicity, we will use the same notation for $(\mathbb{G}_{n,\mu}^{\mu_0})^\star$ and $\mathbb{G}_0^\star$ as the original processes. See Definition 1.9.1 in Van der Vaart and Wellner (2023) for almost uniform and outer almost sure convergence.

REMARK B.1. The almost sure convergence result (58) follows from the definition of outer almost sure convergence, using the supremum distance. Observe that the metric space implicit in the weak convergence (9) is $\ell^\infty(\mathcal{Y})$ endowed with the supremum norm (the metric is implicit in the concept of weak convergence through the boundedness and continuity of the functions).

Now, observe that under H_0,

$$\begin{aligned}
&\left|\int_{\Omega\times\mathbb{R}_{\geq 0}} (\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2 \,\mathrm{d}F_{n,\omega}^{\mu_0}(t)\,\mathrm{d}P_n(\omega) - \int_{\Omega\times\mathbb{R}_{\geq 0}} (\mathbb{G}_0(\omega,t))^2 \,\mathrm{d}F_\omega^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega)\right| \\
&\qquad \leq \left|\int_{\Omega\times\mathbb{R}_{\geq 0}} \left((\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2 - (\mathbb{G}_0(\omega,t))^2\right) \mathrm{d}F_{n,\omega}^{\mu_0}(t)\,\mathrm{d}P_n(\omega)\right| \\
&\qquad + \left|\int_{\Omega\times\mathbb{R}_{\geq 0}} (\mathbb{G}_0(\omega,t))^2 \,\mathrm{d}\left(F_{n,\omega}^{\mu_0}(t)\,P_n(\omega) - F_\omega^{\mu_0}(t)\,\mu_0(\omega)\right)\right|.
\end{aligned} \tag{59}$$

For the first term of the RHS of (59), one has

$$\begin{aligned}\Big|\int_{\Omega\times\mathbb{R}_{\geq 0}} &\big((\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2-(\mathbb{G}_0(\omega,t))^2\big)\,\mathrm{d}F_{n,\omega}^{\mu_0}(t)\,\mathrm{d}P_n(\omega)\Big|\\
&\leq \int_{\Omega\times\mathbb{R}_{\geq 0}} \big|(\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2-(\mathbb{G}_0(\omega,t))^2\big|\,\mathrm{d}F_{n,\omega}^{\mu_0}(t)\,\mathrm{d}P_n(\omega)\\
&\leq \sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \big|(\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2-(\mathbb{G}_0(\omega,t))^2\big|\int_{\Omega\times\mathbb{R}_{\geq 0}} \mathrm{d}F_{n,\omega}^{\mu_0}(t)\,\mathrm{d}P_n(\omega)\\
&= \sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \big|(\mathbb{G}_{n,\mu}^{\mu_0}(\omega,t))^2-(\mathbb{G}_0(\omega,t))^2\big|.\end{aligned}$$

Now, observe that

$$\big\|(\mathbb{G}_{n,\mu}^{\mu_0})^2-\mathbb{G}_0^2\big\|_\infty \leq \big\|\mathbb{G}_{n,\mu}^{\mu_0}-\mathbb{G}_0\big\|_\infty^2+2\,\|\mathbb{G}_0\|_\infty\,\big\|\mathbb{G}_{n,\mu}^{\mu_0}-\mathbb{G}_0\big\|_\infty \xrightarrow{a.s.} 0,$$

by (58) and the fact that $\mathbb{G}_0$ has almost surely bounded sample paths. Boundedness holds because $\mathbb{G}_0$ is a map into $\ell^\infty(\mathcal{Y})$.

Now, focus on the second term of the RHS of (59). The sample paths of the Gaussian process $\mathbb{G}_0(\omega,t)$ are bounded and uniformly continuous almost surely. Uniform continuity holds by Lemma B.6. An application of Lemma B.4 to every sample path gives

$$\int_{\Omega\times\mathbb{R}_{\geq 0}} \mathbb{G}_0(\omega,t)^2\,\mathrm{d}\left(F_{n,\omega}^{\mu_0}(t)\,P_n(\omega)-F_\omega^{\mu_0}(t)\,\mu_0(\omega)\right) \xrightarrow{a.s.} 0 \text{ as } n\to\infty,$$

which is the second term of the RHS of (58). Thus, almost surely,

$$T_n^{\mathrm{CM}} \xrightarrow{\mathcal{L}} \int_{\Omega\times\mathbb{R}_{\geq 0}} \mathbb{G}_0(\omega,t)^2\,\mathrm{d}F_\omega^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega) \text{ as } n\to\infty.$$

This concludes the proof. □

B.3. Proof of Theorem 3.3. This section contains the proof of Theorem 3.3, together with the Lemmas needed in the proof. Observe that Lemmas B.1–B.4 also hold when μ_0 is replaced by $\mu_{\boldsymbol{\theta}_0}$.

Lemma B.7 extends Lemma B.5 for the composite null hypothesis.

LEMMA B.7. *Let $\rho_{2,f}$ and $\rho_{2,\tilde{\mathbb{G}}_0}$ be the $L^2(P)$ pseudometrics induced by the class $\mathcal{F}$ and by the Gaussian process $\tilde{\mathbb{G}}_0$, respectively. That is, for $p\geq 1$ and $(\omega,t),(\omega',t')\in\Omega\times\mathbb{R}_{\geq 0}$, define*

$$\begin{aligned}\rho_{p,f}\big((\omega,t),(\omega',t')\big) &:= \Big(\mathsf{E}\big(\big|f_{\omega,t}(X)-f_{\omega',t'}(X)\big|^p\big)\Big)^{1/p}\\
\rho_{p,\tilde{\mathbb{G}}_0}\big((\omega,t),(\omega',t')\big) &:= \Big(\mathsf{E}\big(\big|\tilde{\mathbb{G}}_0(\omega,t)-\tilde{\mathbb{G}}_0(\omega',t')\big|^p\big)\Big)^{1/p}.\end{aligned}$$

Assume that the map $(\omega,t)\mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$ is uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}$. Assume also conditions (c̃.2), (c̃.3), and (d). Then, there exists a nondecreasing function $\phi:[0,\infty)\to[0,\infty)$ with $\phi(r)\to 0$ as $r\to 0$ such that, for all $(\omega,t),(\omega',t')\in\Omega\times\mathbb{R}_{\geq 0}$,

$$\rho_{2,\tilde{\mathbb{G}}_0}\big((\omega,t),(\omega',t')\big)=\rho_{2,f}\big((\omega,t),(\omega',t')\big)\leq \phi\big(d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big)\big)\,. \tag{60}$$

In particular, any function that is uniformly continuous with respect to $\rho_{2,\tilde{\mathbb{G}}_0}$ is also uniformly continuous with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}$.

PROOF. We first prove the equality

$$\rho_{2,\tilde{\mathbb{G}}_0}\big((\omega,t),(\omega',t')\big) = \rho_{2,f}\big((\omega,t),(\omega',t')\big).$$

Since $\tilde{\mathbb{G}}_0$ is the centered Gaussian process indexed by the class $\mathcal{F}$, and $\mathsf{E}\big(f_{\omega,t}(X)\big) = 0$ for every $(\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}$, its covariance is given by

$$\mathbb{C}\mathrm{ov}\left[\tilde{\mathbb{G}}_0(\omega,t), \tilde{\mathbb{G}}_0(\omega',t')\right] = \mathsf{E}\big(f_{\omega,t}(X) f_{\omega',t'}(X)\big).$$

Hence,

$$\begin{aligned}
\rho_{2,\tilde{\mathbb{G}}_0}\big((\omega,t),(\omega',t')\big)^2 &= \mathsf{E}\Big(\big(\tilde{\mathbb{G}}_0(\omega,t) - \tilde{\mathbb{G}}_0(\omega',t')\big)^2\Big) \\
&= \mathsf{E}\Big(\big(f_{\omega,t}(X) - f_{\omega',t'}(X)\big)^2\Big) \\
&= \rho_{2,f}\big((\omega,t),(\omega',t')\big)^2.
\end{aligned}$$

Now, prove the inequality in (60). Recall that

$$f_{\omega,t}(x) = y_{\omega,t}(x) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) + g_{\omega,t}(x), \qquad g_{\omega,t}(x) = \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} \boldsymbol{\psi}_{\boldsymbol{\theta}_0}(x).$$

Fix $(\omega,t),(\omega',t') \in \Omega \times \mathbb{R}_{\geq 0}$, and set

$$r := d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big).$$

By the triangle inequality,

$$\begin{aligned}
\rho_{2,f}\big((\omega,t),(\omega',t')\big) &= \Big(\mathsf{E}\Big(\big(f_{\omega,t}(X) - f_{\omega',t'}(X)\big)^2\Big)\Big)^{1/2} \\
&\leq \Big(\mathsf{E}\Big(\big(y_{\omega,t}(X) - y_{\omega',t'}(X)\big)^2\Big)\Big)^{1/2} + \big|F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) - F_{\omega'}^{\mu_{\boldsymbol{\theta}_0}}(t')\big| \\
&\quad + \Big(\mathsf{E}\Big(\big(g_{\omega,t}(X) - g_{\omega',t'}(X)\big)^2\Big)\Big)^{1/2}.
\end{aligned}$$

For the first term, Lemma B.5 gives

$$\mathsf{E}\Big(\big(y_{\omega,t}(X) - y_{\omega',t'}(X)\big)^2\Big) \leq 4\bar{\Delta} r.$$

For the second term, since

$$F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) = \mathsf{E}\big(y_{\omega,t}(X)\big),$$

one has

$$\big|F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) - F_{\omega'}^{\mu_{\boldsymbol{\theta}_0}}(t')\big| \leq \mathsf{E}\big(\big|y_{\omega,t}(X) - y_{\omega',t'}(X)\big|\big) = \mathsf{E}\Big(\big(y_{\omega,t}(X) - y_{\omega',t'}(X)\big)^2\Big) \leq 4\bar{\Delta}\, r.$$

Finally, for the third term, observe that

$$g_{\omega,t}(x) - g_{\omega',t'}(x) = \Big(\dot{F}_\omega^{\boldsymbol{\theta}_0}(t) - \dot{F}_{\omega'}^{\boldsymbol{\theta}_0}(t')\Big)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} \boldsymbol{\psi}_{\boldsymbol{\theta}_0}(x).$$

Hence, by Cauchy–Schwarz,

$$\Big(\mathsf{E}\Big(\big(g_{\omega,t}(X) - g_{\omega',t'}(X)\big)^2\Big)\Big)^{1/2} \leq \big\|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t) - \dot{F}_{\omega'}^{\boldsymbol{\theta}_0}(t')\big\|\, \|\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\| \Big(\mathsf{E}\big(\|\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\|^2\big)\Big)^{1/2}.$$

Recall that by Assumption $(\tilde{c}.2)$, $\mathsf{E}\|\psi_{\boldsymbol{\theta}_0}(X)\|^2 < \infty$ and by Assumption $(\tilde{c}.3)$ the matrix $\boldsymbol{V}_{\boldsymbol{\theta}_0}$ is non-singular. Define the following modulus of continuity of the map $(\omega,t) \mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$:

$$\eta_{\dot{F}}(r) := \sup\Big\{\big\|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t) - \dot{F}_{\omega'}^{\boldsymbol{\theta}_0}(t')\big\| : (\omega,t),(\omega',t') \in \Omega\times\mathbb{R}_{\geq 0},\, d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big) \leq r\Big\}.$$

Thus,

$$\Big(\mathsf{E}\big(\big(g_{\omega,t}(X)-g_{\omega',t'}(X)\big)^2\big)\Big)^{1/2}\leq \eta_{\dot F}(r)\big\|\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\big\|\Big(\mathsf{E}\big\|\psi_{\boldsymbol{\theta}_0}(X)\big\|^2\Big)^{1/2}$$

and by the assumed uniform continuity of $(\omega,t)\mapsto \dot F_\omega^{\boldsymbol{\theta}_0}(t)$, $\eta_{\dot F}(r)\to 0$ as $r\to 0$.

Combining the three bounds, we obtain

$$\rho_{2,f}\big((\omega,t),(\omega',t')\big)\leq 2\sqrt{\bar\Delta r}+4\bar\Delta\, r+\|\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\|\Big(\mathsf{E}\big(\|\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\|^2\big)\Big)^{1/2}\eta_{\dot F}(r).$$

Thus, the result holds with

$$\phi(r):=2\sqrt{\bar\Delta r}+4\bar\Delta\, r+\|\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\|\Big(\mathsf{E}\big(\|\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X)\|^2\big)\Big)^{1/2}\eta_{\dot F}(r).$$

It holds that ϕ is nondecreasing and $\phi(r)\to 0$ as $r\to 0$, which concludes the proof. □

Lemma B.8 is the analogue of Lemma B.6 for the composite null hypothesis.

LEMMA B.8. *The process $\tilde{\mathbb{G}}_0(\omega,t)$ has almost surely uniformly continuous sample paths on the metric space $(\Omega\times\mathbb{R}_{\geq 0}, d_{\Omega\times\mathbb{R}_{\geq 0}})$, considering the sample paths as functions from $(\Omega\times\mathbb{R}_{\geq 0}, d_{\Omega\times\mathbb{R}_{\geq 0}})$ to $(\mathbb{R},|\cdot|)$.*

PROOF. The proof is completely analogous to that of Lemma B.6, replacing the class $\mathcal{Y}$ by $\mathcal{F}$, the Gaussian process $\mathbb{G}_0$ by $\tilde{\mathbb{G}}_0$, and Lemma B.5 by Lemma B.7. □

Using Lemmas B.7 and B.8, Theorem 3.3 can be proved.

PROOF OF THEOREM 3.3. We divide the proof into its two parts, *(i)* and *(ii)*.

Proof of (i). As explained in the proof of Theorem 3.1, $G_{n,\mu}^{\mu_{\hat\theta}}$ is, up to an $o_P(1)$ remainder (that converges uniformly in ω and t), the empirical process indexed by $\mathcal{F}$. Therefore, the continuous mapping theorem gives

$$\tilde T_n^{\mathrm{KS}}\xrightarrow{\mathcal{L}}\sup_{\omega\in\Omega,t\geq 0}\big|\tilde{\mathbb{G}}_0(\omega,t)\big|.$$

Proof of (ii). The proof for $\tilde T_n^{\mathrm{CM}}$ is completely analogous to that of T_n^{CM} in Theorem 3.2. It suffices to replace the class $\mathcal{Y}$ by $\mathcal{F}$, the Gaussian process $\mathbb{G}_0$ by $\tilde{\mathbb{G}}_0$, and Lemmas B.5 and B.6 by Lemmas B.7 and B.8, respectively. □

B.4. Proof of Proposition 3.4. This section contains the proof of Proposition 3.4, as well as technical lemmas required to prove this result.

LEMMA B.9 (sample forms an asymptotic δ-net). *Let $\mathcal{L}_n=\{X_1,X_2,\ldots,X_n\}$ be iid draws from a distribution μ such that $\Omega\subset\operatorname{supp}(\mu)$. Then, the sample $\mathcal{L}_n$ is a.s. dense in Ω as n diverges to infinity, i.e.,*

$$\sup_{\omega\in\Omega}\min_{1\leq i\leq n} d(\omega,X_i)\to 0 \text{ a.s. as } n\to\infty.$$

In particular, for every $\varepsilon>0$ the random set $\mathcal{L}_n$ is a.s. a 2ε-net of Ω, for n sufficiently large.

PROOF. Fix $\varepsilon>0$. By total boundedness of Ω (implied by Assumption *(b.1)*) there exists a finite ε-net

$$\Omega\subset\bigcup_{j=1}^m B(\omega_j,\varepsilon)$$

for some points $\omega_1, \dots, \omega_m \in \Omega$.

Define, for each $j = 1, \dots, m$, the event

$$A_{n,j} := \{B(\omega_j, \varepsilon) \cap \mathcal{L}_n = \emptyset\}$$

For each fixed j one has $\mathsf{P}\big(A_{n,j}\big) = \big(1 - \mathsf{P}\left(B(\omega_j, \varepsilon)\right)\big)^n$. Set $c_\varepsilon := \min_{1 \le j \le m} \mathsf{P}\big(B(\omega_j, \varepsilon)\big)$. Since $\omega_j \in \Omega \subset \operatorname{supp}(\mu)$, one has $c_\varepsilon > 0$.

Consider the event

$$A_n := \Big\{\exists j \in \{1, \dots, m\} \text{ such that } B(\omega_j, \varepsilon) \cap \mathcal{L}_n = \emptyset\Big\} = \bigcup_{j=1}^{m} A_{n,j}.$$

It holds that

$$\sum_{n=1}^{\infty} \mathsf{P}(A_n) \le \sum_{n=1}^{\infty} \sum_{j=1}^{m} \mathsf{P}(A_{n,j}) = \sum_{n=1}^{\infty} \sum_{j=1}^{m} \big(1 - \mathsf{P}\big(B(\omega_j, \varepsilon)\big)\big)^n \le \sum_{n=1}^{\infty} m\big(1 - c_\varepsilon\big)^n < \infty,$$

because $1 - c_\varepsilon < 1$. By the Borel–Cantelli lemma, with probability one, only finitely many of the events A_n occur; equivalently, with probability one there exists a (random) integer $N = N(\varepsilon)$ such that for all $n \ge N(\varepsilon)$ no $A_{n,j}$ occurs, i.e. every ball $B(\omega_j, \varepsilon)$ contains at least one sample point. Since $\{\omega_1, \dots, \omega_m\}$ forms an ε-net of Ω, for every $\omega \in \Omega$ we can pick $1 \le j \le m$ with $d(\omega, \omega_j) < \varepsilon$, and hence for any sample element $X_i \in B(\omega_j, \varepsilon)$, the triangle inequality gives $d(\omega, X_i) \le 2\varepsilon$ for all $\omega \in \Omega$ with probability one. Therefore,

$$\mathsf{P}\left(\exists N(\varepsilon) :\ \forall n \ge N(\varepsilon),\ \sup_{\omega \in \Omega} \min_{1 \le i \le n} d(\omega, X_i) \le 2\varepsilon\right) = 1.$$

Equivalently,

$$\mathsf{P}\left(\liminf_{n \to \infty} \left\{\sup_{\omega \in \Omega} \min_{1 \le i \le n} d(\omega, X_i) \le 2\varepsilon\right\}\right) = 1.$$

Observe that $\sup_{\omega \in \Omega} \min_{1 \le i \le n} d(\omega, X_i)$ is nonnegative and nonincreasing as a function of n. Therefore,

$$\mathsf{P}\left(\lim_{n \to \infty} \sup_{\omega \in \Omega} \min_{1 \le i \le n} d(\omega, X_i) \le 2\varepsilon\right) = 1. \tag{61}$$

Finally, let $\varepsilon_k = 1/k$ for $k \in \mathbb{N}$. The above argument can be repeated to obtain (61) for every ε_k. Making $k \to \infty$, this implies that $\sup_{\omega \in \Omega} \min_{1 \le i \le n} d(\omega, X_i) \to 0$ a.s. as $n \to \infty$. □

Using Lemmas B.5 and B.9, Proposition 3.4 can be proved.

PROOF OF PROPOSITION 3.4. We divide the proof into its two parts, *(i)* and *(ii)*.

Proof of (i). Consider the class $\mathcal{Y} = \{y_{\omega,t} : \omega \in \Omega,\ t \ge 0\}$. Under H_0, the class $\mathcal{Y}$ is Donsker. Throughout the proof, we use the $L^2(\mu)$ pseudometric $\rho_{2,y}$ defined in Lemma B.5, identifying $\mathcal{Y}$ with its index set $\Omega \times \mathbb{R}_{\ge 0}$.

Since $\mathcal{Y}$ is Donsker, the empirical process is asymptotically equicontinuous with respect to $\rho_{2,y}$ (page 139 in Van der Vaart and Wellner, 2023): for every $\eta > 0$,

$$\lim_{\delta \to 0^+} \limsup_{n \to \infty} \mathsf{P}^* \left\{ \sup_{\substack{f, g \in \mathcal{Y} \\ \rho_{2,y}(f,g) < \delta}} \left|\mathbb{G}_{n,\mu}^{\mu_0}(f - g)\right| > \eta \right\} = 0.$$

Fix $\tau > 0$ and $\gamma > 0$. By asymptotic equicontinuity, there exists $\delta = \delta(\tau, \gamma) > 0$ such that

$$\limsup_{n\to\infty} \mathsf{P}^* \left\{ \sup_{\substack{f,g\in\mathcal{Y} \\ \rho_{2,y}(f,g)<\delta}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(f) - \mathbb{G}^{\mu_0}_{n,\mu}(g) \right| > \tau \right\} < \gamma.$$

Hence, for n sufficiently large,

$$\mathsf{P}^* \left\{ \sup_{\substack{f,g\in\mathcal{Y} \\ \rho_{2,y}(f,g)<\delta}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(f) - \mathbb{G}^{\mu_0}_{n,\mu}(g) \right| > \tau \right\} < \gamma. \tag{62}$$

For any function $y_{\omega,t} \in \mathcal{Y}$, Lemma B.5 shows that the map $\omega \mapsto y_{\omega,t}$ is uniformly continuous with respect to $\rho_{2,y}$, uniformly in t. Indeed,

$$\rho_{2,y}(y_{\omega,t}, y_{\omega',t}) \leq 2\sqrt{\bar{\Delta}\, d(\omega, \omega')}$$

for all $\omega, \omega' \in \Omega$ and all $t \geq 0$. Therefore, choosing $r > 0$ such that $r < \delta^2/(4\bar{\Delta})$, it follows that for all $\omega, \omega' \in \Omega$ and all $t \geq 0$, $d(\omega, \omega') < r$ implies $\rho_{2,y}(y_{\omega,t}, y_{\omega',t}) < \delta$.

By Lemma B.9, the sample $\mathcal{L}_n$ is dense in Ω, so for n sufficiently large:

$$\mathsf{P}\left(\sup_{\omega\in\Omega} \min_{1\leq i\leq n} d(\omega, X_i) < r \right) \geq 1 - \gamma.$$

Thus, with probability at least $1-\gamma$, for every $\omega \in \Omega$ there exists $X_i \in \mathcal{L}_n$ such that $d(\omega, X_i) < r$, and hence by uniform continuity $\rho_{2,y}(y_{\omega,t}, y_{X_i,t}) < \delta$ for all $t \geq 0$. This means that $\mathcal{Y}_n := \{y_{X_i,t} : i = 1, \ldots, n,\ t \geq 0\}$ forms a δ-net for $\mathcal{Y}$ with probability at least $1-\gamma$ when n is sufficiently large, i.e., for every $y_{\omega,t} \in \mathcal{Y}$ there exists $y_{X_i,t} \in \mathcal{Y}_n$ such that $\rho_{2,y}(y_{\omega,t}, y_{X_i,t}) < \delta$.

By the triangle inequality, one has

$$\left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{\omega,t}) \right| \leq \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right| + \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{\omega,t}) - \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right|.$$

Therefore, if $\mathcal{Y}_n$ is a δ-net of $\mathcal{Y}$, then

$$\sup_{y\in\mathcal{Y}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y) \right| \leq \max_{1\leq i\leq n} \sup_{t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right| + \sup_{\substack{f,g\in\mathcal{Y} \\ \rho_{2,y}(f,g)<\delta}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(f) - \mathbb{G}^{\mu_0}_{n,\mu}(g) \right|.$$

The last term on the right is less than or equal to τ with (outer) probability at least $1-\gamma$ by (62). Trivially, $\max_{1\leq i\leq n} \sup_{t\geq 0} |\mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t})| \leq \sup_{y\in\mathcal{Y}} |\mathbb{G}^{\mu_0}_{n,\mu}(y)|$. Hence, for every $\tau > 0$ and $\gamma > 0$, it holds that

$$\mathsf{P}^* \left\{ \max_{1\leq i\leq n} \sup_{t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right| \leq \sup_{y\in\mathcal{Y}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y) \right| \leq \max_{1\leq i\leq n} \sup_{t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right| + \tau \right\} \geq 1 - 2\gamma.$$

This means that

$$\mathsf{P}^* \left\{ \left| \sup_{y\in\mathcal{Y}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y) \right| - \max_{1\leq i\leq n} \sup_{t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right| \right| > \tau \right\} < 2\gamma$$

for n sufficiently large. Therefore,

$$\sup_{y\in\mathcal{Y}} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y) \right| - \max_{1\leq i\leq n} \sup_{t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(y_{X_i,t}) \right| \xrightarrow{\mathsf{P}^*} 0 \text{ as } n \to \infty,$$

that is,

$$\sup_{\omega\in\Omega,\ t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(\omega, t) \right| = \max_{1\leq i\leq n} \sup_{t\geq 0} \left| \mathbb{G}^{\mu_0}_{n,\mu}(X_i, t) \right| + o_{\mathsf{P}^*}(1).$$

Under Assumption (*b.2*), for each fixed X_i, the standard argument for the Kolmogorov–Smirnov statistic discretizes the supremum over t into a maximum over the distance sample $d(X_i, X_1), \dots, d(X_i, X_n)$. This proves (23).

Proof of (ii). The proof of (24) is analogous to that of (23). One replaces $\mathcal{Y}$ by $\mathcal{F}$, and Lemma B.5 by Lemma B.7. □

APPENDIX C: PROOFS OF SECTION 4

C.1. Fixed alternatives.

C.1.1. *Simple null hypothesis.* Theorem 4.1 states the consistency of the Kolmogorov–Smirnov test statistic under fixed alternatives.

PROOF OF THEOREM 4.1. For $(\omega, t) \in \Omega \times \mathbb{R}_{\geq 0}$, write

$$F^{\mu}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t) = \big(F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big) + \big(F^{\mu}_{\omega}(t) - F^{\mu_0}_{\omega}(t)\big),$$

and define the empirical process under μ ($\mu \neq \mu_0$) by

$$\mathbb{G}^{\mu}_{n,\mu}(\omega, t) := \sqrt{n}\,\big(F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big). \tag{63}$$

Next, define $\Delta(\omega, t) := F^{\mu}_{\omega}(t) - F^{\mu_0}_{\omega}(t)$. Since $\mu \neq \mu_0$, there exists at least one pair $(\omega_0, t_0) \in \Omega \times \mathbb{R}_{\geq 0}$ such that $\Delta(\omega_0, t_0) \neq 0$. Therefore,

$$\delta := \sup_{\omega\in\Omega, t\geq 0} |\Delta(\omega, t)| \;\geq\; |\Delta(\omega_0, t_0)| \;>\; 0.$$

Diego: I present two options to continue the proof from here. Option 1 is simpler. I let you choose.

Option 1. Since $\mathcal{Y}$ is μ-Donsker, it is also μ-Glivenko–Cantelli. Therefore,

$$\sup_{\omega\in\Omega, t\geq 0} \big|F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big| = o_{\mathsf{P}^*}(1).$$

Now, write

$$\frac{T^{\mathrm{KS}}_n}{\sqrt{n}} = \sup_{\omega\in\Omega, t\geq 0} \Big|\Delta(\omega, t) + \big(F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big)\Big|.$$

Using the inequality $\big|\sup|a+b| - \sup|a|\big| \leq \sup|b|$, we obtain

$$\left|\frac{T^{\mathrm{KS}}_n}{\sqrt{n}} - \delta\right| \leq \sup_{\omega\in\Omega, t\geq 0} \big|F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big| = \frac{\|\mathbb{G}^{\mu}_{n,\mu}\|_\infty}{\sqrt{n}} = o_{\mathsf{P}^*}(1).$$

Therefore,

$$\frac{T^{\mathrm{KS}}_n}{\sqrt{n}} \xrightarrow{\mathsf{P}^*} \delta,$$

and in particular,

$$T^{\mathrm{KS}}_n \xrightarrow{\mathsf{P}^*} \infty.$$

Option 2. Using the reverse triangle inequality pointwise in (ω, t) and taking suprema, we obtain

$$T^{\mathrm{KS}}_n \geq \sup_{\omega\in\Omega, t\geq 0} \sqrt{n}\,|\Delta(\omega, t)| - \sup_{\omega\in\Omega, t\geq 0} \sqrt{n}\,\big|F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big| = \sqrt{n}\,\delta - \big\|\mathbb{G}^{\mu}_{n,\mu}\big\|_\infty.$$

Since $\mathcal{Y}$ is Donsker under μ, the empirical process $\mathbb{G}_{n,\mu}^{\mu}$ is asymptotically tight in $\ell^\infty(\Omega \times \mathbb{R}_{\geq 0})$; (see, e.g., page 139 in Van der Vaart and Wellner, 2023). By the continuous mapping theorem for asymptotic tightness (see, e.g., Problem 7(i) in Section 1.3 of Van der Vaart and Wellner, 2023), the sequence $\|\mathbb{G}_{n,\mu}^{\mu}\|_\infty$ is asymptotically tight in $\mathbb{R}$. In particular, for every $\varepsilon > 0$ there exists $M < \infty$ such that

$$\limsup_{n\to\infty} \mathsf{P}_\mu^* \left(\|\mathbb{G}_{n,\mu}^{\mu}\|_\infty > M\right) \leq \varepsilon. \tag{64}$$

Take any $M_0 > 0$ and set $a_n := \sqrt{n}\,\delta - M_0$. Fix $\varepsilon > 0$ and choose M to satisfy (64). Then, there exists $n_0 \in \mathbb{N}$ such that $a_n \geq M$ for all $n \geq n_0$, and hence, for all $n \geq n_0$, $\mathsf{P}_\mu^* \left(\|\mathbb{G}_{n,\mu}^{\mu}\|_\infty \geq a_n\right) \leq \mathsf{P}_\mu^* \left(\|\mathbb{G}_{n,\mu}^{\mu}\|_\infty > M\right)$. Taking $\limsup_{n\to\infty}$ yields

$$\limsup_{n\to\infty} \mathsf{P}_\mu^* \left(\|\mathbb{G}_{n,\mu}^{\mu}\|_\infty \geq a_n\right) \leq \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that

$$\mathsf{P}_\mu^* \left(\|\mathbb{G}_{n,\mu}^{\mu}\|_\infty \geq \sqrt{n}\,\delta - M_0\right) \to 0, \qquad n \to \infty.$$

Hence, $\mathsf{P}_\mu^*(T_n^{\mathrm{KS}} \leq M_0) \to 0$ for every $M_0 > 0$. This proves

$$T_n^{\mathrm{KS}} \xrightarrow{\mathsf{P}^*} \infty.$$

□

We now turn to the Cramér–von Mises test statistic. Theorem 4.2 states that the consistency of the Cramér–von Mises test statistic under fixed alternatives. To prove this result, Lemmas C.1, C.2 and C.3 are needed.

LEMMA C.1. *Let $M = \mathrm{diam}(\Omega)$ and define $\Delta(\omega, t) := F_\omega^\mu(t) - F_\omega^{\mu_0}(t)$ for $(\omega, t) \in \Omega \times [0, M]$. There exist constants $\bar{\Delta}_\mu < \infty$ and $\bar{\Delta}_{\mu_0} < \infty$ (independent of t) such that, for any $\omega, \omega' \in \Omega$ and each $t \in [0, M]$,*

$$|\Delta(\omega, t) - \Delta(\omega', t)| \leq (\bar{\Delta}_\mu + \bar{\Delta}_{\mu_0})\, d(\omega, \omega').$$

That is, for each $t \in [0, M]$ the map $\omega \mapsto \Delta(\omega, t)$ is Lipschitz on Ω with Lipschitz constant bounded uniformly over $t \in [0, M]$.

PROOF. Fix $t \in [0, M]$ and $\omega, \omega' \in \Omega$, and define $r := d(\omega, \omega')$. For every $x \in \Omega$, the triangle inequality yields

$$d(\omega', x) \leq d(\omega', \omega) + d(\omega, x) = r + d(\omega, x).$$

Hence, $\{d(\omega, x) \leq t\} \subset \{d(\omega', x) \leq t + r\}$, and so $F_\omega^\mu(t) \leq F_{\omega'}^\mu(t + r)$. Similarly, $F_{\omega'}^\mu(t) \leq F_\omega^\mu(t + r)$, so

$$|F_\omega^\mu(t) - F_{\omega'}^\mu(t)| \leq \max\{F_{\omega'}^\mu(t + r) - F_{\omega'}^\mu(t),\, F_\omega^\mu(t + r) - F_\omega^\mu(t)\}.$$

By (*b.2*), the densities $\{f_\omega^\mu\}$ are uniformly bounded above by some $\bar{\Delta}_\mu < \infty$, hence

$$F_\omega^\mu(t + r) - F_\omega^\mu(t) = \int_t^{t+r} f_\omega^\mu(u)\, \mathrm{d}u \leq \bar{\Delta}_\mu\, r,$$

and similarly for ω'. Therefore,

$$|F_\omega^\mu(t) - F_{\omega'}^\mu(t)| \leq \bar{\Delta}_\mu\, d(\omega, \omega').$$

The same argument applied to μ_0 yields $|F_\omega^{\mu_0}(t) - F_{\omega'}^{\mu_0}(t)| \leq \bar{\Delta}_{\mu_0}\, d(\omega, \omega')$ for a finite $\bar{\Delta}_{\mu_0}$. Finally,

$$|\Delta(\omega, t) - \Delta(\omega', t)| \leq |F_\omega^\mu(t) - F_{\omega'}^\mu(t)| + |F_\omega^{\mu_0}(t) - F_{\omega'}^{\mu_0}(t)| \leq (\bar{\Delta}_\mu + \bar{\Delta}_{\mu_0})\, d(\omega, \omega'),$$

which gives the result. □

LEMMA C.2. *Assume that (b.2) holds. If there exist $\omega_0 \in \Omega$ and $t_0 \in [0, M]$ such that $\Delta(\omega_0, t_0) \neq 0$, then there exist $\varepsilon_\omega > 0$, $\varepsilon_t > 0$, and $c_0 > 0$ such that*

$$|\Delta(\omega, t)| \geq c_0 \qquad \forall\, \omega \in B(\omega_0, \varepsilon_\omega),\ \forall\, t \in (t_0 - \varepsilon_t, t_0 + \varepsilon_t) \cap [0, M].$$

PROOF. Let $\delta_0 := |\Delta(\omega_0, t_0)| > 0$. For each $\omega \in \Omega$, the map $t \mapsto \Delta(\omega, t)$ is continuous on $[0, M]$ by assumption (b.2). Thus, there exists $\varepsilon_t > 0$ such that

$$|\Delta(\omega_0, t) - \Delta(\omega_0, t_0)| < \frac{\delta_0}{2} \quad \text{for all } t \in [0, M] \text{ such that } |t - t_0| < \varepsilon_t,$$

which implies $|\Delta(\omega_0, t)| \geq \frac{\delta_0}{2}$. Next, by Lemma C.1, for all $t \in [0, M]$ one has

$$|\Delta(\omega, t) - \Delta(\omega_0, t)| \leq (\bar{\Delta}_\mu + \bar{\Delta}_{\mu_0})\, d(\omega, \omega_0).$$

Observe that the right-hand side does not depend on t.

Choose $\varepsilon_\omega := \frac{\delta_0}{4(\bar{\Delta}_\mu + \bar{\Delta}_{\mu_0})}$ and set $c_0 := \frac{\delta_0}{4}$. Then, for all $\omega \in B(\omega_0, \varepsilon_\omega)$ and all $t \in (t_0 - \varepsilon_t, t_0 + \varepsilon_t) \cap [0, M]$, it holds that

$$|\Delta(\omega, t)| \geq |\Delta(\omega_0, t)| - |\Delta(\omega, t) - \Delta(\omega_0, t)| \geq \delta_0/2 - (\bar{\Delta}_\mu + \bar{\Delta}_{\mu_0})\varepsilon_\omega = \delta_0/4 := c_0 > 0.$$

This concludes the result. $\square$

LEMMA C.3. *Let $\omega_0 \in \operatorname{supp}(\mu)$. Let $B_0 := B(\omega_0, \varepsilon_\omega)$ for some $\varepsilon_\omega > 0$, and let $I_0 \subset [0, M]$ be (any) nonempty open interval (relative to $[0, M]$). If X, Y are iid with law μ, then*

$$\mathsf{P}\big(X \in B_0,\ d(X, Y) \in I_0\big) > 0.$$

PROOF. Fix $\omega \in \Omega$. Let μ_ω denote the law of $d(\omega, Y)$. Under (b.2), the random variable $d(\omega, Y)$ admits a continuous density f^μ_ω on $[0, M]$ and satisfies $\underline{\Delta}_\omega := \inf_{t \in \operatorname{supp}(\mu_\omega)} f^\mu_\omega(t) > 0$. Set

$$A_\omega := \{t \in [0, M] : f^\mu_\omega(t) > 0\}.$$

Since f^μ_ω is continuous on $[0, M]$, the set A_ω is (relatively) open in $[0, M]$.

We first show that $A_\omega \subset \operatorname{supp}(\mu_\omega)$. Let $t \in A_\omega$, so that $f^\mu_\omega(t) > 0$. By continuity of f^μ_ω on $[0, M]$, there exists $\eta > 0$ such that

$$f^\mu_\omega(s) \geq \frac{f^\mu_\omega(t)}{2} > 0 \qquad \text{for all } s \in (t - \eta, t + \eta) \cap [0, M].$$

Let $U \subset [0, M]$ be a relatively open set with $t \in U$. Then there exists $\eta' > 0$ such that

$$(t - \eta', t + \eta') \cap [0, M] \subset U.$$

Setting $\tilde{\eta} := \min\{\eta, \eta'\}$, we obtain

$$\mu_\omega(U) = \int_U f^\mu_\omega(s)\, \mathrm{d}s \geq \int_{(t - \tilde{\eta}, t + \tilde{\eta}) \cap [0, M]} f^\mu_\omega(s)\, \mathrm{d}s > 0.$$

Hence, by the definition of support, $t \in \operatorname{supp}(\mu_\omega)$.

Next, we show that $\operatorname{supp}(\mu_\omega) \subset A_\omega$. If $t \in \operatorname{supp}(\mu_\omega)$, then by Assumption (b.2) one has $f^\mu_\omega(t) \geq \underline{\Delta}_\omega > 0$, so $t \in A_\omega$. Therefore, $A_\omega = \operatorname{supp}(\mu_\omega)$, which is relatively closed in $[0, M]$. Since A_ω is also relatively open in $[0, M]$, and $A_\omega \neq \varnothing$ (because $d(\omega, Y)$ admits a density on $[0, M]$), the connectedness of $[0, M]$ implies that $A_\omega = [0, M]$, and thus $f^\mu_\omega(t) > 0$ for all $t \in [0, M]$. Consequently, for any nonempty open interval $I_0 \subset [0, M]$,

$$\mathsf{P}\big(d(\omega, Y) \in I_0\big) = \int_{I_0} f_\omega^\mu(t)\, dt > 0. \tag{65}$$

Finally, by the tower property,

$$\mathsf{P}\big(X \in B_0,\ d(X,Y) \in I_0\big) = \mathsf{E}_\mu\left[1_{\{X\in B_0\}} 1_{\{d(X,Y)\in I_0\}}\right] = \mathsf{E}_\mu\left[1_{\{X\in B_0\}}\, \mathsf{P}_\mu\left(d(X,Y) \in I_0 \,|\, X\right)\right]$$

and for every $\omega \in \Omega$, since Y is independent of X,

$$\mathsf{P}_\mu\left(d(X,Y) \in I_0 \,|\, X = \omega\right) = \mathsf{P}\big(d(\omega, Y) \in I_0\big).$$

Therefore,

$$\mathsf{P}\big(X \in B_0,\ d(X,Y) \in I_0\big) = \int_{B_0} \mathsf{P}\big(d(\omega, Y) \in I_0\big)\, \mathrm{d}\mu(\omega).$$

The integrand is strictly positive for every $\omega \in B_0$ by (65), and $\mu(B_0) > 0$ because $\omega_0 \in \operatorname{supp}(\mu)$. Hence, $\mathsf{P}\left(X \in B_0,\ D \in I_0\right) > 0$. □

Using Lemmas C.1, C.2 and C.3, Theorem 4.2 can be proved.

PROOF OF THEOREM 4.2. Since either *(a.2)* holds, or *(a.1)* holds and $\operatorname{supp}(\mu) = \Omega$, under H_1, there exist $\omega_0 \in \operatorname{supp}(\mu)$ and $t_0 \in (0, M]$ such that $F_{\omega_0}^\mu(t_0) \neq F_{\omega_0}^{\mu_0}(t_0)$, i.e. $\Delta(\omega_0, t_0) \neq 0$. Apply Lemma C.2 to obtain $\varepsilon_\omega > 0$, $\varepsilon_t > 0$, and $c_0 > 0$ such that

$$|\Delta(\omega, t)| \geq c_0 \quad \forall\, (\omega, t) \in B_0 \times I_0,$$

where $B_0 = B(\omega_0, \varepsilon_\omega)$ and $I_0 = (t_0 - \varepsilon_t, t_0 + \varepsilon_t) \cap [0, M]$. Since $\omega_0 \in \operatorname{supp}(\mu)$, Lemma C.3 yields

$$\mathsf{P}\big(X \in B_0,\ d(X,Y) \in I_0\big) > 0, \tag{66}$$

for iid $X, Y \sim \mu$.

Define

$$k(x,y) := \big(F_x^\mu(d(x,y)) - F_x^{\mu_0}(d(x,y))\big)^2 = \Delta\big(x, d(x,y)\big)^2, \quad x, y \in \Omega,$$

and let $J := \mathbb{E}_\mu\big[k(X,Y)\big]$, for a pair of iid random objects $X, Y \sim \mu$.

By Lemma C.2, one has

$$k(X,Y)\, 1_{\{X\in B_0,\ d(X,Y)\in I_0\}} \ \geq\ c_0^2\, 1_{\{X\in B_0,\ d(X,Y)\in I_0\}}.$$

Taking expectations and using (66) yields

$$J = \mathsf{E}_\mu[k(X,Y)] \geq \mathsf{E}_\mu\left[k(X,Y)\, 1_{\{X\in B_0,\ d(X,Y)\in I_0\}}\right] \geq c_0^2\, \mathsf{P}\big(X \in B_0,\ d(X,Y) \in I_0\big) > 0. \tag{67}$$

Next, define

$$e_n := \sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \big|F_{n,\omega}^\mu(t) - F_\omega^\mu(t)\big|.$$

By Glivenko–Cantelli, one has $e_n \xrightarrow{\mathsf{P}} 0$. Observe that one can write

$$\frac{1}{n} T_n^{\mathrm{CM}} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \big(F_{n,X_i}^\mu(d(X_i, X_j)) - F_{X_i}^{\mu_0}(d(X_i, X_j))\big)^2.$$

Define $k_n(x,y) := \left(F^{\mu}_{n,x}(d(x,y)) - F^{\mu_0}_{x}(d(x,y))\right)^2$, so that $\frac{1}{n}T_n^{\mathrm{CM}} = \frac{1}{n^2}\sum_{i,j} k_n(X_i,X_j)$. For any $x,y \in \Omega$ write

$$F^{\mu}_{n,x}(d(x,y)) - F^{\mu_0}_{x}(d(x,y)) = \left(F^{\mu}_{x}(d(x,y)) - F^{\mu_0}_{x}(d(x,y))\right) + \left(F^{\mu}_{n,x}(d(x,y)) - F^{\mu}_{x}(d(x,y))\right).$$

The second term is bounded in absolute value by e_n, uniformly in x,y. The first term is bounded by 1 in absolute value. Therefore, using $|(a+b)^2 - a^2| \le 2|a||b| + b^2$ for any $a,b \in \mathbb{R}$, it holds that

$$\sup_{x,y\in\Omega} |k_n(x,y) - k(x,y)| \le 2e_n + e_n^2.$$

Consequently,

$$\left|\frac{T_n^{\mathrm{CM}}}{n} - V_n\right| \le 2e_n + e_n^2 \xrightarrow{\mathsf{P}_\mu} 0, \tag{68}$$

where

$$V_n := \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} k(X_i,X_j).$$

It remains to show $V_n \xrightarrow{\mathsf{P}_\mu} J$. Since k is a positive function uniformly bounded by 1, it is integrable. Decompose

$$V_n = \frac{1}{n^2}\sum_{i\neq j} k(X_i,X_j) + \frac{1}{n^2}\sum_{i=1}^{n} k(X_i,X_i),$$

For the second term, one has $0 \le n^{-2}\sum_i k(X_i,X_i) \le n^{-1} \to 0$ a.s. Define the symmetrized kernel $\tilde{k}(x,y) := \frac{1}{2}(k(x,y)+k(y,x))$, so $\tilde{k}(x,y) = \tilde{k}(y,x)$ and $0 \le \tilde{k} \le 1$. Then

$$\sum_{i\neq j} k(X_i,X_j) = 2\sum_{1\le i<j\le n} \tilde{k}(X_i,X_j),$$

and hence

$$\frac{1}{n^2}\sum_{i\neq j} k(X_i,X_j) = \left(1 - \frac{1}{n}\right)U_n, \quad \text{where } U_n := \frac{1}{\binom{n}{2}}\sum_{1\le i<j\le n} \tilde{k}(X_i,X_j).$$

By the strong law of large numbers for U-statistics of order two applied to $\tilde{k}$ (see, e.g., Theorem 5.4A of Serfling, 1980), one has $U_n \xrightarrow{\text{a.s.}} \mathbb{E}_\mu\left(\tilde{k}(X,Y)\right)$. Since X,Y are iid, $\mathbb{E}_\mu\left(k(X,Y)\right) = \mathbb{E}_\mu\left(k(Y,X)\right)$, thus from the definition of $\tilde{k}$ it follows that $\mathbb{E}_\mu\left(\tilde{k}(X,Y)\right) = \mathbb{E}_\mu\left(k(X,Y)\right) = J$. Therefore, $V_n \xrightarrow{\text{a.s.}} J$, and in particular $V_n \xrightarrow{\mathsf{P}} J$. Combining this with (68) yields $\frac{1}{n}T_n^{\mathrm{CM}} \xrightarrow{\mathsf{P}_\mu} J$, and hence $T_n^{\mathrm{CM}} \xrightarrow{\mathsf{P}_\mu} \infty$. □

C.1.2. *Composite null.* Theorem 4.3 states the consistency of the Kolmogorov–Smirnov test statistic under fixed alternatives, when the null hypothesis is composite and a parameter is estimated.

PROOF OF THEOREM 4.3. For $(\omega,t) \in \Omega \times \mathbb{R}_{\ge 0}$, write

$$\sqrt{n}\left(F^{\mu}_{n,\omega}(t) - F^{\mu_{\hat{\theta}}}_{\omega}(t)\right) = \mathbb{G}^{\mu}_{n,\mu}(\omega,t) + \sqrt{n}\left(F^{\mu}_{\omega}(t) - F^{\mu_{\hat{\theta}}}_{\omega}(t)\right),$$

with $\mathbb{G}^{\mu}_{n,\mu}$ as in (63). Let $\mu_{\boldsymbol{\theta}_1}$ be the probability measure corresponding to $\boldsymbol{\theta}_1$. Define

$$\Delta^{\star}(\omega,t) := F^{\mu}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t), \qquad \delta^{\star} := \sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \big|\Delta^{\star}(\omega,t)\big|.$$

Since $\mu \notin \{\mu_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \boldsymbol{\Theta}\}$, in particular $\mu \neq \mu_{\boldsymbol{\theta}_1}$, and hence there exists $(\omega_0,t_0) \in \Omega \times \mathbb{R}_{\geq 0}$ such that $\Delta^{\star}(\omega_0,t_0) \neq 0$. Therefore, $\delta^{\star} \geq |\Delta^{\star}(\omega_0,t_0)| > 0$.

Define

$$r_n := \sup_{\omega\in\Omega, t\geq 0} \big|F^{\mu_{\hat{\boldsymbol{\theta}}}}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t)\big|.$$

By (25), for every $\varepsilon > 0$ there exists $\eta > 0$ such that

$$\|\boldsymbol{\theta} - \boldsymbol{\theta}_1\| < \eta \quad \Longrightarrow \quad \sup_{\omega,t}\big|F^{\mu_{\boldsymbol{\theta}}}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t)\big| < \varepsilon.$$

Since $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_1$, it follows that $\mathsf{P}^*(r_n > \varepsilon) \leq \mathsf{P}(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_1\| \geq \eta) \to 0$. Thus, $r_n \xrightarrow{\mathsf{P}^*} 0$.

Using the triangle inequality pointwise in (ω,t) and taking suprema,

$$\begin{aligned}
\tilde{T}^{\mathrm{KS}}_n &= \sup_{\omega,t} \sqrt{n}\,\big|F^{\mu}_{n,\omega}(t) - F^{\mu_{\hat{\boldsymbol{\theta}}}}_{\omega}(t)\big| \\
&\geq \sup_{\omega,t} \sqrt{n}\,\big|F^{\mu}_{\omega}(t) - F^{\mu_{\hat{\boldsymbol{\theta}}}}_{\omega}(t)\big| - \sup_{\omega,t} \sqrt{n}\,\big|F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big| \\
&= \sup_{\omega,t} \sqrt{n}\,\big|F^{\mu}_{\omega}(t) - F^{\mu_{\hat{\boldsymbol{\theta}}}}_{\omega}(t)\big| - \|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} \\
&\geq \sup_{\omega,t} \sqrt{n}\,\big|F^{\mu}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t)\big| - \sup_{\omega,t} \sqrt{n}\,\big|F^{\mu_{\hat{\boldsymbol{\theta}}}}_{\omega}(t) - F^{\mu_{\boldsymbol{\theta}_1}}_{\omega}(t)\big| - \|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} \\
&\geq \sqrt{n}\,\delta^{\star} - \sqrt{n}\,r_n - \|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty}.
\end{aligned}$$

Fix any $M_0 > 0$ and define $a_n := \sqrt{n}\,\delta^{\star} - M_0$. Then,

$$\mathsf{P}^*(\tilde{T}^{\mathrm{KS}}_n \leq M_0) \leq \mathsf{P}^*\left(\|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} + \sqrt{n}\,r_n \geq a_n\right).$$

Since the class $\mathcal{Y}$ is μ-Donsker, it is also μ-Glivenko–Cantelli. Hence,

$$\sup_{\omega\in\Omega, t\geq 0} \big|F^{\mu}_{n,\omega}(t) - F^{\mu}_{\omega}(t)\big| = o_{\mathsf{P}^*}(1).$$

Equivalently,

$$\frac{1}{\sqrt{n}}\|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} = o_{\mathsf{P}^*}(1).$$

Therefore,

$$\mathsf{P}^*\left(\|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} \geq \frac{a_n}{2}\right) = \mathsf{P}^*\left(\frac{1}{\sqrt{n}}\|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} \geq \frac{\delta^{\star}}{2} - \frac{M_0}{2\sqrt{n}}\right) \to 0.$$

Also, since $r_n \to 0$ in P^*, we have $\mathsf{P}^*(\sqrt{n}\,r_n \geq a_n/2) \to 0$. To see this, note that

$$\mathsf{P}^*\left(\sqrt{n}\,r_n \geq \frac{a_n}{2}\right) = \mathsf{P}^*\left(r_n \geq \frac{a_n}{2\sqrt{n}}\right) = \mathsf{P}^*\left(r_n \geq \frac{\delta^{\star}}{2} - \frac{M_0}{2\sqrt{n}}\right).$$

Since $\frac{M_0}{2\sqrt{n}} \to 0$, there exists n_0 such that $\frac{\delta^{\star}}{2} - \frac{M_0}{2\sqrt{n}} \geq \frac{\delta^{\star}}{4}$ for all $n \geq n_0$, and hence, for $n \geq n_0$,

$$\mathsf{P}^*\left(\sqrt{n}\,r_n \geq \frac{a_n}{2}\right) \leq \mathsf{P}^*\left(r_n \geq \frac{\delta^{\star}}{4}\right) \to 0.$$

Therefore,

$$\mathsf{P}^*\left(\|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} + \sqrt{n}\,r_n \geq a_n\right) \leq \mathsf{P}^*\left(\|\mathbb{G}^{\mu}_{n,\mu}\|_{\infty} \geq \frac{a_n}{2}\right) + \mathsf{P}^*\left(\sqrt{n}\,r_n \geq \frac{a_n}{2}\right) \to 0.$$

Hence, $\mathsf{P}^*(\tilde{T}^{\mathrm{KS}}_n \leq M_0) \to 0$ for every $M_0 > 0$, which proves $\tilde{T}^{\mathrm{KS}}_n \xrightarrow{\mathsf{P}^*} \infty$. $\square$

Theorem 4.4 proves the consistency of the Cramér–von Mises test statistic under fixed alternatives, when the null hypothesis is composite and a parameter is estimated.

PROOF OF THEOREM 4.4. Let $M = \operatorname{diam}(\Omega)$ and define $\Delta^\star(\omega,t) := F^\mu_\omega(t) - F^{\mu_{\boldsymbol{\theta}_1}}_\omega(t)$ for $(\omega,t) \in \Omega \times [0,M]$. Since either *(a.2)* holds, or *(a.1)* holds and $\operatorname{supp}(\mu) = \Omega$, and since $\mu \neq \mu_{\boldsymbol{\theta}_1}$, there exist $\omega_0 \in \operatorname{supp}(\mu)$ and $t_0 \in (0,M]$ such that $\Delta^\star(\omega_0,t_0) \neq 0$. Applying Lemma C.2 (with $\boldsymbol{\theta}_0$ replaced by $\boldsymbol{\theta}_1$), we obtain $\varepsilon_\omega > 0$, $\varepsilon_t > 0$, and $c_0 > 0$ such that

$$|\Delta^\star(\omega,t)| \geq c_0 \quad \text{for all } (\omega,t) \in B_0 \times I_0, \tag{69}$$

where $B_0 = B(\omega_0,\varepsilon_\omega)$ and $I_0 = (t_0 - \varepsilon_t, t_0 + \varepsilon_t) \cap [0,M]$. Since $\omega_0 \in \operatorname{supp}(\mu)$, Lemma C.3 yields

$$\mathsf{P}\big(X \in B_0,\ d(X,Y) \in I_0\big) > 0 \tag{70}$$

for iid $X, Y \sim \mu$.

Define the kernel

$$k^\star(x,y) := \big(F^\mu_x(d(x,y)) - F^{\mu_{\boldsymbol{\theta}_1}}_x(d(x,y))\big)^2 = \Delta^\star(x, d(x,y))^2, \qquad x,y \in \Omega,$$

and let $J^\star := \mathsf{E}_\mu[k^\star(X,Y)]$ for iid $X,Y \sim \mu$. From (69) and (70), one gets $J^\star > 0$ in the same way (67) was obtained in the proof of Theorem 4.2.

Next, define the stochastic errors

$$e_n := \sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \big|F^\mu_{n,\omega}(t) - F^\mu_\omega(t)\big|, \qquad r_n := \sup_{(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}} \big|F^{\mu_{\hat{\boldsymbol{\theta}}}}_\omega(t) - F^{\mu_{\boldsymbol{\theta}_1}}_\omega(t)\big|.$$

By Glivenko–Cantelli, $e_n \xrightarrow{\mathsf{P}} 0$. Moreover, by (25) and $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_1$, the same argument as in the proof of Theorem 4.3 gives $r_n \xrightarrow{\mathsf{P}^\star} 0$.

One can express the Cramér–von Mises statistic as

$$\frac{1}{n}\tilde{T}^{\mathrm{CM}}_n = \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n \big(F^\mu_{n,X_i}(d(X_i,X_j)) - F^{\mu_{\hat{\boldsymbol{\theta}}}}_{X_i}(d(X_i,X_j))\big)^2,$$

write, for any $x,y \in \Omega$,

$$\begin{aligned} F^\mu_{n,x}(d(x,y)) - F^{\mu_{\hat{\boldsymbol{\theta}}}}_x(d(x,y)) =& \big(F^\mu_x(d(x,y)) - F^{\mu_{\boldsymbol{\theta}_1}}_x(d(x,y))\big) \\ &+ \big(F^\mu_{n,x}(d(x,y)) - F^\mu_x(d(x,y))\big) \\ &- \big(F^{\mu_{\hat{\boldsymbol{\theta}}}}_x(d(x,y)) - F^{\mu_{\boldsymbol{\theta}_1}}_x(d(x,y))\big). \end{aligned}$$

The second term is bounded by e_n uniformly in x,y, and the third term is bounded by r_n uniformly in x,y. Let

$$k^\star_n(x,y) := \big(F^\mu_{n,x}(d(x,y)) - F^{\mu_{\hat{\boldsymbol{\theta}}}}_x(d(x,y))\big)^2.$$

Now, use $|(a+b)^2 - a^2| \leq 2|a||b| + b^2$ for any $a,b \in \mathbb{R}$, taking $a := \big(F^\mu_x(d(x,y)) - F^{\mu_{\boldsymbol{\theta}_1}}_x(d(x,y))\big)$ and $b := \big(F^\mu_{n,x}(d(x,y)) - F^\mu_x(d(x,y))\big) - \big(F^{\mu_{\hat{\boldsymbol{\theta}}}}_x(d(x,y)) - F^{\mu_{\boldsymbol{\theta}_1}}_x(d(x,y))\big)$. Since $|a| \leq 1$, we obtain

$$\sup_{x,y\in\Omega}\big|k^\star_n(x,y) - k^\star(x,y)\big| \leq 2(e_n + r_n) + (e_n + r_n)^2 \xrightarrow{\mathsf{P}^\star} 0.$$

Therefore, with

$$V^\star_n := \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n k^\star(X_i,X_j), \qquad \frac{1}{n}\tilde{T}^{\mathrm{CM}}_n = \frac{1}{n^2}\sum_{i,j} k^\star_n(X_i,X_j),$$

we have $\left|\frac{1}{n}\tilde{T}_n^{\mathrm{CM}} - V_n^\star\right| \xrightarrow{\mathsf{P}^\star} 0$.

Finally, $k^\star$ is bounded and nonnegative, hence integrable, and the same U-statistic LLN argument as in the proof of Theorem 4.2 yields $V_n^\star \xrightarrow{\mathsf{P}^\star} J^\star$. Thus, $\frac{1}{n}\tilde{T}_n^{\mathrm{CM}} \xrightarrow{\mathsf{P}^\star} J^\star > 0$, and consequently $\tilde{T}_n^{\mathrm{CM}} \xrightarrow{\mathsf{P}^\star} \infty$. □

C.2. Local alternatives.

C.2.1. *Simple null hypothesis.* This section contains the proofs of Proposition 4.1 and Theorems 4.5 and 4.6, which establish the behavior of the Kolmogorov–Smirnov and Cramér–von Mises test statistics under local alternatives.

PROOF OF PROPOSITION 4.1. Define the empirical process under Q_n by $\mathbb{G}_{n,Q_n}^{Q_n}(f) := \sqrt{n}(P_n - Q_n)f$, $f \in \mathcal{F}$. Consider two mutually independent iid sequences U_i and V_i, $i \geq 1$, where $U_i \sim \mu_0$ and $V_i \sim G$. For each $n \in \mathbb{N}$, consider an iid sequence $B_{n,i}$, $i \geq 1$, independent of $(U_i)_{i\geq 1}$ and $(V_i)_{i\geq 1}$, where $B_{n,i} \sim \mathrm{Bernoulli}(n^{-1/2})$. Define $X_{n,i} := U_i$ if $B_{n,i} = 0$ and $X_{n,i} := V_i$ if $B_{n,i} = 1$. Then $X_{n,1}, \ldots, X_{n,n}$ are iid with common distribution $Q_n = \big(1 - \frac{1}{\sqrt{n}}\big)\mu_0 + \frac{1}{\sqrt{n}}G$. Observe that for any $f \in \mathcal{F}$, it holds that $f(X_{n,i}) = (1 - B_{n,i})f(U_i) + B_{n,i}f(V_i)$. Write $N_{n,1} := \sum_{i=1}^n B_{n,i}$ and $N_{n,0} := n - N_{n,1}$. One has that

$$\begin{aligned}
\mathbb{G}_{n,Q_n}^{Q_n}(f) &= \frac{1}{\sqrt{n}}\sum_{i=1}^n\Big((1 - B_{n,i})f(U_i) + B_{n,i}f(V_i) - Q_nf\Big)\\
&= \frac{1}{\sqrt{n}}\sum_{i=1}^n(1 - B_{n,i})\Big(f(U_i) - \mu_0 f\Big) + \frac{1}{\sqrt{n}}\sum_{i=1}^n B_{n,i}\Big(f(V_i) - Gf\Big)\\
&\quad + \frac{1}{\sqrt{n}}\Big(N_{n,0}\,\mu_0 f + N_{n,1}\,Gf - nQ_nf\Big)
\end{aligned}$$

$$= \frac{1}{\sqrt{n}}\sum_{i=1}^n(1 - B_{n,i})\Big(f(U_i) - \mu_0 f\Big) + \frac{1}{\sqrt{n}}\sum_{i=1}^n B_{n,i}\Big(f(V_i) - Gf\Big) \tag{71}$$

$$+ \sqrt{n}\left(\frac{N_{n,0}}{n} - \left(1 - \frac{1}{\sqrt{n}}\right)\right)\mu_0 f + \sqrt{n}\left(\frac{N_{n,1}}{n} - \frac{1}{\sqrt{n}}\right)Gf, \tag{72}$$

Since $\frac{N_{n,0}}{n} - \big(1 - \frac{1}{\sqrt{n}}\big) = -\big(\frac{N_{n,1}}{n} - \frac{1}{\sqrt{n}}\big)$, (72) combines into

$$R_n(f) := \frac{N_{n,1} - \sqrt{n}}{\sqrt{n}}(G - \mu_0)f.$$

Since $N_{n,1} \sim \mathrm{Bin}(n, n^{-1/2})$, it follows that $\frac{\mathsf{E}((N_{n,1} - \sqrt{n})^2)}{n} = \frac{\mathrm{Var}(N_{n,1})}{n} = \frac{1}{\sqrt{n}}\left(1 - \frac{1}{\sqrt{n}}\right) \to 0$. Thus, since $\sup_{f\in\mathcal{F}}\big|(G - \mu_0)f\big| < \infty$ by assumption, it holds that

$$\sup_{f\in\mathcal{F}}|R_n(f)| \xrightarrow{\mathsf{P}} 0. \tag{73}$$

Next, consider the second sum in (71). Define, for $n \geq 0$, the empirical process based on $V_i \sim G$ by

$$\mathbb{G}_{n,G}^{G}(f) := \begin{cases} \frac{1}{\sqrt{n}}\sum_{i=1}^n\Big(f(V_i) - Gf\Big), & n \geq 1,\\ 0, & n = 0.\end{cases}$$

Let $I_{n,1} := \{i \in \{1,\dots,n\} : B_{n,i} = 1\}$ and, on the event $\{N_{n,1} \geq 1\}$, write $I_{n,1} = \{j_{n,1} < \cdots < j_{n,N_{n,1}}\}$. Then,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} B_{n,i}\Big(f(V_i) - Gf\Big) = \frac{1}{\sqrt{n}} \sum_{\ell=1}^{N_{n,1}} \Big(f(V_{j_{n,\ell}}) - Gf\Big).$$

Since $(V_i)_{i\geq 1}$ are iid and independent of $(B_{n,i})_{i=1}^n$ for every n, then conditional on $B_{n,1},\dots,B_{n,n}$, if $N_{n,1} = m$, then $(V_{j_{n,\ell}})_{\ell=1}^m$ has the same law as $(V_\ell)_{\ell=1}^m$. Hence, the following equality in distribution also holds unconditionally:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} B_{n,i}\Big(f(V_i) - Gf\Big) \overset{d}{=} \sqrt{\frac{N_{n,1}}{n}}\, \mathbb{G}^G_{N_{n,1},G}(f),$$

as random elements in $\ell^\infty(\mathcal{F})$. By the Donsker property, $\mathbb{G}^G_{n,G} \xrightarrow{\mathcal{L}} \mathbb{G}_G$ in $\ell^\infty(\mathcal{F})$ for some tight limit $\mathbb{G}_G$. Since $\mathcal{F}$ is Donsker under G, the empirical process $\mathbb{G}^G_{n,G}$ is asymptotically tight in $\ell^\infty(\mathcal{F})$; (see, e.g., page 139 in Van der Vaart and Wellner, 2023). By the continuous mapping theorem for asymptotic tightness (see, e.g., Problem 7(i) in Section 1.3 of Van der Vaart and Wellner, 2023), the sequence $\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{n,G}(f)|$ is asymptotically tight in $\mathbb{R}$. In particular, for every $\varepsilon > 0$ there exists $M_\varepsilon < \infty$ such that

$$\limsup_{n\to\infty} \mathsf{P}\left(\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{n,G}(f)| > M_\varepsilon\right) \leq \frac{\varepsilon}{2}. \tag{74}$$

For every fixed $m > 0$, one has

$$\mathsf{P}\left(\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{N_{n,1},G}(f)| > M_\varepsilon\right) \leq \mathsf{P}(N_{n,1} \leq m) + \sup_{k\geq m} \mathsf{P}\left(\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{k,G}(f)| > M_\varepsilon\right) \tag{75}$$

By (74), there exists $m_0 \in \mathbb{N}$ such that

$$\sup_{k\geq m_0} \mathsf{P}\left(\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{k,G}(f)| > M_\varepsilon\right) \leq \varepsilon.$$

Applying (75) with $m = m_0$, and since $N_{n,1} \xrightarrow{\mathsf{P}} \infty$, we obtain

$$\limsup_{n\to\infty} \mathsf{P}\left(\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{N_{n,1},G}(f)| > M_\varepsilon\right) \leq \limsup_{n\to\infty} \mathsf{P}(N_{n,1} \leq m_0) + \varepsilon = \varepsilon.$$

Therefore,

$$\sup_{f\in\mathcal{F}} |\mathbb{G}^G_{N_{n,1},G}(f)| = O_{\mathsf{P}}(1).$$

Moreover, $\frac{N_{n,1}}{n} \xrightarrow{\mathsf{P}} 0$ because $\mathsf{E}\big(\frac{N_{n,1}}{n}\big) = n^{-1/2}$. Consequently,

$$\sup_{f\in\mathcal{F}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^{n} B_{n,i}\Big(f(V_i) - Gf\Big)\right| \overset{d}{=} \sqrt{\frac{N_{n,1}}{n}} \sup_{f\in\mathcal{F}} |\mathbb{G}^G_{N_{n,1},G}(f)| \xrightarrow{\mathsf{P}} 0. \tag{76}$$

It remains to handle the first term in (71). Define, for $n \geq 0$, the empirical process based on $U_i \sim \mu_0$ by

$$\mathbb{G}^{\mu_0}_{n,\mu_0}(f) := \begin{cases} \frac{1}{\sqrt{n}} \sum_{i=1}^n \big(f(U_i) - \mu_0 f\big), & n \geq 1, \\ 0, & n = 0. \end{cases}$$

Then, one has

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(1-B_{n,i})\Big(f(U_i)-\mu_0 f\Big)=\mathbb{G}_{n,\mu_0}^{\mu_0}(f)-\frac{1}{\sqrt{n}}\sum_{i=1}^{n}B_{n,i}\Big(f(U_i)-\mu_0 f\Big).$$

For the second term on the RHS of the equality, by the definition of $j_{n,\ell}$, one has

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}B_{n,i}\Big(f(U_i)-\mu_0 f\Big)=\frac{1}{\sqrt{n}}\sum_{\ell=1}^{N_{n,1}}\Big(f(U_{j_{n,\ell}})-\mu_0 f\Big).$$

Since $(U_i)_{i\geq 1}$ are iid and independent of $(B_{n,i})_{i=1}^{n}$, then conditional on $B_{n,1},\ldots,B_{n,n}$, $(U_{j_{n,\ell}})_{\ell=1}^{m}$ has the same law as $(U_\ell)_{\ell=1}^{m}$, if $N_{n,1}=m$. Hence, the following equality in distribution also holds unconditionally:

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}B_{n,i}\Big(f(U_i)-\mu_0 f\Big)\overset{d}{=}\sqrt{\frac{N_{n,1}}{n}}\,\mathbb{G}_{N_{n,1},\mu_0}^{\mu_0}(f)$$

as random elements in $\ell^\infty(\mathcal{F})$. Since $\mathcal{F}$ is μ_0-Donsker, the same argument as above yields $\sup_{f\in\mathcal{F}}|\mathbb{G}_{N_{n,1},\mu_0}^{\mu_0}(f)|=O_{\mathsf{P}}(1)$, and hence

$$\sup_{f\in\mathcal{F}}\Big|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}B_{n,i}\Big(f(U_i)-\mu_0 f\Big)\Big|\overset{d}{=}\sqrt{\frac{N_{n,1}}{n}}\sup_{f\in\mathcal{F}}|\mathbb{G}_{N_{n,1},\mu_0}^{\mu_0}(f)|\overset{\mathsf{P}}{\to}0. \tag{77}$$

Finally, observe that applying the triangle inequality and taking suprema, it follows from (71)–(72) that

$$\begin{aligned}\sup_{f\in\mathcal{F}}\big|\mathbb{G}_{n,Q_n}^{Q_n}(f)-\mathbb{G}_{n,\mu_0}^{\mu_0}(f)\big|\leq{}&\sup_{f\in\mathcal{F}}\Big|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}B_{n,i}\Big(f(U_i)-\mu_0 f\Big)\Big|\\&+\sup_{f\in\mathcal{F}}\Big|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}B_{n,i}\Big(f(V_i)-Gf\Big)\Big|\\&+\sup_{f\in\mathcal{F}}|R_n(f)|.\end{aligned}$$

Combining (73), (76), and (77) gives (30). □

Theorem 4.5 gives the weak limit of the Kolmogorov–Smirnov statistic under local alternatives, when the null hypothesis is simple.

PROOF OF THEOREM 4.5. Observe that since $0\leq y_{\omega,t}\leq 1$ for all $(\omega,t)\in\Omega\times\mathbb{R}_{\geq 0}$, it holds that $\sup_{\omega,t}\big|(G-\mu_0)y_{\omega,t}\big|\leq 2$, so the conditions of Proposition 4.1 are satisfied for $\mathcal{Y}$. Under the local alternative defined in (28), an application of Corollary 4.1 to the class of functions $\mathcal{Y}$ yields $\mathbb{G}_{n,Q_n}^{Q_n}\overset{\mathcal{L}}{\to}\mathbb{G}_0$ in $\ell^\infty(\mathcal{Y})$. Combined with (29), this gives

$$\sqrt{n}\Big(F_{n,\omega}^{Q_n}(t)-F_{\omega}^{\mu_0}(t)\Big)\overset{\mathcal{L}}{\to}\mathbb{G}_0(\omega,t)+h(\omega,t)\qquad\text{in }\ell^\infty(\mathcal{Y}). \tag{78}$$

Finally, by an application of the continuous mapping theorem, we obtain

$$T_n^{\mathrm{KS}}\overset{\mathcal{L}}{\to}\sup_{\omega\in\Omega,t\geq 0}|\mathbb{G}_0(\omega,t)+h(\omega,t)|\,.$$

This concludes the result. □

Lemma C.4 is the analogue of Lemma B.4 for the local alternative. Let $M = \operatorname{diam}(\Omega)$. For $0 < B < \infty$ and a modulus of continuity $w : [0,\infty) \to [0,\infty)$, let $\mathcal{G}_{B,w}$ denote the class of functions on $\Omega \times [0, M]$ that are uniformly bounded by B and have modulus of continuity w, that is,

$$\mathcal{G}_{B,w} := \left\{ g : \Omega \times [0,M] \to \mathbb{R} : \|g\|_\infty \le B,\ |g(z) - g(z')| \le w\big(d_{\Omega\times\mathbb{R}_{\ge 0}}(z,z')\big) \text{ for all } z, z' \in \Omega \times [0,M] \right\}.$$

Here we only obtain convergence in probability, whereas Lemma B.4 yields almost sure convergence. This is why we prove below a uniform version, which is the extra ingredient needed in the proof of Theorem 4.6.

LEMMA C.4. *Under the local alternative* (28)*, it holds that*

$$\sup_{g\in\mathcal{G}_{B,w}} \left| \int_{\Omega\times[0,M]} g(\omega,t)\,\mathrm{d}F^{Q_n}_{n,\omega}(t)\,\mathrm{d}P_n(\omega) - \int_{\Omega\times[0,M]} g(\omega,t)\,\mathrm{d}F^{\mu_0}_{\omega}(t)\,\mathrm{d}\mu_0(\omega) \right| = o_P(1).$$

PROOF. For $g \in \mathcal{G}_{B,w}$, define

$$r_{n,g}(\omega) := \int_{[0,M]} g(\omega,t)\,\mathrm{d}F^{Q_n}_{n,\omega}(t), \qquad r_g(\omega) := \int_{[0,M]} g(\omega,t)\,\mathrm{d}F^{\mu_0}_{\omega}(t).$$

Then,

$$\sup_{g\in\mathcal{G}_{B,w}} \left| \int_\Omega r_{n,g}(\omega)\,\mathrm{d}P_n(\omega) - \int_\Omega r_g(\omega)\,\mathrm{d}\mu_0(\omega) \right| \le A_n + B_n, \tag{79}$$

where

$$A_n := \sup_{g\in\mathcal{G}_{B,w}} \sup_{\omega\in\Omega} |r_{n,g}(\omega) - r_g(\omega)|$$

and

$$B_n := \sup_{g\in\mathcal{G}_{B,w}} \left| \int_\Omega r_g(\omega)\,\mathrm{d}(P_n - \mu_0)(\omega) \right|.$$

We first control A_n. Fix $\varepsilon > 0$, and choose $\delta > 0$ such that $w(\delta) \le \varepsilon$. Repeating the regularization argument in the proof of Lemma B.2, with

$$g_\varepsilon(\omega,t) := \inf_{s\in[0,M]} \left\{ g(\omega,s) + \frac{2B}{\delta}|t-s| \right\},$$

gives, uniformly over $g \in \mathcal{G}_{B,w}$,

$$A_n \le C\varepsilon + C_{B,\delta,M} \sup_{\omega\in\Omega, t\in[0,M]} \left| F^{Q_n}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t) \right|,$$

for constants C and $C_{B,\delta,M}$ independent of n. It remains to show that

$$\sup_{\omega\in\Omega, t\in[0,M]} \left| F^{Q_n}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t) \right| \xrightarrow{\mathsf{P}} 0. \tag{80}$$

The decomposition

$$F^{Q_n}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t) = (P_n - Q_n)y_{\omega,t} + (Q_n - \mu_0)y_{\omega,t}$$

gives

$$\sup_{\omega\in\Omega, t\in[0,M]} \left| F^{Q_n}_{n,\omega}(t) - F^{\mu_0}_{\omega}(t) \right| \le \frac{1}{\sqrt{n}} \sup_{\omega\in\Omega, t\in[0,M]} \left| \mathbb{G}^{Q_n}_{n,Q_n}(\omega,t) \right| + \frac{1}{\sqrt{n}} \sup_{\omega\in\Omega, t\in[0,M]} |(G-\mu_0)y_{\omega,t}|.$$

By Corollary 4.1 applied to $\mathcal{Y}$, one has $\mathbb{G}_{n,Q_n}^{Q_n} \xrightarrow{\mathcal{L}} \mathbb{G}_0$ in $\ell^\infty(\mathcal{Y})$, and therefore

$$\sup_{\omega\in\Omega, t\in[0,M]} \left|\mathbb{G}_{n,Q_n}^{Q_n}(\omega,t)\right| = O_P(1).$$

Moreover, one has $\sup_{\omega,t} |(G-\mu_0)y_{\omega,t}| \leq 2$. Thus, (80) holds and hence $A_n = o_P(1)$.

We next control B_n. Write

$$B_n \leq \sup_{g\in\mathcal{G}_{B,w}} |(P_n - Q_n)r_g| + \sup_{g\in\mathcal{G}_{B,w}} |(Q_n-\mu_0)r_g|. \tag{81}$$

Since $|r_g(\omega)| \leq B$ for every $g \in \mathcal{G}_{B,w}$ and $\omega \in \Omega$, one has

$$\sup_{g\in\mathcal{G}_{B,w}} |(Q_n-\mu_0)r_g| \leq \frac{2B}{\sqrt{n}}.$$

For the first term in (81), note that

$$r_g(\omega) = \int_\Omega g(\omega, d(\omega,x))\,\mathrm{d}\mu_0(x).$$

Hence, for $\omega,\omega' \in \Omega$,

$$|r_g(\omega) - r_g(\omega')| \leq \int_\Omega \left|g(\omega,d(\omega,x)) - g(\omega',d(\omega',x))\right|\,\mathrm{d}\mu_0(x).$$

The triangle inequality gives

$$d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,d(\omega,x)),(\omega',d(\omega',x))\big) \leq 2d(\omega,\omega'),$$

and therefore

$$|r_g(\omega) - r_g(\omega')| \leq w\big(2d(\omega,\omega')\big),$$

uniformly in $g \in \mathcal{G}_{B,w}$. Thus, the family $\{r_g : g \in \mathcal{G}_{B,w}\}$ is uniformly bounded by B and equicontinuous with common modulus $u \mapsto w(2u)$.

Fix $\eta > 0$. Choose $\delta > 0$ such that $w(2\delta) \leq \eta/3$, and let $\omega_1,\ldots,\omega_m$ be a finite δ-net of Ω. Consider the set

$$S := \big\{\big(r_g(\omega_1),\ldots,r_g(\omega_m)\big) : g \in \mathcal{G}_{B,w}\big\} \subset [-B,B]^m.$$

Since S is totally bounded, it admits a finite $\eta/3$-net with centers belonging to S. Hence, there exist $g_1,\ldots,g_N \in \mathcal{G}_{B,w}$ such that, for every $g \in \mathcal{G}_{B,w}$, there exists $k \in \{1,\ldots,N\}$ satisfying

$$\max_{1\leq j\leq m} |r_g(\omega_j) - r_{g_k}(\omega_j)| \leq \eta/3.$$

If $d(\omega,\omega_j) < \delta$, then

$$|r_g(\omega) - r_{g_k}(\omega)| \leq |r_g(\omega) - r_g(\omega_j)| + |r_g(\omega_j) - r_{g_k}(\omega_j)| + |r_{g_k}(\omega_j) - r_{g_k}(\omega)| \leq \eta.$$

Thus, $\|r_g - r_{g_k}\|_\infty \leq \eta$, and so

$$\sup_{g\in\mathcal{G}_{B,w}} |(P_n-Q_n)r_g| \leq \max_{1\leq k\leq N} |(P_n-Q_n)r_{g_k}| + 2\eta.$$

For each fixed k,

$$\mathsf{Var}((P_n-Q_n)r_{g_k}) = \frac{\mathsf{Var}(r_{g_k}(X_1))}{n} \leq \frac{B^2}{n},$$

so $(P_n-Q_n)r_{g_k} = o_P(1)$ by Chebyshev's inequality. Since N is finite,

$$\max_{1\leq k\leq N} |(P_n-Q_n)r_{g_k}| = o_P(1).$$

Letting $\eta \downarrow 0$, we obtain

$$\sup_{g\in\mathcal{G}_{B,w}} |(P_n - Q_n)r_g| = o_P(1).$$

Therefore, $B_n = o_P(1)$, and the result follows. □

PROOF OF THEOREM 4.6. The proof is analog to that of Theorem 3.2.
From (29), we have

$$T_n^{\mathrm{CM}} = \int_{\Omega\times\mathbb{R}_{\geq 0}} \left(\mathbb{G}_{n,Q_n}^{Q_n}(\omega,t) + h(\omega,t)\right)^2 \mathrm{d}F_{n,\omega}^{Q_n}(t)\,\mathrm{d}P_n(\omega).$$

By Corollary 4.1,

$$\mathbb{G}_{n,Q_n}^{Q_n} + h \xrightarrow{\mathcal{L}} \mathbb{G}_0 + h \quad \text{in } \ell^\infty(\mathcal{Y}).$$

In the same way as in the proof of Theorem 3.2, it can be shown that $\mathbb{G}_0 + h$ is separable, and therefore the Skorohod representation theorem yields a probability space on which it holds that

$$\sup_{\omega\in\Omega,t\geq 0} \left|\mathbb{G}_{n,Q_n}^{Q_n}(\omega,t) - \mathbb{G}_0(\omega,t)\right| \xrightarrow{a.s.} 0. \tag{82}$$

Now, write

$$\left|\int \left(\mathbb{G}_{n,Q_n}^{Q_n} + h\right)^2 \mathrm{d}F_{n,\omega}^{Q_n}\,\mathrm{d}P_n - \int \left(\mathbb{G}_0 + h\right)^2 \mathrm{d}F_{\omega}^{\mu_0}\,\mathrm{d}\mu_0\right|$$

$$\leq \left|\int \left(\mathbb{G}_{n,Q_n}^{Q_n} + h\right)^2 - \left(\mathbb{G}_0 + h\right)^2 \mathrm{d}F_{n,\omega}^{Q_n}(t)\,\mathrm{d}P_n(\omega)\right| \tag{83}$$

$$+ \left|\int \left(\mathbb{G}_0(\omega,t) + h(\omega,t)\right)^2 \mathrm{d}F_{n,\omega}^{Q_n}(t)\,\mathrm{d}P_n(\omega) - \int \left(\mathbb{G}_0(\omega,t) + h(\omega,t)\right)^2 \mathrm{d}F_{\omega}^{\mu_0}(t)\,\mathrm{d}\mu_0(\omega)\right|. \tag{84}$$

For (83), one has:

$$\left|\int \left(\left(\mathbb{G}_{n,Q_n}^{Q_n} + h\right)^2 - \left(\mathbb{G}_0 + h\right)^2\right) \mathrm{d}F_{n,\omega}^{Q_n}(t)\,\mathrm{d}P_n(\omega)\right| \leq \sup_{\omega,t}\Big|\left(\mathbb{G}_{n,Q_n}^{Q_n}(\omega,t) + h(\omega,t)\right)^2$$
$$- \left(\mathbb{G}_0(\omega,t) + h(\omega,t)\right)^2\Big| \xrightarrow{a.s.} 0.$$

The convergence to 0 follows from the identity $(a+h)^2 - (b+h)^2 = (a-b)(a+b+2h)$, the almost sure convergence (82), and the facts that $|h| \leq 1$ and $\sup_{\omega,t}|\mathbb{G}_0(\omega,t)| < \infty$ almost surely.

To control (84), observe first that $\mathbb{G}_0$ has bounded and uniformly continuous sample paths almost surely on $\Omega\times[0,M]$, by Lemma B.6. Moreover, h is deterministic and Lipschitz on $\Omega\times[0,M]$.

Fix $\eta > 0$. Since $\|\mathbb{G}_0\|_\infty < \infty$ almost surely, there exists $L < \infty$ such that

$$\mathsf{P}\left(\|\mathbb{G}_0\|_\infty \leq L\right) \geq 1 - \eta/2.$$

For $\delta > 0$, define

$$W(\delta) := \sup_{d_{\Omega\times\mathbb{R}_{\geq 0}}((\omega,t),(\omega',t'))\leq\delta} |\mathbb{G}_0(\omega,t) - \mathbb{G}_0(\omega',t')|.$$

Since $\mathbb{G}_0$ has uniformly continuous sample paths almost surely, $W(\delta) \to 0$ in probability as $\delta \downarrow 0$. Hence, one may choose a decreasing sequence $(\delta_m)_{m\geq 1}$ such that

$$\mathsf{P}\big(W(\delta_m) > 2^{-m}\big) \leq \eta\, 2^{-m-1}, \qquad m \geq 1.$$

Define $w_0 : [0,\infty) \to [0,\infty)$ by

$$w_0(u) := \begin{cases} 0, & u = 0, \\ 2^{-m}, & \delta_{m+1} < u \leq \delta_m, \\ \max\{2L, 2^{-1}\}, & u > \delta_1. \end{cases}$$

Then w_0 is nondecreasing and $w_0(u) \to 0$ as $u \downarrow 0$. Let

$$A_\eta := \{\|\mathbb{G}_0\|_\infty \leq L\} \cap \bigcap_{m\geq 1} \left\{W(\delta_m) \leq 2^{-m}\right\},$$

which satisfies $\mathsf{P}(A_\eta) \geq 1 - \eta$. On A_η, one has

$$|\mathbb{G}_0(\omega,t) - \mathbb{G}_0(\omega',t')| \leq w_0\big(d_{\Omega\times\mathbb{R}_{\geq 0}}((\omega,t),(\omega',t'))\big)$$

for all $(\omega,t),(\omega',t') \in \Omega \times [0,M]$.

By (57) applied with the measure $\nu \in \{\mu_0, G\}$, there exists $C_\nu < \infty$ such that

$$\big|F_\omega^\nu(t) - F_{\omega'}^\nu(t')\big| = \big|\nu\big(y_{\omega,t} - y_{\omega',t'}\big)\big| \leq \nu\big(\big(y_{\omega,t} - y_{\omega',t'}\big)^2\big) \leq C_\nu\, d_{\Omega\times\mathbb{R}_{\geq 0}}\big((\omega,t),(\omega',t')\big).$$

Hence, $(\omega,t) \mapsto F_\omega^\nu(t)$ is Lipschitz for each $\nu \in \{\mu_0, G\}$, and therefore $h(\omega,t) = F_\omega^G(t) - F_\omega^{\mu_0}(t)$ is Lipschitz. Let L_h be a Lipschitz constant of h. On A_η, $\|(\mathbb{G}_0 + h)^2\|_\infty \leq (L + \|h\|_\infty)^2$, and, using $|a^2 - b^2| \leq (|a| + |b|)|a - b|$, it holds that

$$\begin{aligned}|(\mathbb{G}_0 + h)^2(\omega,t) - (\mathbb{G}_0 + h)^2(\omega',t')| \leq 2(L + \|h\|_\infty)\Big(&w_0\big(d_{\Omega\times\mathbb{R}_{\geq 0}}((\omega,t),(\omega',t'))\big) \\ &+ L_h\, d_{\Omega\times\mathbb{R}_{\geq 0}}((\omega,t),(\omega',t'))\Big)\end{aligned}$$

for every $(\omega,t),(\omega',t') \in \Omega \times [0,M]$. Therefore, defining

$$B := (L + \|h\|_\infty)^2, \qquad w(u) := 2(L + \|h\|_\infty)\big(w_0(u) + L_h u\big),$$

then

$$\mathsf{P}\big((\mathbb{G}_0 + h)^2 \in \mathcal{G}_{B,w}\big) \geq 1 - \eta.$$

Therefore, for every $\varepsilon > 0$,

$$\begin{aligned}&\mathsf{P}\left(\left|\int (\mathbb{G}_0 + h)^2\, \mathrm{d}F_{n,\omega}^{Q_n}(t)\, \mathrm{d}P_n(\omega) - \int (\mathbb{G}_0 + h)^2\, \mathrm{d}F_\omega^{\mu_0}(t)\, \mathrm{d}\mu_0(\omega)\right| > \varepsilon\right) \leq \mathsf{P}\big((\mathbb{G}_0 + h)^2 \notin \mathcal{G}_{B,w}\big) \\ &\quad + \mathsf{P}\left(\sup_{g\in\mathcal{G}_{B,w}} \left|\int g(\omega,t)\, \mathrm{d}F_{n,\omega}^{Q_n}(t)\, \mathrm{d}P_n(\omega) - \int g(\omega,t)\, \mathrm{d}F_\omega^{\mu_0}(t)\, \mathrm{d}\mu_0(\omega)\right| > \varepsilon\right).\end{aligned}$$

The first term is bounded by η, and the second converges to 0 by Lemma C.4. Since $\eta > 0$ is arbitrary, the quantity in (84) is $o_P(1)$. This concludes the proof. $\square$

C.2.2. *Composite null hypothesis.* This section contains the proof of Proposition 4.2, which establishes the local convergence of $\hat{\boldsymbol{\theta}}_n$ and $\tilde{\boldsymbol{\theta}}_n$ to $\boldsymbol{\theta}_0$. Remark C.1 shows a useful fact that will be applied in the proof.

REMARK C.1. By *(c̃.3)*, as $\boldsymbol{\theta} \to \boldsymbol{\theta}_0$, one has

$$\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}} \;=\; \boldsymbol{V}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}-\boldsymbol{\theta}_0) + o(\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|). \tag{85}$$

Let $c := \frac{1}{2}\|\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\|_2^{-1}$, with $\boldsymbol{V}_{\boldsymbol{\theta}_0}$ as defined in *(c̃.3)* and $\|\cdot\|_2$ the spectral norm. By *(c̃.3)*, $\boldsymbol{V}_{\boldsymbol{\theta}_0}$ is nonsingular, hence $c > 0$. It holds that $\|\boldsymbol{V}_{\boldsymbol{\theta}_0}\boldsymbol{v}\| \geq 2c\|\boldsymbol{v}\|$ for all $\boldsymbol{v} \in \mathbb{R}^p$.

In addition, by definition of $o(\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|)$, taking U sufficiently small, it holds that

$$\big\|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}} - \boldsymbol{V}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)\big\| \leq c\,\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|. \tag{86}$$

PROOF OF PROPOSITION 4.2. We divide the proof into its two parts, *(i)* and *(ii)*.

Proof of (i). By the reverse triangle inequality, one has

$$\|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}\| \geq \|\boldsymbol{V}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)\| - \big\|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}} - \boldsymbol{V}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta}-\boldsymbol{\theta}_0)\big\| \geq 2c\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\| - c\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\| = c\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\| \tag{87}$$

in a neighborhood of $\boldsymbol{\theta}_0$.

By (33), for every $\boldsymbol{\theta} \in U$,

$$\|G\psi_{\boldsymbol{\theta}}\| \leq \|G\psi_{\boldsymbol{\theta}_0}\| + \|G(\psi_{\boldsymbol{\theta}} - \psi_{\boldsymbol{\theta}_0})\| \leq \|G\psi_{\boldsymbol{\theta}_0}\| + G\|\psi_{\boldsymbol{\theta}} - \psi_{\boldsymbol{\theta}_0}\| \leq \|G\psi_{\boldsymbol{\theta}_0}\| + GL\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|.$$

Since $GL < \infty$ by Cauchy–Schwarz and U is bounded, it follows that

$$M_G := \sup_{\boldsymbol{\theta}\in U}\|G\psi_{\boldsymbol{\theta}}\| < \infty. \tag{88}$$

Since $Q_n\psi_{\tilde{\boldsymbol{\theta}}_n} = 0$, by definition of Q_n one has

$$\left(1 - \frac{1}{\sqrt{n}}\right)\|\mu_{\boldsymbol{\theta}_0}\psi_{\tilde{\boldsymbol{\theta}}_n}\| = \frac{\|G\psi_{\tilde{\boldsymbol{\theta}}_n}\|}{\sqrt{n}} \leq \frac{M_G}{\sqrt{n}}.$$

Hence, for n sufficiently large,

$$\|\mu_{\boldsymbol{\theta}_0}\psi_{\tilde{\boldsymbol{\theta}}_n}\| \leq \frac{M_G}{\sqrt{n}\left(1 - \frac{1}{\sqrt{n}}\right)} \leq \frac{2M_G}{\sqrt{n}},$$

and combining with (87) at $\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}}_n$ gives $\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| \to 0$ as $n \to \infty$.

Proof of (ii). For each $j = 1, \ldots, p$, let $\mathcal{F}_j := \{\psi_{\boldsymbol{\theta},j} : \boldsymbol{\theta} \in U\}$. We show that $\mathcal{F}_j$ satisfies the conditions of Corollary 4.1. Each class $\mathcal{F}_j$ is $\mu_{\boldsymbol{\theta}_0}$- and G-Donsker by (33) and *(c̃.2)*, because $\mu_{\boldsymbol{\theta}_0}L^2 < \infty$, $\mu_{\boldsymbol{\theta}_0}\|\psi_{\boldsymbol{\theta}_0}\|^2 < \infty$, $GL^2 < \infty$, and $G\|\psi_{\boldsymbol{\theta}_0}\|^2 < \infty$ (see Example 19.7 in van der Vaart (1998)). Moreover, for every $\boldsymbol{\theta} \in U$,

$$\big|(G - \mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta},j}\big| \leq \big\|(G - \mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta}}\big\| \leq \|G\psi_{\boldsymbol{\theta}}\| + \mu_{\boldsymbol{\theta}_0}\|\psi_{\boldsymbol{\theta}} - \psi_{\boldsymbol{\theta}_0}\| \leq M_G + \mu_{\boldsymbol{\theta}_0}L\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|,$$

so $\sup_{\boldsymbol{\theta}\in U}\big|(G - \mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta},j}\big| < \infty$ because U is bounded.

An application of Corollary 4.1 to each $\mathcal{F}_j$ and the continuous mapping theorem gives $\sup_{\boldsymbol{\theta}\in U}|(P_n - Q_n)\psi_{\boldsymbol{\theta},j}| = O_{Q_n}(n^{-1/2})$. Since $p < \infty$,

$$\sup_{\boldsymbol{\theta}\in U}\|(P_n - Q_n)\psi_{\boldsymbol{\theta}}\| \leq \sqrt{p}\max_{1\leq j\leq p}\sup_{\boldsymbol{\theta}\in U}|(P_n - Q_n)\psi_{\boldsymbol{\theta},j}| = O_{Q_n}(n^{-1/2}). \tag{89}$$

Since $P_n\psi_{\hat{\boldsymbol{\theta}}} = 0$, (89) implies that, on the event $\{\hat{\boldsymbol{\theta}} \in U\}$,

$$\|Q_n\psi_{\hat{\boldsymbol{\theta}}}\| = \|(P_n - Q_n)\psi_{\hat{\boldsymbol{\theta}}}\| \leq \sup_{\boldsymbol{\theta}\in U}\|(P_n - Q_n)\psi_{\boldsymbol{\theta}}\| = O_{Q_n}(n^{-1/2}), \tag{90}$$

and in particular $\|Q_n\psi_{\hat{\boldsymbol{\theta}}}\| = o_{Q_n}(1)$ on $\{\hat{\boldsymbol{\theta}} \in U\}$.
By (87),

$$\inf_{\boldsymbol{\theta}\in U,\, \|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|\geq\varepsilon} \|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}\| \geq c\varepsilon. \tag{91}$$

Moreover,

$$\|(Q_n-\mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta}}\| = \frac{1}{\sqrt{n}}\|(G-\mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta}}\| \leq \frac{1}{\sqrt{n}}\left(\|G\psi_{\boldsymbol{\theta}}\|+\|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}\|\right) \leq \frac{1}{\sqrt{n}}\Big(M_G+\sup_{\boldsymbol{\theta}\in U}\|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}\|\Big),$$

so $\sup_{\boldsymbol{\theta}\in U}\|(Q_n-\mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta}}\| = O(n^{-1/2}) = o(1)$. The fact that $\sup_{\boldsymbol{\theta}\in U}\|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}\| < \infty$ follows from (33). Therefore, for n sufficiently large,

$$\sup_{\boldsymbol{\theta}\in U}\|(Q_n-\mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta}}\| < \frac{c\varepsilon}{2}. \tag{92}$$

Combining (91) and (92), for n sufficiently large and all $\boldsymbol{\theta}\in U$ with $\|\boldsymbol{\theta}-\boldsymbol{\theta}_0\|\geq\varepsilon$,

$$\|Q_n\psi_{\boldsymbol{\theta}}\| \geq \|\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}\| - \|(Q_n-\mu_{\boldsymbol{\theta}_0})\psi_{\boldsymbol{\theta}}\| > c\varepsilon - \frac{c\varepsilon}{2} = \frac{c\varepsilon}{2}.$$

The preceding bound applied at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$ on the event $\{\hat{\boldsymbol{\theta}}\in U\}$ gives

$$\mathsf{P}\left(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|\geq\varepsilon\right) \leq \mathsf{P}(\hat{\boldsymbol{\theta}}\notin U) + \mathsf{P}\Big(\|Q_n\psi_{\hat{\boldsymbol{\theta}}}\| \geq \frac{c\varepsilon}{2},\ \hat{\boldsymbol{\theta}}\in U\Big) \to 0,$$

because $\mathsf{P}(\hat{\boldsymbol{\theta}}\notin U)\to 0$ and $\|Q_n\psi_{\hat{\boldsymbol{\theta}}}\| = O_{Q_n}(n^{-1/2})$ by (90) if $\hat{\boldsymbol{\theta}}\in U$. This implies $\mathsf{P}(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|\geq\varepsilon)\to 0$ for every $\varepsilon>0$, i.e. $\hat{\boldsymbol{\theta}} \xrightarrow{\mathsf{P}} \boldsymbol{\theta}_0$. □

We introduce Lemmas C.6 and C.5, which are auxiliary results used in the proof of Theorem 4.7.

LEMMA C.5. *Assume conditions (c̃.1) and (c̃.2), $GL^2<\infty$, and $G\|\psi_{\boldsymbol{\theta}_0}\|^2<\infty$. Let $\tilde{\boldsymbol{\theta}}_n\in\Theta$ be such that $Q_n\psi_{\tilde{\boldsymbol{\theta}}_n} = 0$ and $\tilde{\boldsymbol{\theta}}_n\to\boldsymbol{\theta}_0$. Let also $\hat{\boldsymbol{\theta}}\in\Theta$ be such that $P_n\psi_{\hat{\boldsymbol{\theta}}} = o_P(n^{-1/2})$ and $\hat{\boldsymbol{\theta}}\xrightarrow{\mathsf{P}}\boldsymbol{\theta}_0$. Under the local alternative defined in (32), it holds that*

$$(P_n-Q_n)\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right) = o_P(n^{-1/2})$$

PROOF. Define, for a law P and a measurable function f taking values in $\mathbb{R}^p$, the pseudo-norms

$$\rho_P(f) := \left(P\|f-Pf\|^2\right)^{\frac{1}{2}}, \quad \|f\|_{P,2} := \left(P\|f\|^2\right)^{\frac{1}{2}}.$$

Fix $\rho>0$ such that $B(\boldsymbol{\theta}_0,\rho)$ is contained in a neighborhood of $\boldsymbol{\theta}_0$ on which (c̃.1) holds, and define

$$A_n := \{\hat{\boldsymbol{\theta}}\in B(\boldsymbol{\theta}_0,\rho),\ \tilde{\boldsymbol{\theta}}_n\in B(\boldsymbol{\theta}_0,\rho)\}.$$

Since $\hat{\boldsymbol{\theta}}\xrightarrow{\mathsf{P}}\boldsymbol{\theta}_0$ and $\tilde{\boldsymbol{\theta}}_n\to\boldsymbol{\theta}_0$, $\mathsf{P}(A_n)\to 1$. Thus, it holds that

$$\rho_{\mu_{\boldsymbol{\theta}_0}}(f) \leq \|f\|_{\mu_{\boldsymbol{\theta}_0},2} \leq \left(1-\frac{1}{\sqrt{n}}\right)^{-1/2}\|f\|_{Q_n,2}. \tag{93}$$

On A_n, by (c̃.1), one has

$$\|\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\|_{Q_n,2} \leq \|L\|_{Q_n,2}\|\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n\| \xrightarrow{\mathsf{P}} 0,$$

Since $\mathsf{P}\big(A_n\big) \to 1$, this bound holds with probability increasing to one as $n \to \infty$. Hence, for every $\delta > 0$, one has

$$\begin{aligned}
\|\mathbb{G}^{Q_n}_{n,Q_n}\big(\psi_{\hat{\boldsymbol\theta}} - \psi_{\tilde{\boldsymbol\theta}_n}\big)\| &\le \sup_{\substack{\boldsymbol\theta_1,\boldsymbol\theta_2 \in B(\boldsymbol\theta_0,\rho)\\ \|\psi_{\boldsymbol\theta_1}-\psi_{\boldsymbol\theta_2}\|_{Q_n,2}<\delta}} \|\mathbb{G}^{Q_n}_{n,Q_n}\big(\psi_{\boldsymbol\theta_1} - \psi_{\boldsymbol\theta_2}\big)\| + o_P(1)\\
&\le \sup_{\substack{\boldsymbol\theta_1,\boldsymbol\theta_2 \in B(\boldsymbol\theta_0,\rho)\\ \|\psi_{\boldsymbol\theta_1}-\psi_{\boldsymbol\theta_2}\|_{Q_n,2}<\delta}} \|\mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}\big(\psi_{\boldsymbol\theta_1} - \psi_{\boldsymbol\theta_2}\big)\| \\
&\quad + 2 \sup_{\boldsymbol\theta \in B(\boldsymbol\theta_0,\rho)} \left\|\mathbb{G}^{Q_n}_{n,Q_n}\psi_{\boldsymbol\theta} - \mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}\psi_{\boldsymbol\theta}\right\| + o_P(1).
\end{aligned} \tag{94}$$

To show $\mathbb{G}^{Q_n}_{n,Q_n}(\psi_{\hat{\boldsymbol\theta}} - \psi_{\tilde{\boldsymbol\theta}_n}) \xrightarrow{\mathsf{P}} 0$, we first show that

(95)

$$\sup_{\boldsymbol\theta \in B(\boldsymbol\theta_0,\rho)} \left\|\mathbb{G}^{Q_n}_{n,Q_n}(\psi_{\boldsymbol\theta}) - \mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}(\psi_{\boldsymbol\theta})\right\| \le \sqrt{p} \max_{1\le j\le p} \sup_{\boldsymbol\theta \in B(\boldsymbol\theta_0,\rho)} \left|\mathbb{G}^{Q_n}_{n,Q_n}(\psi_{\boldsymbol\theta,j}) - \mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}(\psi_{\boldsymbol\theta,j})\right| \xrightarrow{\mathsf{P}} 0.$$

For each $j = 1, \dots, p$, let $\mathcal{F}_{\rho,j} := \{\psi_{\boldsymbol\theta,j} : \boldsymbol\theta \in B(\boldsymbol\theta_0, \rho)\}$. The assumptions of Proposition 4.1 hold for each $\mathcal{F}_{\rho,j}$ because each $\mathcal{F}_{\rho,j}$ is $\mu_{\boldsymbol\theta_0}$- and G-Donsker, and

$$\sup_{f \in \mathcal{F}_{\rho,j}} |(G - \mu_{\boldsymbol\theta_0})f| \le \sup_{\boldsymbol\theta \in B(\boldsymbol\theta_0,\rho)} \|(G - \mu_{\boldsymbol\theta_0})\psi_{\boldsymbol\theta}\| \le 2\Big(\|G\psi_{\boldsymbol\theta_0}\| + \rho\,(GL + \mu_{\boldsymbol\theta_0}L)\Big) < \infty.$$

The fact that each $\mathcal{F}_{\rho,j}$ is $\mu_{\boldsymbol\theta_0}$- and G-Donsker follows from *($\tilde{c}$.1)* and *($\tilde{c}$.2)*, $\mu_{\boldsymbol\theta_0}L^2 < \infty$, $GL^2 < \infty$, and $G\|\psi_{\boldsymbol\theta_0}\|^2 < \infty$ (see Example 19.7 in van der Vaart, 1998). By Proposition 4.1 applied to the class $\bigcup_{j=1}^p \mathcal{F}_{\rho,j}$, there exists a version of $\mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}$ such that the RHS of (95) converges to zero in probability.

For the first term on the RHS of (94), since each class $\mathcal{F}_{\rho,j}$ is $\mu_{\boldsymbol\theta_0}$-Donsker, it is asymptotically equicontinuous under $\mu_{\boldsymbol\theta_0}$ (see pp. 138–139 in van der Vaart, 1998). Hence, for every $\varepsilon > 0$ and every $j = 1, \dots, p$,

$$\lim_{\delta\downarrow 0} \limsup_{n\to\infty} \mathsf{P}\left(\sup_{\substack{\boldsymbol\theta_1,\boldsymbol\theta_2 \in B(\boldsymbol\theta_0,\rho)\\ \rho_{\mu_{\boldsymbol\theta_0}}(\psi_{\boldsymbol\theta_1,j}-\psi_{\boldsymbol\theta_2,j})<\delta}} \left|\mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}\big(\psi_{\boldsymbol\theta_1,j} - \psi_{\boldsymbol\theta_2,j}\big)\right| > \varepsilon\right) = 0. \tag{96}$$

Moreover, applying (93), one obtains

$$\sup_{\substack{\boldsymbol\theta_1,\boldsymbol\theta_2 \in B(\boldsymbol\theta_0,\rho)\\ \|\psi_{\boldsymbol\theta_1}-\psi_{\boldsymbol\theta_2}\|_{Q_n,2}<\delta}} \left\|\mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}\big(\psi_{\boldsymbol\theta_1} - \psi_{\boldsymbol\theta_2}\big)\right\| \le \sqrt{p} \max_{1\le j\le p} \sup_{\substack{\boldsymbol\theta_1,\boldsymbol\theta_2 \in B(\boldsymbol\theta_0,\rho)\\ \rho_{\mu_{\boldsymbol\theta_0}}(\psi_{\boldsymbol\theta_1,j}-\psi_{\boldsymbol\theta_2,j})<2\delta}} \left|\mathbb{G}^{\mu_{\boldsymbol\theta_0}}_{n,\mu_{\boldsymbol\theta_0}}\big(\psi_{\boldsymbol\theta_1,j} - \psi_{\boldsymbol\theta_2,j}\big)\right|.$$

Since (94) holds for every $\delta > 0$, taking $\limsup_{n\to\infty}$ and then letting $\delta \downarrow 0$ yields $\mathbb{G}^{Q_n}_{n,Q_n}(\psi_{\hat{\boldsymbol\theta}} - \psi_{\tilde{\boldsymbol\theta}_n}) \xrightarrow{\mathsf{P}} 0$, and hence $(P_n - Q_n)\left(\psi_{\hat{\boldsymbol\theta}} - \psi_{\tilde{\boldsymbol\theta}_n}\right) = o_P(n^{-1/2})$. □

LEMMA C.6. *Assume conditions ($\tilde{c}$.1) and ($\tilde{c}$.2), $GL^2 < \infty$, and $G\|\psi_{\boldsymbol\theta_0}\|^2 < \infty$. Let $\tilde{\boldsymbol\theta}_n \in \Theta$ be such that $Q_n\psi_{\tilde{\boldsymbol\theta}_n} = 0$ and $\tilde{\boldsymbol\theta}_n \to \boldsymbol\theta_0$. Under the local alternative defined in* (32)*, it holds that*

$$(P_n - Q_n)\,\psi_{\tilde{\boldsymbol\theta}_n} = O_P(n^{-1/2}). \tag{97}$$

PROOF. Fix $\rho > 0$ such that $B(\boldsymbol\theta_0, \rho)$ is contained in a neighborhood of $\boldsymbol\theta_0$ on which *($\tilde{c}$.1)* holds. Since $\tilde{\boldsymbol\theta}_n \to \boldsymbol\theta_0$ by assumption, there exists $n(\rho) > 0$ such that $\tilde{\boldsymbol\theta}_n \in B(\boldsymbol\theta_0, \rho)$ for all $n > n(\rho)$. Observe that also $\|\sqrt{n}(P_n - Q_n)\psi_{\tilde{\boldsymbol{\theta}}_n}\| \leq \sup_{\boldsymbol{\theta}\in B(\boldsymbol{\theta}_0,\rho)} \|\sqrt{n}(P_n - Q_n)\psi_{\boldsymbol{\theta}}\|$ for $n > n(\rho)$.

For each $j = 1, \ldots, p$, let $\mathcal{F}_{\rho,j} := \{\psi_{\boldsymbol{\theta},j} : \boldsymbol{\theta} \in B(\boldsymbol{\theta}_0, \rho)\}$. The assumptions of Corollary 4.1 hold by the same arguments as in Lemma C.5. An application of Corollary 4.1 to each $\mathcal{F}_{\rho,j}$ gives $\sup_{\boldsymbol{\theta}\in B(\boldsymbol{\theta}_0,\rho)} |\sqrt{n}(P_n - Q_n)\psi_{\boldsymbol{\theta},j}| = O_P(1)$. Hence, since $p < \infty$,

$$\sup_{\boldsymbol{\theta}\in B(\boldsymbol{\theta}_0,\rho)} \|\sqrt{n}(P_n - Q_n)\psi_{\boldsymbol{\theta}}\| \leq \sqrt{p} \max_{1\leq j\leq p} \sup_{\boldsymbol{\theta}\in B(\boldsymbol{\theta}_0,\rho)} |\sqrt{n}(P_n - Q_n)\psi_{\boldsymbol{\theta},j}| = O_P(1),$$

and hence $(P_n - Q_n)\,\psi_{\tilde{\boldsymbol{\theta}}_n} = O_P(n^{-1/2})$. This concludes the result. □

Using Lemmas C.6 and C.5, we can now prove Theorem 4.7.

PROOF OF THEOREM 4.7. We divide the proof int its two parts, *(i)* and *(ii)*.

Proof of (i). Since $\boldsymbol{\theta} \mapsto \mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}}$ is differentiable at $\boldsymbol{\theta}_0$ by *($\tilde{c}$.3)*, it holds that

$$\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}} = \mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}_0} + \boldsymbol{V}_{\boldsymbol{\theta}_0}(\boldsymbol{\theta} - \boldsymbol{\theta}_0) + o(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|).$$

Since $\tilde{\boldsymbol{\theta}}_n \to \boldsymbol{\theta}_0$, evaluating this expansion at $\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}}_n$ gives

$$\mu_{\boldsymbol{\theta}_0}\psi_{\tilde{\boldsymbol{\theta}}_n} = \mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}_0} + \boldsymbol{V}_{\boldsymbol{\theta}_0}(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) + o(\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|).$$

Together with $Q_n\psi_{\tilde{\boldsymbol{\theta}}_n} = 0$, this gives

$$0 = \left(1 - \frac{1}{\sqrt{n}}\right)\left(\boldsymbol{V}_{\boldsymbol{\theta}_0}(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) + o(\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|)\right) + \frac{1}{\sqrt{n}}G\psi_{\tilde{\boldsymbol{\theta}}_n},$$

and thus

$$\boldsymbol{V}_{\boldsymbol{\theta}_0}(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) = -\frac{1}{\sqrt{n}\left(1 - \frac{1}{\sqrt{n}}\right)}G\psi_{\tilde{\boldsymbol{\theta}}_n} + o(\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|). \tag{98}$$

Now, by *($\tilde{c}$.1)* and the assumption that $\tilde{\boldsymbol{\theta}}_n \to \boldsymbol{\theta}_0$, it holds that

$$\|G\psi_{\tilde{\boldsymbol{\theta}}_n}\| \leq \|G\psi_{\boldsymbol{\theta}_0}\| + GL\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|$$

for n sufficiently large, where $GL < \infty$ because $GL^2 < \infty$. Plugging into (98) gives

$$\left(1 - \frac{1}{\sqrt{n}}\right)\|\boldsymbol{V}_{\boldsymbol{\theta}_0}(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)\| \leq \frac{1}{\sqrt{n}}\|G\psi_{\boldsymbol{\theta}_0}\| + \frac{GL}{\sqrt{n}}\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| + o(\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|).$$

By Remark C.1, $\|\boldsymbol{V}_{\boldsymbol{\theta}_0}(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0)\| \geq 2c\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|$. Therefore, for every $\varepsilon > 0$, there exists $n(\varepsilon) > 0$ such that, for every $n > n(\varepsilon)$, one has

$$2c\left(1 - \frac{1}{\sqrt{n}}\right)\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| \leq \frac{1}{\sqrt{n}}\|G\psi_{\boldsymbol{\theta}_0}\| + \frac{GL}{\sqrt{n}}\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| + \varepsilon\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|.$$

Thus,

$$\left(2c\left(1 - \frac{1}{\sqrt{n}}\right) - \frac{GL}{\sqrt{n}} - \varepsilon\right)\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| \leq \frac{1}{\sqrt{n}}\|G\psi_{\boldsymbol{\theta}_0}\|.$$

Take $\varepsilon = \frac{c}{2}$ to obtain

$$\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| \leq \frac{2}{c\sqrt{n}}\|G\psi_{\boldsymbol{\theta}_0}\| = O\big(n^{-\frac{1}{2}}\big), \tag{99}$$

for n sufficiently large. The bound $\|G\psi_{\boldsymbol{\theta}_0}\| < \infty$ follows from $G\|\psi_{\boldsymbol{\theta}_0}\|^2 < \infty$.

Now, rewrite (98) as

(100)
$$\boldsymbol{V}_{\boldsymbol{\theta}_0}(\tilde{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0)=-\frac{1}{\sqrt{n}\left(1-\frac{1}{\sqrt{n}}\right)}G\psi_{\boldsymbol{\theta}_0}-\frac{1}{\sqrt{n}\left(1-\frac{1}{\sqrt{n}}\right)}G\left(\psi_{\tilde{\boldsymbol{\theta}}_n}-\psi_{\boldsymbol{\theta}_0}\right)+o(\|\tilde{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0\|).$$

By condition *(c̃.1)*, one has

$$\frac{1}{\sqrt{n}\left(1-\frac{1}{\sqrt{n}}\right)}\|G(\psi_{\tilde{\boldsymbol{\theta}}_n}-\psi_{\boldsymbol{\theta}_0})\|\leq\frac{1}{\sqrt{n}\left(1-\frac{1}{\sqrt{n}}\right)}GL\|\tilde{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0\|=O(n^{-1})$$

by (99). Thus, (100) yields

$$\tilde{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0=-\frac{1}{\sqrt{n}\left(1-\frac{1}{\sqrt{n}}\right)}\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}G\psi_{\boldsymbol{\theta}_0}+o\left(n^{-\frac{1}{2}}\right),$$

which gives (34), applying the assumption $\|G\psi_{\boldsymbol{\theta}_0}\|<\infty$.

Proof of (ii). Now, we turn to (35). Since $P_n\psi_{\hat{\boldsymbol{\theta}}}=o_P(n^{-1/2})$ and $Q_n\psi_{\tilde{\boldsymbol{\theta}}_n}=0$, one has

$$0=(P_n-Q_n)\,\psi_{\hat{\boldsymbol{\theta}}}+Q_n\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right)+o_P(n^{-1/2}). \tag{101}$$

To control the first summand in (101), write

$$(P_n-Q_n)\,\psi_{\hat{\boldsymbol{\theta}}}=(P_n-Q_n)\,\psi_{\tilde{\boldsymbol{\theta}}_n}+(P_n-Q_n)\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right).$$

By Lemma C.5, it holds that $(P_n-Q_n)\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right)=o_P(n^{-1/2})$. Therefore,

$$(P_n-Q_n)\,\psi_{\hat{\boldsymbol{\theta}}}=(P_n-Q_n)\,\psi_{\tilde{\boldsymbol{\theta}}_n}+o_P(n^{-1/2}). \tag{102}$$

Focus now on the second summand of (101), i.e, $Q_n\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right)$. By definition of Q_n, one has

$$Q_n\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right)=\left(1-\frac{1}{\sqrt{n}}\right)\mu_{\boldsymbol{\theta}_0}\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right)+\frac{1}{\sqrt{n}}G\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right).$$

By (85) applied at $\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}_n$, *(c̃.1)* and $GL<\infty$, one has

(103)
$$\begin{aligned}Q_n\left(\psi_{\hat{\boldsymbol{\theta}}}-\psi_{\tilde{\boldsymbol{\theta}}_n}\right)&=\left(1-\frac{1}{\sqrt{n}}\right)\left(\boldsymbol{V}_{\boldsymbol{\theta}_0}(\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n)+o_P(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|)+o(\|\tilde{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0\|)\right)+O_P\left(\frac{1}{\sqrt{n}}\|\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n\|\right)\\&=\boldsymbol{V}_{\boldsymbol{\theta}_0}(\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n)+O_P\left(\frac{1}{\sqrt{n}}\|\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n\|\right)+o_P(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|)+o(\|\tilde{\boldsymbol{\theta}}_n-\boldsymbol{\theta}_0\|)\\&=\boldsymbol{V}_{\boldsymbol{\theta}_0}(\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n)+O_P\left(\frac{1}{\sqrt{n}}\|\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n\|\right)+o_P(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|)+o_P(n^{-\frac{1}{2}}),\end{aligned}$$

where the last equality follows from (99).

Combining (102) and (103) into (101) gives

(104)
$$0=(P_n-Q_n)\psi_{\tilde{\boldsymbol{\theta}}_n}+\boldsymbol{V}_{\boldsymbol{\theta}_0}(\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n)+O_P\left(\frac{1}{\sqrt{n}}\|\hat{\boldsymbol{\theta}}-\tilde{\boldsymbol{\theta}}_n\|\right)+o_P(\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}_0\|)+o_P(n^{-\frac{1}{2}})$$

We show that $\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\| = O_P(n^{-1/2})$. Using Lemma C.6 in (104), we obtain

$$\left\|\boldsymbol{V}_{\boldsymbol{\theta}_0}(\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n)\right\| \le O_P(n^{-\frac{1}{2}}) + O_P\left(\frac{1}{\sqrt{n}}\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\|\right) + o_P(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|) + o_P(n^{-\frac{1}{2}})$$
$$= O_P(n^{-1/2}) + o_P(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|).$$

The last equality uses $\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\| = o_P(1)$. By Remark C.1, $\|\boldsymbol{V}_{\boldsymbol{\theta}_0}(\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n)\| \ge 2c\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\|$. Then, applying the triangle inequality to $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|$ and using $\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| = O(n^{-1/2})$ from (99) gives

$$2c\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\| \le O_P(n^{-\frac{1}{2}}) + o_P(\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\|).$$

By Lemma C.7, this implies that $\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\| = O_P(n^{-1/2})$. Observe also that since $\|\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n\| = O_P(n^{-1/2})$ and $\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\| = O(n^{-1/2})$, then the triangle inequality gives $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| = O_P(n^{-1/2})$. Plugging back into (104) and using $Q_n\psi_{\tilde{\boldsymbol{\theta}}_n} = 0$ gives

$$\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}_n = -\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} P_n\psi_{\tilde{\boldsymbol{\theta}}_n} + o_P(n^{-\frac{1}{2}}),$$

which combined with (34) gives (35). □

PROOF OF THEOREM 4.8. First, observe that under the local alternative (32) we can write, for every $(\omega, t) \in \Omega \times \mathbb{R}_{\ge 0}$,

$$F_{n,\omega}^{Q_n}(t) - F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) = \left(F_{n,\omega}^{Q_n}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right) - \left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right). \tag{105}$$

The asymptotic behavior of the first term in the RHS of (105) was given in (78) with μ_0 replaced by $\mu_{\boldsymbol{\theta}_0}$.

It remains to analyze $F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)$. By Assumption *(d)*, it holds that

$$\sqrt{n}\left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right) = \sqrt{n}\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_P\left(\sqrt{n}\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|\right). \tag{106}$$

where the remainder is uniform in $(\omega, t) \in \Omega \times \mathbb{R}_{\ge 0}$.

Recall that by Theorem 4.7, it holds that

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\sqrt{n}\,P_n\psi_{\tilde{\boldsymbol{\theta}}_n} - \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}G\psi_{\boldsymbol{\theta}_0} + o_P(1), \tag{107}$$

Since $Q_n\psi_{\tilde{\boldsymbol{\theta}}_n} = 0$, we have $\sqrt{n}\,P_n\psi_{\tilde{\boldsymbol{\theta}}_n} = \sqrt{n}(P_n - Q_n)\psi_{\tilde{\boldsymbol{\theta}}_n} = \mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\tilde{\boldsymbol{\theta}}_n})$. Moreover,

$$\mathsf{E}\left\|\mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\tilde{\boldsymbol{\theta}}_n} - \psi_{\boldsymbol{\theta}_0})\right\|^2 \le Q_n\left\|\psi_{\tilde{\boldsymbol{\theta}}_n} - \psi_{\boldsymbol{\theta}_0}\right\|^2 \le Q_nL^2\,\|\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|^2 \to 0,$$

because Q_nL^2 is bounded and $\tilde{\boldsymbol{\theta}}_n \to \boldsymbol{\theta}_0$. Hence, by Markov's inequality, $\mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\tilde{\boldsymbol{\theta}}_n} - \psi_{\boldsymbol{\theta}_0}) = o_P(1)$. Therefore,

$$\sqrt{n}P_n\psi_{\tilde{\boldsymbol{\theta}}_n} = \mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\boldsymbol{\theta}_0}) + o_P(1). \tag{108}$$

Plugging (108) into (107) gives

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = -\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\,\mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\boldsymbol{\theta}_0}) - \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}G\psi_{\boldsymbol{\theta}_0} + o_P(1). \tag{109}$$

Since $Q_n\|\psi_{\boldsymbol{\theta}_0}\|^2$ is bounded, $\mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\boldsymbol{\theta}_0}) = O_P(1)$; hence (109) gives $\sqrt{n}\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| = O_P(1)$, and the remainder in (106) is $o_P(1)$ uniformly in $(\omega, t) \in \Omega \times \mathbb{R}_{\ge 0}$. Substituting (109) into (106), we obtain

(110)
$$\sqrt{n}\left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right) = -\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\,\mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\boldsymbol{\theta}_0}) - \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}G\psi_{\boldsymbol{\theta}_0} + o_P(1).$$

Combining (105) with (110), the rest of the proof follows analogously to that of Theorem 3.1. Define

$$g_{\omega,t}(x) := -\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\psi_{\boldsymbol{\theta}_0}(x), \qquad f_{\omega,t}(x) := y_{\omega,t}(x) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) - g_{\omega,t}(x),$$

so that $\mu_{\boldsymbol{\theta}_0} f_{\omega,t} = 0$.

Combining (105), (32), and (110), and using that $\mathbb{G}_{n,Q_n}^{Q_n}(g_{\omega,t}) = -\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\mathbb{G}_{n,Q_n}^{Q_n}(\psi_{\boldsymbol{\theta}_0})$, one obtains

$$\begin{aligned}\sqrt{n}\Big(F_{n,\omega}^{Q_n}(t) - F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t)\Big) &= \mathbb{G}_{n,Q_n}^{Q_n}(y_{\omega,t}) - \mathbb{G}_{n,Q_n}^{Q_n}(g_{\omega,t}) + h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} G\psi_{\boldsymbol{\theta}_0} + o_P(1)\\ &= \mathbb{G}_{n,Q_n}^{Q_n}(f_{\omega,t}) + h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} G\psi_{\boldsymbol{\theta}_0} + o_P(1), \end{aligned} \tag{111}$$

where we used that $\mathbb{G}_{n,Q_n}^{Q_n}(F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)) = 0$.

It remains to prove the weak convergence of $\mathbb{G}_{n,Q_n}^{Q_n}$ to $\tilde{\mathbb{G}}_0$ in $\ell^\infty(\mathcal{F})$, where $\mathcal{F} := \{f_{\omega,t} : (\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}\}$. As in the proof of Theorem 3.1, write $\mathcal{F} \subset \mathcal{Y} - \mathcal{G}$, where

$$\mathcal{Y} := \{y_{\omega,t} - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) : (\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}\}, \qquad \mathcal{G} := \{g_{\omega,t} : (\omega,t) \in \Omega \times \mathbb{R}_{\geq 0}\}.$$

The class $\mathcal{Y}$ is $\mu_{\boldsymbol{\theta}_0}$- and G-Donsker by Theorem 5.1 of Dubey, Chen and Müller (2024). For the class $\mathcal{G}$, note that

$$g_{\omega,t}(x) = \sum_{j=1}^p \alpha_j(\omega,t)\,\psi_{\boldsymbol{\theta}_0,j}(x), \qquad \boldsymbol{\alpha}(\omega,t)^\top := -\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1},$$

so $\mathcal{G} \subset \operatorname{span}\{\psi_{\boldsymbol{\theta}_0,1},\ldots,\psi_{\boldsymbol{\theta}_0,p}\}$ is finite-dimensional. Moreover, by assumptions *($\tilde{c}$.2)*, *($\tilde{c}$.3)* and *(d)*, and by $G\|\psi_{\boldsymbol{\theta}_0}\|^2 < \infty$, the envelope of $\mathcal{G}$ is in $L^2(\mu_{\boldsymbol{\theta}_0})$ and $L^2(G)$. Thus, $\mathcal{G}$ is $\mu_{\boldsymbol{\theta}_0}$-Donsker and G-Donsker. Moreover, $\sup_{h\in\mathcal{Y}\cup\mathcal{G}} \mu_{\boldsymbol{\theta}_0} h^2 < \infty$ and $\sup_{h\in\mathcal{Y}\cup\mathcal{G}} Gh^2 < \infty$, hence $\mathcal{Y} - \mathcal{G}$ is $\mu_{\boldsymbol{\theta}_0}$-Donsker and G-Donsker, and therefore so is $\mathcal{F}$. If $\sup_{f\in\mathcal{F}} |(G - \mu_{\boldsymbol{\theta}_0})f| < \infty$ holds, then Corollary 4.1 with μ_0 replaced by $\mu_{\boldsymbol{\theta}_0}$ applies and yields

$$\mathbb{G}_{n,Q_n}^{Q_n} \xrightarrow{\mathcal{L}} \tilde{\mathbb{G}}_0 \qquad \text{in } \ell^\infty(\mathcal{F}). \tag{112}$$

Combining (111) and (112) and using Slutsky's theorem gives (36).

It is only left to verify the condition $\sup_{f\in\mathcal{F}} |(G - \mu_{\boldsymbol{\theta}_0})f| < \infty$, in order to apply Corollary 4.1. Since $0 \leq y_{\omega,t} \leq 1$, one has $\sup_{\omega,t} |(G - \mu_{\boldsymbol{\theta}_0})y_{\omega,t}| \leq 2$. Moreover, $(G - \mu_{\boldsymbol{\theta}_0})F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) = 0$ and $\mu_{\boldsymbol{\theta}_0}\psi_{\boldsymbol{\theta}_0} = 0$, hence

$$|(G - \mu_{\boldsymbol{\theta}_0})g_{\omega,t}| = \big|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} G\psi_{\boldsymbol{\theta}_0}\big| \leq \sup_{\omega,t} \|\dot{F}_\omega^{\boldsymbol{\theta}_0}(t)\|\, \|\boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1}\|\, \|G\psi_{\boldsymbol{\theta}_0}\|.$$

By *(d)*, *($\tilde{c}$.3)*, and $G\|\psi_{\boldsymbol{\theta}_0}\|^2 < \infty$, it holds that $\sup_{f\in\mathcal{F}} |(G - \mu_{\boldsymbol{\theta}_0})f| < \infty$. □

PROOF OF THEOREM 4.9. The convergence of the Kolmogorov–Smirnov statistic follows from Theorem 4.8 and the continuous mapping theorem.

For the Cramér–von Mises statistic, the result follows by the same argument as in Theorem 4.6, replacing μ_0 by $\mu_{\boldsymbol{\theta}_0}$ and $\mathbb{G}_0 + h$ by $\tilde{\mathbb{G}}_0(\omega,t) + h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} G\psi_{\boldsymbol{\theta}_0}$. As in Theorem 4.8, the only caveat is to verify boundedness and uniform continuity of the deterministic drift, which in this case is given by $h(\omega,t) + \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} G\psi_{\boldsymbol{\theta}_0}$.

The uniform continuity and boundedness of h was established in the proof of Theorem 4.6. Uniform continuity of $(\omega,t) \mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top \boldsymbol{V}_{\boldsymbol{\theta}_0}^{-1} G\psi_{\boldsymbol{\theta}_0}$ with respect to $d_{\Omega\times\mathbb{R}_{\geq 0}}$ follows from the assumed uniform continuity of $(\omega,t) \mapsto \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)$. Boundedness follows from assumptions *(d)*, the invertibility of $\boldsymbol{V}_{\boldsymbol{\theta}_0}$ by *($\tilde{c}$.3)*, and $G\|\psi_{\boldsymbol{\theta}_0}\|^2 < \infty$. □

LEMMA C.7. *For $n \geq 1$, let A_n, B_n, and C_n be nonnegative random variables such that $A_n \leq B_n + C_n$ for all n. Assume that $C_n = o_P(A_n)$, i.e., for every $\epsilon > 0$, $\mathsf{P}(C_n > \epsilon A_n) \to 0$. Then, $A_n = O_P(B_n)$.*

PROOF. Since $C_n = o_P(A_n)$, one has $\mathsf{P}(C_n > \frac{1}{2}A_n) \to 0$, that is, $\mathsf{P}\left(C_n \leq \frac{1}{2}A_n\right) \to 1$. If $C_n \leq \frac{1}{2}A_n$, then $A_n \leq B_n + C_n \leq B_n + \frac{1}{2}A_n$, so $A_n \leq 2B_n$. Therefore, for every n,

$$\mathsf{P}(A_n > 2B_n) \leq \mathsf{P}\left(A_n > 2B_n,\ C_n \leq \tfrac{1}{2}A_n\right) + \mathsf{P}\left(C_n > \tfrac{1}{2}A_n\right) = \mathsf{P}\left(C_n > \tfrac{1}{2}A_n\right),$$

where the equality follows from $A_n > 2B_n$. As $\mathsf{P}(C_n > \frac{1}{2}A_n) \to 0$, we conclude that $\mathsf{P}(A_n > 2B_n) \to 0$, hence $A_n = O_P(B_n)$. □

APPENDIX D: PROOFS OF SECTION 5

PROOF OF THEOREM 5.1. We divide the proof into its two parts, *(i)* and *(ii)*.

Proof of (i). It was shown in Section 5 that

$$\mathbb{G}_n^*(\omega, t) = \frac{m}{\tau}\sqrt{n}(P_n^* - P_n)y_{\omega,t}.$$

Since $\mathcal{Y}$ is μ_0-Donsker, Theorem 2.6 in Kosorok (2008) gives

$$\mathbb{G}_n^* \overset{P}{\underset{*}{\to}} \mathbb{G}_0 \quad \text{in } \ell^\infty(\mathcal{Y}).$$

This concludes the result.

Proof of (ii). First observe that *($\tilde{c}$.2)* and Theorem 2.6 in Kosorok (2008), applied to the finite-dimensional class generated by the components of $\boldsymbol{\psi}_{\boldsymbol{\theta}_0}$, give

$$\frac{m}{\tau}\frac{1}{\sqrt{n}}\sum_{i=1}^n \left(\frac{\xi_i}{\bar{\xi}_n} - 1\right)\boldsymbol{\psi}_{\boldsymbol{\theta}_0}(X_i) = O_{P_*}(1).$$

Hence, by the bootstrap Bahadur representation (39), it holds that $\sqrt{n}(\hat{\boldsymbol{\theta}}^* - \hat{\boldsymbol{\theta}}) = O_{P_*}(1)$. Moreover, (12) gives $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = O_P(1)$, and hence $\sqrt{n}(\hat{\boldsymbol{\theta}}^* - \boldsymbol{\theta}_0) = O_{P_*}(1)$. Therefore, by *(d)*,

$$\sqrt{n}\left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right) = \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_P(1),$$

$$\sqrt{n}\left(F_\omega^{\mu_{\hat{\boldsymbol{\theta}}^*}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t)\right) = \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top\sqrt{n}(\hat{\boldsymbol{\theta}}^* - \boldsymbol{\theta}_0) + o_{P_*}(1).$$

The remainders $o_P(1)$ and $o_{P_*}(1)$ are both uniform in $(\omega, t) \in \Omega \times \mathbb{R}_{\geq 0}$. Subtracting these two expansions and using (39) in the definition of $\tilde{\mathbb{G}}_n^*$ gives

$$\sup_{\omega\in\Omega, t\geq 0}\left|\tilde{\mathbb{G}}_n^*(\omega, t) - \frac{m}{\tau}\sqrt{n}(P_n^* - P_n)f_{\omega,t}\right| = o_{P_*}(1). \tag{113}$$

Since $\mathcal{F}$ is $\mu_{\boldsymbol{\theta}_0}$-Donsker, Theorem 2.6 in Kosorok (2008) implies that

$$\frac{m}{\tau}\sqrt{n}(P_n^* - P_n) \overset{P}{\underset{*}{\to}} \tilde{\mathbb{G}}_0 \quad \text{in } \ell^\infty(\mathcal{F}).$$

Let $\varphi \in \mathrm{BL}_1(\ell^\infty(\mathcal{F}))$. Define

$$D_n^* := \sup_{\omega\in\Omega, t\geq 0}\left|\tilde{\mathbb{G}}_n^*(\omega, t) - \frac{m}{\tau}\sqrt{n}(P_n^* - P_n)f_{\omega,t}\right|.$$

By (113), $D_n^* = o_{P_*}(1)$. Moreover, for every $\varepsilon > 0$,

$$\left|\mathsf{E}_*\varphi(\tilde{\mathbb{G}}_n^*) - \mathsf{E}_*\varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)\right| \leq \mathsf{E}_*\left|\varphi(\tilde{\mathbb{G}}_n^*) - \varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)\right| \tag{114}$$

$$\leq \varepsilon\,\mathsf{P}_*(D_n^* \leq \varepsilon) + 2\,\mathsf{P}_*(D_n^* > \varepsilon) \tag{115}$$

$$\leq \varepsilon + 2\,\mathsf{P}_*(D_n^* > \varepsilon). \tag{116}$$

Here, we have used that, if $D_n^* \le \varepsilon$, the Lipschitz property of φ gives

$$\left|\varphi(\tilde{\mathbb{G}}_n^*) - \varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)\right| \le D_n^* \le \varepsilon.$$

Also, if $D_n^* > \varepsilon$, then using $\|\varphi\|_\infty \le 1$, this absolute difference can be bounded by 2. Since $\mathsf{P}_*(D_n^* > \varepsilon) \xrightarrow{\mathsf{P}} 0$ by (113), and ε is arbitrary, the LHS of (114) converges to zero in probability.

It remains to verify the asymptotic measurability condition. Since

$$\frac{m}{\tau}\sqrt{n}(P_n^* - P_n) \underset{*}{\xrightarrow{P}} \tilde{\mathbb{G}}_0 \quad \text{in } \ell^\infty(\mathcal{F}),$$

Theorem 2.6 in Kosorok (2008) gives, for every $\varphi \in \mathrm{BL}_1(\ell^\infty(\mathcal{F}))$,

$$\mathsf{E}_*\overline{\varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)} - \mathsf{E}_*\underline{\varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)} \xrightarrow{\mathsf{P}} 0.$$

Since φ is bounded by one and Lipschitz with constant at most one,

$$\left|\varphi(\tilde{\mathbb{G}}_n^*) - \varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)\right| \le \min\{D_n^*, 2\}.$$

Taking majorants and minorants gives

$$\begin{aligned}\mathsf{E}_*\overline{\varphi(\tilde{\mathbb{G}}_n^*)} - \mathsf{E}_*\underline{\varphi(\tilde{\mathbb{G}}_n^*)} \le \mathsf{E}_*\overline{\varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)} &- \mathsf{E}_*\underline{\varphi\left(\frac{m}{\tau}\sqrt{n}(P_n^* - P_n)\right)} \\ &+ 2\mathsf{E}_*\overline{\min\{D_n^*, 2\}}.\end{aligned}$$

Moreover, for every $\varepsilon > 0$,

$$\mathsf{E}_*\overline{\min\{D_n^*, 2\}} \le \varepsilon + 2\mathsf{P}_*(D_n^* > \varepsilon) \xrightarrow{\mathsf{P}} \varepsilon$$

by (113). Since ε is arbitrary,

$$\mathsf{E}_*\overline{\varphi(\tilde{\mathbb{G}}_n^*)} - \mathsf{E}_*\underline{\varphi(\tilde{\mathbb{G}}_n^*)} \xrightarrow{\mathsf{P}} 0.$$

This concludes the proof. □

PROOF OF THEOREM 5.2. The proof is divided into *(i)* and *(ii)*.

Proof of (i). The asymptotic distribution of the Kolmogorov–Smirnov statistic follows from Theorem 5.1*(i)* by the same argument used in the proof of Theorem 3.2.

For the Cramér–von Mises statistic under the simple null hypothesis, the difference with the proof of Theorem 3.2 is that the convergence $\mathbb{G}_{n,\mu}^{\mu_0} \xrightarrow{\mathcal{L}} \mathbb{G}_0$ is replaced by $\mathbb{G}_n^* \underset{*}{\xrightarrow{P}} \mathbb{G}_0$ in $\ell^\infty(\mathcal{Y})$. The only point that is not verbatim is the application of Skorohod's construction. To justify it, consider the sequence $(\mathbb{G}_n^*)_{n\ge1}$ as random elements in $\ell^\infty(\mathcal{Y})$, and let $(n_j)_{j\ge1}$ be an arbitrary subsequence of positive integers. There exists a further subsequence $(n_{j_k})_{k\ge1}$ such that, for almost every realization of the data,

$$\sup_{\varphi\in\mathrm{BL}_1(\ell^\infty(\mathcal{Y}))}\left|\mathsf{E}_*\varphi(\mathbb{G}_{n_{j_k}}^*) - \mathsf{E}\varphi(\mathbb{G}_0)\right| \to 0.$$

Fix a realization of the data for which this convergence holds and for which Lemma B.4 applies along the same subsequence. Conditionally on this data sequence, the law of $\mathbb{G}_{n_{j_k}}^*$ converges weakly to the law of $\mathbb{G}_0$ in $\ell^\infty(\mathcal{Y})$. Therefore, the same Skorohod construction used in the proof of Theorem 3.2 can be applied. From this point, the same argument as in the proof of Theorem 3.2 gives, conditionally on the fixed data realization, $T^{*\mathrm{CM}}_{n_{j_k}} \xrightarrow{\mathcal{L}} T_0$, where

$$T_0 := \int_{\Omega\times\mathbb{R}_{\geq 0}} \mathbb{G}_0(\omega,t)^2 \,\mathrm{d}F^{\mu_0}_{\omega}(t)\,\mathrm{d}\mu_0(\omega).$$

Equivalently,

$$\sup_{\varphi\in\mathrm{BL}_1(\mathbb{R})} \left|\mathsf{E}_*\varphi(T^{*\mathrm{CM}}_{n_{j_k}}) - \mathsf{E}\varphi(T_0)\right| \to 0. \tag{117}$$

This result holds almost surely with respect to the data. Since the original subsequence $(n_j)_{j\geq 1}$ was arbitrary, every subsequence of the bounded Lipschitz distance has a further subsequence converging to zero almost surely. Therefore,

$$\sup_{\varphi\in\mathrm{BL}_1(\mathbb{R})} \left|\mathsf{E}_*\varphi(T^{*\mathrm{CM}}_{n}) - \mathsf{E}\varphi(T_0)\right| \xrightarrow{\mathsf{P}} 0.$$

Additionally, observe that conditionally on the data, $T^{*\mathrm{CM}}_n$ is a finite sum of measurable functions of the multipliers. Hence, for every $\varphi\in\mathrm{BL}_1(\mathbb{R})$, the asymptotic measurability condition

$$\mathsf{E}_*\overline{\varphi(T^{*\mathrm{CM}}_n)} - \mathsf{E}_*\underline{\varphi(T^{*\mathrm{CM}}_n)} = 0$$

is automatically satisfied. Hence,

$$T^{*\mathrm{CM}}_n \underset{*}{\xrightarrow{P}} \int_{\Omega\times\mathbb{R}_{\geq 0}} \mathbb{G}_0(\omega,t)^2 \,\mathrm{d}F^{\mu_0}_{\omega}(t)\,\mathrm{d}\mu_0(\omega).$$

Proof of (ii). For the Kolmogorov–Smirnov statistic, the result follows from Theorem 5.1(*ii*) using the same argument as in the proof of Theorem 3.3.

The asymptotic distribution of the Cramér–von Mises statistic follows by the same argument as in the proof of (*i*), replacing Theorem 3.2 by Theorem 3.3, and using $\tilde{\mathbb{G}}^*_n \underset{*}{\xrightarrow{P}} \tilde{\mathbb{G}}_0$ in $\ell^\infty(\mathcal{F})$, which follows from Theorem 5.1(*ii*). Conditionally on the data, $\hat{\boldsymbol{\theta}}^*$ is measurable as a function of the multipliers and, for every (ω,t), the map $\boldsymbol{\theta}\mapsto F^{\mu_{\boldsymbol{\theta}}}_{\omega}(t)$ is Borel measurable. Hence, $\tilde{T}^{*\mathrm{CM}}_n$ is a finite sum of measurable functions of the multipliers. Therefore, the asymptotic measurability condition

$$\mathsf{E}_*\overline{\varphi(\tilde{T}^{*\mathrm{CM}}_n)} - \mathsf{E}_*\underline{\varphi(\tilde{T}^{*\mathrm{CM}}_n)} = 0$$

for every $\varphi\in\mathrm{BL}_1(\mathbb{R})$ holds. □

APPENDIX E: PROOFS FOR SPECIFIC PARAMETRIC MODELS

E.1. Proof of Proposition 3.1.

PROOF OF PROPOSITION 3.1. We now show that the expansion (13) satisfies the requirements of Theorem 5.41 of van der Vaart (1998). First, since $\boldsymbol{\xi}_0\neq 0$, the map $\boldsymbol{\xi}\mapsto\psi_{\boldsymbol{\xi}}(\boldsymbol{x})$ is twice continuously differentiable in a neighborhood of $\boldsymbol{\xi}_0$, for every $\boldsymbol{x}\in\mathbb{S}^q$. Moreover, since $\mathsf{E}_{\boldsymbol{\xi}_0}(\boldsymbol{X}) = A_q(\kappa_0)\boldsymbol{\mu}_0$, it follows that $\mathsf{E}_{\boldsymbol{\xi}_0}\left(\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})\right)=0$.

Second, for a given sample $\mathcal{L}_n=\{\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n\}$, the MLE is a consistent estimator and satisfies

$$\sum_{i=1}^n \psi_{\hat{\boldsymbol{\xi}}_n}(\boldsymbol{X}_i)=0.$$

To see that $\mathsf{E}\left(\|\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})\|^2\right) < \infty$, observe that

$$\mathsf{E}\left(\|\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})\|^2\right) = \mathsf{E}\left(\|\boldsymbol{X} - A_q(\kappa_0)\boldsymbol{\mu}_0\|^2\right) = 1 - A_q(\kappa_0)^2 \leq 1 < \infty. \tag{118}$$

Since $\psi_{\boldsymbol{\xi}}$ is the derivative of the log-likelihood function with respect to $\boldsymbol{\xi}$, it holds that $\mathsf{P}_{\boldsymbol{\xi}_0}(\dot{\psi}_{\boldsymbol{\xi}_0}) = -\mathcal{I}_{\boldsymbol{\xi}_0}$, where $\mathcal{I}_{\boldsymbol{\xi}_0}$ is the Fisher information matrix. For a $\mathrm{vMF}(\boldsymbol{\mu}, \kappa)$, this matrix is positive-definite (thus non-singular).

It is only left to check that there exists an integrable function $g(\boldsymbol{x})$ such that $\|\ddot{\psi}(\boldsymbol{x})\| \leq g(\boldsymbol{x})$ for every $\boldsymbol{\xi}$ in a neighborhood of $\boldsymbol{\xi}_0$. Observe that the second-order derivatives obtained by differentiating (15) with respect to $\boldsymbol{\xi}$ do not depend on $\boldsymbol{x}$ and can therefore be bounded by a constant in a neighborhood of $\boldsymbol{\xi}_0$. To show that this constant is finite, we show the boundedness of the four quantities involved: $\|\boldsymbol{\xi}\|$, $A_q(\|\boldsymbol{\xi}\|)$, $A_q'(\|\boldsymbol{\xi}\|)$, and $A_q''(\|\boldsymbol{\xi}\|)$.

Since $\|\boldsymbol{\xi}_0\| > 0$, there exists a neighborhood U of $\boldsymbol{\xi}_0$ and constants $0 < c \leq C < \infty$ such that $c \leq \|\boldsymbol{\xi}\| \leq C$ for every $\boldsymbol{\xi} \in U$. Since A_q, A_q', and A_q'' are continuous on $(0, \infty)$, they are bounded on $[c, C]$. □

E.2. Proof of Proposition 3.2.

PROOF OF PROPOSITION 3.2. In all the models considered, the map $\boldsymbol{\theta} \mapsto F_\omega^{\mu_{\boldsymbol{\theta}}}(t)$ is differentiable in a neighborhood of the true parameter, for every $(\omega, t) \in \Omega \times \mathbb{R}_{\geq 0}$. We apply the fundamental theorem of calculus. For $\boldsymbol{\theta}$ sufficiently close to $\boldsymbol{\theta}_0$, let $\boldsymbol{\theta}_s = \boldsymbol{\theta}_0 + s(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$, $s \in [0, 1]$. Then

$$F_\omega^{\mu_{\boldsymbol{\theta}}}(t) - F_\omega^{\mu_{\boldsymbol{\theta}_0}}(t) - \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)^\top(\boldsymbol{\theta} - \boldsymbol{\theta}_0) = (\boldsymbol{\theta} - \boldsymbol{\theta}_0)^\top \int_0^1 \left(\dot{F}_\omega^{\boldsymbol{\theta}_s}(t) - \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)\right) \mathrm{d}s.$$

Consequently, to verify Assumption *(d)*, it is enough to prove that, in a neighborhood of the true parameter, the map $\boldsymbol{\theta} \mapsto \dot{F}_\omega^{\boldsymbol{\theta}}(t)$ is locally Lipschitz in $\boldsymbol{\theta}$, uniformly over (ω, t). That is,

$$\sup_{\omega, t} \left\| \dot{F}_\omega^{\boldsymbol{\theta}}(t) - \dot{F}_\omega^{\boldsymbol{\theta}_0}(t) \right\| \leq L\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|,$$

for some $L < \infty$.

Consider $X \sim \mathcal{N}(\mu, \sigma_0^2)$, where $\sigma_0 > 0$ is fixed and $\mu \in \mathbb{R}$ is unknown. For any $\omega \in \mathbb{R}$, we have

$$\dot{F}_\omega^{\mu}(t) := \frac{\partial}{\partial \mu} F_\omega^{\mu}(t) = -\frac{1}{\sigma_0}\phi\left(\frac{\omega - \mu + t}{\sigma_0}\right) + \frac{1}{\sigma_0}\phi\left(\frac{\omega - \mu - t}{\sigma_0}\right). \tag{119}$$

Let $\mu_s = \mu_0 + s(\mu - \mu_0)$. By the expression for $\dot{F}_\omega^{\mu}(t)$ in (119) and the boundedness of ϕ',

$$\sup_{\omega \in \mathbb{R}, t \geq 0} \left| \dot{F}_\omega^{\mu_s}(t) - \dot{F}_\omega^{\mu_0}(t) \right| \leq L|\mu_s - \mu_0| \leq L|\mu - \mu_0|,$$

where

$$L := \frac{2}{\sigma_0^2} \sup_{u \in \mathbb{R}} |\phi'(u)| = \frac{2\,\phi(1)}{\sigma_0^2} = \frac{2}{\sigma_0^2\sqrt{2\pi e}}.$$

Hence,

$$\sup_{\omega \in \mathbb{R}, t \geq 0} \left| F_\omega^{\mu}(t) - F_\omega^{\mu_0}(t) - \dot{F}_\omega^{\mu_0}(t)(\mu - \mu_0) \right| \leq L|\mu - \mu_0|^2 = o(|\mu - \mu_0|),$$

so Assumption *(d)* holds.

If $X \sim \mathcal{N}(\mu_0, \sigma^2)$, where $\mu_0 \in \mathbb{R}$ is fixed and $\sigma > 0$ is unknown, let $\sigma_s = \sigma_0 + s(\sigma - \sigma_0)$, and take a neighborhood of σ_0 with $\sigma \geq \underline{\sigma} > 0$. For any $\omega \in \mathbb{R}$, we have

$$\dot{F}_\omega^\sigma(t) := \frac{\partial}{\partial \sigma} F_\omega^\sigma(t) = -\frac{\omega - \mu_0 + t}{\sigma^2} \phi\left(\frac{\omega - \mu_0 + t}{\sigma}\right) + \frac{\omega - \mu_0 - t}{\sigma^2} \phi\left(\frac{\omega - \mu_0 - t}{\sigma}\right). \tag{120}$$

Write $h(u) = u\phi(u)$, so that (120) becomes

$$\dot{F}_\omega^\sigma(t) = -\frac{1}{\sigma} h\left(\frac{\omega - \mu_0 + t}{\sigma}\right) + \frac{1}{\sigma} h\left(\frac{\omega - \mu_0 - t}{\sigma}\right).$$

Moreover,

$$\frac{\partial}{\partial \sigma}\left\{\sigma^{-1} h(u)\right\} = -\sigma^{-2}\left\{h(u) + u h'(u)\right\}, \qquad u = u(\sigma) := \frac{\omega - \mu_0 \pm t}{\sigma}.$$

Since h and $uh'(u)$ are bounded on $\mathbb{R}$, it follows that

$$\sup_{\omega \in \mathbb{R}, t \geq 0} \left|\dot{F}_\omega^{\sigma_s}(t) - \dot{F}_\omega^{\sigma_0}(t)\right| \leq L|\sigma_s - \sigma_0| \leq L|\sigma - \sigma_0|,$$

where

$$L := \frac{2}{\underline{\sigma}^2} \sup_{u \in \mathbb{R}} |h(u) + u h'(u)| = \frac{2}{\underline{\sigma}^2} \sup_{u \in \mathbb{R}} |(2u - u^3)\phi(u)| < \infty.$$

Therefore, Assumption *(d)* holds.

Consider $X \sim \mathcal{N}(\mu, \sigma^2)$, where both $\mu \in \mathbb{R}$ and $\sigma > 0$ are unknown, and let $\boldsymbol{\theta} = (\mu, \sigma)^\top$. Take a neighborhood U of $\boldsymbol{\theta}_0 = (\mu_0, \sigma_0)^\top$ on which $\sigma \geq \underline{\sigma} > 0$. We have already proved that the second-order partial derivatives $\partial_{\mu\mu} F_\omega^{\boldsymbol{\theta}}(t)$ and $\partial_{\sigma\sigma} F_\omega^{\boldsymbol{\theta}}(t)$ are uniformly bounded on $\mathbb{R} \times \mathbb{R}_{\geq 0} \times U$. It remains to control the mixed partial derivatives. Using (119) and differentiating with respect to σ, we obtain

$$\partial_{\mu\sigma} F_\omega^{\boldsymbol{\theta}}(t) = \partial_{\sigma\mu} F_\omega^{\boldsymbol{\theta}}(t) = \frac{1}{\sigma^2} h'\left(\frac{\omega - \mu + t}{\sigma}\right) - \frac{1}{\sigma^2} h'\left(\frac{\omega - \mu - t}{\sigma}\right).$$

Hence,

$$\sup_{\omega \in \mathbb{R},\, t \geq 0,\, \boldsymbol{\theta} \in U} \left|\partial_{\mu\sigma} F_\omega^{\boldsymbol{\theta}}(t)\right| \leq \frac{2}{\underline{\sigma}^2} \sup_{u \in \mathbb{R}} \left|h'(u)\right| < \infty.$$

Therefore, all entries of the Hessian matrix are uniformly bounded. Consequently, using the same reasoning as in the previous two cases, there exists $L < \infty$ such that

$$\sup_{\omega \in \mathbb{R}, t \geq 0} \left\|\dot{F}_\omega^{\boldsymbol{\theta}_s}(t) - \dot{F}_\omega^{\boldsymbol{\theta}_0}(t)\right\| \leq L\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|,$$

and Assumption *(d)* follows.

Consider the canonical parameter $\boldsymbol{\xi} = \kappa\boldsymbol{\mu} \in \mathbb{R}^{q+1} \setminus \{0\}$. For any $(\boldsymbol{\omega}, t)$, compute $\dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}}(t)$ as follows:

$$\begin{aligned}
\dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}}(t) &= \nabla_{\boldsymbol{\xi}} \mathsf{E}(y_{\boldsymbol{\omega},t}(\boldsymbol{X})) \\
&= \int_{\mathbb{S}^q} y_{\boldsymbol{\omega},t}(\boldsymbol{x}) \nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}}(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x} \\
&= \int_{\mathbb{S}^q} y_{\boldsymbol{\omega},t}(\boldsymbol{x}) f_{\boldsymbol{\xi}}(\boldsymbol{x}) \nabla_{\boldsymbol{\xi}} \log\left(f_{\boldsymbol{\xi}}(\boldsymbol{x})\right) \mathrm{d}\boldsymbol{x} \\
&= \mathsf{E}\big(y_{\boldsymbol{\omega},t}(\boldsymbol{X}) \psi_{\boldsymbol{\xi}}(\boldsymbol{X})\big).
\end{aligned} \tag{121}$$

The second equality in (121) follows by an application of the Dominated Convergence Theorem (DCT) (see, e.g., Exercise 51 in De Barra, 2003). To show that the DCT applies, it suffices to show that $\nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}}(\boldsymbol{x})$ is locally uniformly (in $\boldsymbol{\xi}$) dominated by an integrable function g (say). Obviously, $|y_{\boldsymbol{\omega},t}(\boldsymbol{x})| \leq 1$ for all $\boldsymbol{x}$. It holds that $\nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}}(\boldsymbol{x}) = f_{\boldsymbol{\xi}}(\boldsymbol{x})\,(\boldsymbol{x} - A_q(\kappa)\boldsymbol{\mu})$, so $\|\nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}}(\boldsymbol{x})\| \leq 2 f_{\boldsymbol{\xi}}(\boldsymbol{x})$. Take any $\boldsymbol{\xi}_0 \in \mathbb{R}^{q+1} \setminus \{0\}$. For $\boldsymbol{\xi}$ in a small neighborhood of $\boldsymbol{\xi}_0$, there exists $0 < C < \infty$ such that $f_{\boldsymbol{\xi}}(\boldsymbol{x}) \leq C f_{\boldsymbol{\xi}_0}(\boldsymbol{x})$, so $g(\boldsymbol{x}) = 2C f_{\boldsymbol{\xi}_0}(\boldsymbol{x})$, which is integrable. Thus, $\nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}}(\boldsymbol{x})$ is locally uniformly (in $\boldsymbol{\xi}$) bounded by g, and the DCT applies.

Let $\boldsymbol{\xi}_0$ be the true population parameter and let K be a compact neighborhood of $\boldsymbol{\xi}_0$ contained in $\mathbb{R}^{q+1} \setminus \{0\}$. For $\boldsymbol{\xi}$ sufficiently close to $\boldsymbol{\xi}_0$, $\boldsymbol{\xi}_s = \boldsymbol{\xi}_0 + s(\boldsymbol{\xi} - \boldsymbol{\xi}_0)$ belongs to K for all $s \in [0,1]$. By the formula for $\dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}}(t)$ in (121),

$$\dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}_s}(t) - \dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}_0}(t) = \int_{\mathbb{S}^q} y_{\boldsymbol{\omega},t}(\boldsymbol{x}) \left(\nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}_s}(\boldsymbol{x}) - \nabla_{\boldsymbol{\xi}} f_{\boldsymbol{\xi}_0}(\boldsymbol{x})\right) \mathrm{d}\boldsymbol{x}.$$

Since $K \times \mathbb{S}^q$ is compact and $f_{\boldsymbol{\xi}}(\boldsymbol{x})$ is twice continuously differentiable in $(\boldsymbol{\xi}, \boldsymbol{x})$ on $K \times \mathbb{S}^q$, one has

$$\sup_{\boldsymbol{\xi} \in K, \boldsymbol{x} \in \mathbb{S}^q} \left\|\nabla_{\boldsymbol{\xi}}^2 f_{\boldsymbol{\xi}}(\boldsymbol{x})\right\| < \infty.$$

Using also that $|y_{\boldsymbol{\omega},t}(\boldsymbol{x})| \leq 1$, there exists $L < \infty$ such that

$$\sup_{\boldsymbol{\omega} \in \mathbb{S}^q, t \geq 0} \left\|\dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}_s}(t) - \dot{F}_{\boldsymbol{\omega}}^{\boldsymbol{\xi}_0}(t)\right\| \leq L\|\boldsymbol{\xi}_s - \boldsymbol{\xi}_0\| \leq L\|\boldsymbol{\xi} - \boldsymbol{\xi}_0\|.$$

Therefore, Assumption *(d)* holds. □

E.3. Proof of Proposition 3.3.

PROOF OF PROPOSITION 3.3. We divide the proof into its four parts, *(i)*, *(ii)*, *(iii)*, and *(iv)*.

Proof of (i). Consider $X \sim \mathcal{N}(\mu_0, \sigma^2)$ for a fixed $\sigma > 0$. For any $\omega \in \mathbb{R}$, we have

$$\dot{F}_{\omega}^{\mu_0}(t) := \frac{\partial}{\partial \mu} F_{\omega}^{\mu}(t)\bigg|_{\mu=\mu_0} = -\frac{1}{\sigma}\phi\left(\frac{\omega - \mu_0 + t}{\sigma}\right) + \frac{1}{\sigma}\phi\left(\frac{\omega - \mu_0 - t}{\sigma}\right),$$

where ϕ is the PDF of a standard normal distribution.

The MLE for X given a sample $\mathcal{L}_n$ is $\hat{\mu} = \bar{X}$, and $\psi_{\mu_0}(x) = \frac{x - \mu_0}{\sigma^2}$. Observe that $\mathsf{E}(\psi_{\mu_0}(X)) = 0$, $\mathsf{E}(\psi_{\mu_0}(X)^2) = \frac{1}{\sigma^2} = \mathcal{I}_{\mu_0}$.

Now, to compute the remaining addends in the covariance process (17), a simple calculation gives

$$\begin{aligned}
\mathsf{E}\left(y_{\omega,t}(X)\psi_{\mu_0}(X)\right) &= \frac{1}{\sigma^2}\int_{\frac{\omega-\mu_0-t}{\sigma}}^{\frac{\omega-\mu_0+t}{\sigma}} u\phi(u)\sigma\,\mathrm{d}u \\
&= -\frac{1}{\sigma}\phi\left(\frac{\omega - \mu_0 + t}{\sigma}\right) + \frac{1}{\sigma}\phi\left(\frac{\omega - \mu_0 - t}{\sigma}\right) \\
&= \dot{F}_{\omega}^{\mu_0}(t).
\end{aligned}$$

Therefore,

$$\mathbb{C}\mathrm{ov}\left[\mathbb{G}(f_{\omega_1,t_1}), \mathbb{G}(f_{\omega_2,t_2})\right] = \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} - \sigma^2 \dot{F}_{\omega_1}^{\mu_0}(t_1)\dot{F}_{\omega_2}^{\mu_0}(t_2).$$

Proof of (ii). Consider $X \sim \mathcal{N}(\mu, \sigma_0^2)$ for a fixed $\mu \in \mathbb{R}$. For any $\omega \in \mathbb{R}$, we have

$$\dot{F}_{\omega}^{\sigma_0}(t) := \frac{\partial}{\partial \sigma} F_{\omega}^{\sigma}(t)\bigg|_{\sigma=\sigma_0} = -\frac{\omega - \mu + t}{\sigma_0^2}\phi\left(\frac{\omega - \mu + t}{\sigma_0}\right) + \frac{\omega - \mu - t}{\sigma_0^2}\phi\left(\frac{\omega - \mu - t}{\sigma_0}\right),$$

The MLE for σ given a sample $\mathcal{L}_n$ is $\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^n (X_i - \mu)^2}$, and $\psi_{\sigma_0}(x) = \frac{(x-\mu)^2 - \sigma_0^2}{\sigma_0^3}$. Observe that $\mathsf{E}(\psi_{\sigma_0}(X)) = 0$, $\mathsf{E}(\psi_{\sigma_0}(X)^2) = \frac{2}{\sigma_0^2}$.

Doing some algebra, one obtains

$$\begin{aligned}\mathsf{E}\left(y_{\omega,t}(X)\psi_{\sigma_0}(X)\right) &= \frac{1}{\sigma_0}\int_{\frac{\omega-\mu-t}{\sigma_0}}^{\frac{\omega-\mu+t}{\sigma_0}} (u^2-1)\phi(u)\,\mathrm{d}u \\ &= -\frac{\omega-\mu+t}{\sigma_0^2}\phi\left(\frac{\omega-\mu+t}{\sigma_0}\right) + \frac{\omega-\mu-t}{\sigma_0^2}\phi\left(\frac{\omega-\mu-t}{\sigma_0}\right) = \dot{F}_\omega^{\sigma_0}(t).\end{aligned}$$

Therefore,

$$\mathbb{C}\mathrm{ov}\left[\mathbb{G}_0(f_{\omega_1,t_1}), \mathbb{G}_0(f_{\omega_2,t_2})\right] = \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} - \frac{\sigma_0^2}{2}\dot{F}_{\omega_1}^{\sigma_0}(t_1)\dot{F}_{\omega_2}^{\sigma_0}(t_2).$$

Proof of (iii). Consider $X \sim \mathcal{N}(\mu_0, \sigma_0^2)$, where both μ_0 and σ_0 are unknown. The Fisher information matrix is given by $\mathcal{I}_{\mu_0,\sigma_0} = \mathrm{diag}\left(\frac{1}{\sigma_0^2}, \frac{2}{\sigma_0^2}\right)$, a diagonal matrix with $\frac{1}{\sigma_0^2}$ and $\frac{2}{\sigma_0^2}$ in the diagonal. Writing

$$\dot{F}_\omega^{\mu_0,\sigma_0}(t) := \begin{pmatrix} \dot{F}_\omega^{\mu_0}(t) \\ \dot{F}_\omega^{\sigma_0}(t) \end{pmatrix},$$

and $\psi_{\mu_0,\sigma_0} := (\psi_{\mu_0}, \psi_{\sigma_0})^\top$, one has $\mathsf{E}(y_{\omega,t}(X)\psi_{\mu_0,\sigma_0}(X)) = \dot{F}_\omega^{\mu_0,\sigma_0}(t)$ and $\boldsymbol{V}_{\mu_0,\sigma_0} = -\mathcal{I}_{\mu_0,\sigma_0}$. The covariance of the limiting Gaussian process is given by

$$\mathbb{C}\mathrm{ov}\left[\mathbb{G}_0(f_{\omega_1,t_1}), \mathbb{G}_0(f_{\omega_2,t_2})\right] = \mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} - \left(\dot{F}_{\omega_1}^{\mu_0,\sigma_0}(t_1)\right)^\top \mathcal{I}_{\mu_0,\sigma_0}^{-1}\dot{F}_{\omega_2}^{\mu_0,\sigma_0}(t_2).$$

Proof of (iv). We provide the explicit form of the covariance process (17) for $\boldsymbol{X} \sim \mathrm{vMF}(\boldsymbol{\mu}_0, \kappa_0)$ distribution on $\mathbb{S}^q$. Consider the estimation of the canonical parameter $\boldsymbol{\xi}_0 = \kappa_0\boldsymbol{\mu}_0$ through maximum likelihood. The log-likelihood (for one observation) is given by

$$\ell(\boldsymbol{\xi}, \boldsymbol{x}) := \log\left(c_q^{\mathrm{vMF}}(\|\boldsymbol{\xi}\|)\right) + \boldsymbol{\xi}^\top\boldsymbol{x}, \quad \text{where } c_q^{\mathrm{vMF}}(\|\boldsymbol{\xi}\|) = \frac{\|\boldsymbol{\xi}\|^{\frac{q-1}{2}}}{(2\pi)^{\frac{q+1}{2}}\mathcal{I}_{\frac{q-1}{2}}(\|\boldsymbol{\xi}\|)},$$

The score function is

$$\psi_{\boldsymbol{\xi}}(\boldsymbol{x}) := \frac{\partial}{\partial\boldsymbol{\xi}}\ell(\boldsymbol{\xi}, \boldsymbol{x}) = \nabla_{\boldsymbol{\xi}}\log\left(c_q^{\mathrm{vMF}}(\|\boldsymbol{\xi}\|)\right) + \boldsymbol{x} = \boldsymbol{x} - A_q(\|\boldsymbol{\xi}\|)\frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|} = \boldsymbol{x} - \mathsf{E}_{\boldsymbol{\xi}}(\boldsymbol{X}),$$

where $A_q(\|\boldsymbol{\xi}\|) = \frac{\mathcal{I}_{\frac{q+1}{2}}(\|\boldsymbol{\xi}\|)}{\mathcal{I}_{\frac{q-1}{2}}(\|\boldsymbol{\xi}\|)}$. Because the vMF is an exponential family with canonical parameter $\boldsymbol{\xi}$, it holds that $\mathsf{E}_{\boldsymbol{\xi}}(\psi_{\boldsymbol{\xi}}(\boldsymbol{X})) = 0$. The likelihood equation for the MLE is

$$A_q(\|\hat{\boldsymbol{\xi}}\|)\frac{\hat{\boldsymbol{\xi}}}{\|\hat{\boldsymbol{\xi}}\|} = \bar{\boldsymbol{X}};$$

equivalently, when $\bar{\boldsymbol{X}} \neq 0$,

$$\hat{\boldsymbol{\mu}} = \frac{\bar{\boldsymbol{X}}}{\|\bar{\boldsymbol{X}}\|}, \qquad \hat{\kappa} = A_q^{-1}(\|\bar{\boldsymbol{X}}\|), \qquad \hat{\boldsymbol{\xi}} = A_q^{-1}(\|\bar{\boldsymbol{X}}\|)\frac{\bar{\boldsymbol{X}}}{\|\bar{\boldsymbol{X}}\|}.$$

The first term in (17), that is, $\mathcal{C}_{(\boldsymbol{\omega}_1,t_1),(\boldsymbol{\omega}_2,t_2)}$, can be calculated directly from the expression of the distance profile for the vMF distribution under the chordal and geodesic distance (see Section 2.3).

The second quantity of interest is $\dot{F}^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}_2}(t_2)^\top \mathsf{E}\left(y_{\boldsymbol{\omega}_1,t_1}(\boldsymbol{X})\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})\right)$. Observe that

$$\begin{aligned}\mathsf{E}\left(y_{\boldsymbol{\omega},t}(\boldsymbol{X})\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})\right) &= \mathsf{E}(y_{\boldsymbol{\omega},t}(\boldsymbol{X})\boldsymbol{X}) - A_q(\kappa_0)\boldsymbol{\mu}_0 F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}}(t)\\ &= -A_q(\kappa_0)\boldsymbol{\mu}_0 F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}}(t) + F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}}(t)\mathsf{E}\left(\boldsymbol{X} \mid d(\boldsymbol{\omega},\boldsymbol{X}) \le t\right)\\ &= F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}}(t)\left(\mathsf{E}\left(\boldsymbol{X} \mid d(\boldsymbol{\omega},\boldsymbol{X}) \le t\right) - A_q(\kappa_0)\boldsymbol{\mu}_0\right),\end{aligned} \tag{122}$$

with $A_q(\kappa_0) = \frac{\mathcal{I}_{\frac{q+1}{2}}(\kappa_0)}{\mathcal{I}_{\frac{q-1}{2}}(\kappa_0)}$. By (121) and (122),

$$\dot{F}^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}}(t) = F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}}(t)\left(\mathsf{E}\left(\boldsymbol{X} \mid d(\boldsymbol{\omega},\boldsymbol{X}) \le t\right) - A_q(\kappa_0)\boldsymbol{\mu}_0\right).$$

It is only left to calculate the Fisher information matrix $\mathcal{I}_{\boldsymbol{\xi}_0}$:

$$\begin{aligned}\mathcal{I}_{\boldsymbol{\xi}_0} &= \mathsf{E}(\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})\psi_{\boldsymbol{\xi}_0}(\boldsymbol{X})^\top) = \mathsf{E}((\boldsymbol{X}-\mathsf{E}(\boldsymbol{X}))(\boldsymbol{X}-\mathsf{E}(\boldsymbol{X}))^\top) = \mathsf{Var}(\boldsymbol{X})\\ &= A_q'(\kappa_0)\boldsymbol{\mu}_0\boldsymbol{\mu}_0^\top + \frac{A_q(\kappa_0)}{\kappa_0}\left(\boldsymbol{I}_{q+1} - \boldsymbol{\mu}_0\boldsymbol{\mu}_0^\top\right)\\ &= \frac{A_q(\kappa_0)}{\kappa_0}\mathbf{I}_{q+1} + \left(1 - A_q(\kappa_0)^2 - \frac{q+1}{\kappa_0}A_q(\kappa_0)\right)\boldsymbol{\mu}_0\boldsymbol{\mu}_0^\top.\end{aligned} \tag{123}$$

Here, $A_q'(\kappa_0)$ is the derivative of $A_q(\kappa)$ with respect to κ evaluated at κ_0, which satisfies $A_q'(\kappa_0) = 1 - A_q(\kappa_0)^2 - \frac{q}{\kappa_0}A_q(\kappa_0)$. See Section G for a brief note on the variance of a vMF distribution.

Finally, combine (10), (122), (121), and (123) with $\boldsymbol{V}_{\boldsymbol{\xi}_0} = \mathsf{E}(\dot{\psi}_{\boldsymbol{\xi}_0}(\boldsymbol{X})) = -\mathcal{I}_{\boldsymbol{\xi}_0} = -\mathsf{Var}(\boldsymbol{X})$ to produce

$$\begin{aligned}\mathbb{C}\mathrm{ov}\left[\mathbb{G}(f_{\boldsymbol{\omega}_1,t_1}), \mathbb{G}(f_{\boldsymbol{\omega}_2,t_2})\right] &= \mathsf{P}\left(d(\boldsymbol{\omega}_1,\boldsymbol{X}) \le t_1, d(\boldsymbol{\omega}_2,\boldsymbol{X}) \le t_2\right)\\ &\quad - F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}_1}(t_1)F^{\boldsymbol{\xi}_0}_{\boldsymbol{\omega}_2}(t_2)\left(1 + \boldsymbol{m}_1^\top \mathsf{Var}(\boldsymbol{X})^{-1}\boldsymbol{m}_2\right),\end{aligned}$$

where $\boldsymbol{m}_i = \mathsf{E}(\boldsymbol{X} \mid d(\boldsymbol{X},\boldsymbol{\omega}_i) \le t_i) - A_q(\kappa_0)\boldsymbol{\mu}_0$, $i = 1,2$, and $\mathsf{Var}(\boldsymbol{X}) = \mathcal{I}_{\boldsymbol{\xi}_0}$ is given in (123). □

APPENDIX F: THE COVARIANCE PROCESS UNDER THE SIMPLE NULL

DRAFT

This section complements Proposition 3.3 by recording exact formulas for the covariance process (10) of the limiting Gaussian process $\mathbb{G}$ under the simple null in some tractable cases. For the normal example, we restrict attention to the one-dimensional setting. For the vMF and HvMF examples, we work on $\mathbb{S}^2$ and $\mathbb{H}^2$, respectively; in these cases the conditioning argument reduces the joint probability in (10) to a one-dimensional integral involving an angular probability on the circle.

PROPOSITION F.1. *Let $\mathcal{C}$ denote the covariance process in* (10). *The following formulas hold:*

(i) If $X \sim \mathcal{N}(\mu,\sigma^2)$ on $\mathbb{R}$, then

$$\mathcal{C}_{(\omega_1,t_1),(\omega_2,t_2)} = 1_{\{L\le U\}}\left[\Phi\left(\frac{U-\mu}{\sigma}\right) - \Phi\left(\frac{L-\mu}{\sigma}\right)\right] - F_{\omega_1}(t_1)F_{\omega_2}(t_2), \tag{124}$$

where

$$L := \max(\omega_1 - t_1, \omega_2 - t_2), \qquad U := \min(\omega_1 + t_1, \omega_2 + t_2),$$

and

$$F_\omega(t) = \Phi\left(\frac{\omega + t - \mu}{\sigma}\right) - \Phi\left(\frac{\omega - t - \mu}{\sigma}\right).$$

(*ii*) *Let* $\boldsymbol{X} \sim \mathrm{vMF}(\boldsymbol{\mu}, \kappa)$ *on* $\mathbb{S}^2$. *Consider either the chordal distance or the geodesic distance, and let* $0 \le t_1, t_2 \le 2$ *in the chordal case and* $0 \le t_1, t_2 \le \pi$ *in the geodesic case. Define*

$$a_i := \begin{cases} 1 - \frac{t_i^2}{2}, & \textit{for the chordal distance}, \\ \cos(t_i), & \textit{for the geodesic distance}, \end{cases} \qquad U := \boldsymbol{\omega}_1^\top \boldsymbol{X},$$

where f_U *is the density of* U *given by* (7), *and*

$$\mathcal{C}_{(\boldsymbol{\omega}_1,t_1),(\boldsymbol{\omega}_2,t_2)} = \int_{a_1}^1 \mathsf{P}\big(\boldsymbol{\omega}_2^\top \boldsymbol{X} \ge a_2 \mid U = u\big) f_U(u)\,\mathrm{d}u - F_{\boldsymbol{\omega}_1}(t_1) F_{\boldsymbol{\omega}_2}(t_2), \tag{125}$$

(*iii*) *Let* $\boldsymbol{X} \sim \mathrm{HvMF}(\boldsymbol{\mu}, \kappa)$ *on* $\mathbb{H}^2$, *endowed with the geodesic distance, and assume that* $\rho := d_{\mathbb{H}^2}(\boldsymbol{\omega}_1, \boldsymbol{\omega}_2) > 0$. *Let*

$$R := d_{\mathbb{H}^2}(\boldsymbol{\omega}_1, \boldsymbol{X}),$$

where f_R *is the density of* R *given by* (8). *Then*

$$\mathcal{C}_{(\boldsymbol{\omega}_1,t_1),(\boldsymbol{\omega}_2,t_2)} = \int_0^{t_1} \mathsf{P}\left(d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X}) \le t_2 \mid R = r\right) f_R(r)\,\mathrm{d}r - F_{\boldsymbol{\omega}_1}(t_1) F_{\boldsymbol{\omega}_2}(t_2), \tag{126}$$

PROOF. We divide the proof into three parts.

Proof of (i). The event $\{|X - \omega_i| \le t_i\}$ is the interval $[\omega_i - t_i, \omega_i + t_i]$, $i = 1, 2$. Therefore,

$$\{|X - \omega_1| \le t_1,\ |X - \omega_2| \le t_2\} = \{L \le X \le U\}$$

when $L \le U$, and it is empty otherwise. This gives

$$\mathsf{P}\left(|X - \omega_1| \le t_1,\ |X - \omega_2| \le t_2\right) = 1_{\{L \le U\}} \left[\Phi\left(\frac{U - \mu}{\sigma}\right) - \Phi\left(\frac{L - \mu}{\sigma}\right)\right].$$

The expression for $F_\omega(t)$ is immediate. Combining both identities with (10) yields (124).

Proof of (ii). Set

$$\rho := \boldsymbol{\omega}_1^\top \boldsymbol{\omega}_2, \qquad m_1 := \boldsymbol{\mu}^\top \boldsymbol{\omega}_1, \qquad \lambda(u) := \kappa\sqrt{(1 - u^2)(1 - m_1^2)}.$$

For both the chordal and the geodesic distance,

$$\{d(\boldsymbol{\omega}_i, \boldsymbol{X}) \le t_i\} = \{\boldsymbol{\omega}_i^\top \boldsymbol{X} \ge a_i\}, \qquad i = 1, 2.$$

Hence,

$$\mathsf{P}\left(d(\boldsymbol{\omega}_1, \boldsymbol{X}) \le t_1,\ d(\boldsymbol{\omega}_2, \boldsymbol{X}) \le t_2\right) = \mathsf{P}\left(\boldsymbol{\omega}_1^\top \boldsymbol{X} \ge a_1,\ \boldsymbol{\omega}_2^\top \boldsymbol{X} \ge a_2\right).$$

If $\rho = 1$, then $\boldsymbol{\omega}_2 = \boldsymbol{\omega}_1$, so

$$\mathsf{P}\left(\boldsymbol{\omega}_1^\top \boldsymbol{X} \ge a_1,\ \boldsymbol{\omega}_2^\top \boldsymbol{X} \ge a_2\right) = \mathsf{P}\left(U \ge \max(a_1, a_2)\right).$$

If $\rho = -1$, then $\boldsymbol{\omega}_2 = -\boldsymbol{\omega}_1$, so

$$\mathsf{P}\left(\boldsymbol{\omega}_1^\top \boldsymbol{X} \ge a_1,\ \boldsymbol{\omega}_2^\top \boldsymbol{X} \ge a_2\right) = \mathsf{P}\left(a_1 \le U \le -a_2\right).$$

In both cases, (125) holds.

Assume now that $\rho \in (-1,1)$, and define

$$\boldsymbol{\beta} := \frac{\boldsymbol{\omega}_2 - \rho\boldsymbol{\omega}_1}{\sqrt{1-\rho^2}}.$$

Then $\boldsymbol{\omega}_2 = \rho\boldsymbol{\omega}_1 + \sqrt{1-\rho^2}\,\boldsymbol{\beta}$. Write

$$\boldsymbol{X} = U\boldsymbol{\omega}_1 + \sqrt{1-U^2}\,\boldsymbol{Y},$$

where $\boldsymbol{Y} \in \mathbb{S}^1 \cap \boldsymbol{\omega}_1^\perp$. By (7) with $q=2$ and $\boldsymbol{\omega} = \boldsymbol{\omega}_1$, the variable U has density f_U.

If $|m_1| < 1$, let

$$\widetilde{\boldsymbol{\mu}}_1 := \frac{\boldsymbol{\mu} - m_1\boldsymbol{\omega}_1}{\sqrt{1-m_1^2}}.$$

The proof of Proposition 2.1 shows that, conditionally on $U=u$, the density of $\boldsymbol{Y}$ on $\mathbb{S}^1 \cap \boldsymbol{\omega}_1^\perp$ is proportional to

$$\exp\left\{\lambda(u)\widetilde{\boldsymbol{\mu}}_1^\top \boldsymbol{Y}\right\}.$$

Choose $\boldsymbol{\beta}_\perp \in \mathbb{S}^1 \cap \boldsymbol{\omega}_1^\perp$ orthogonal to $\boldsymbol{\beta}$ so that

$$\widetilde{\boldsymbol{\mu}}_1 = \cos(\alpha)\,\boldsymbol{\beta} + \sin(\alpha)\,\boldsymbol{\beta}_\perp.$$

Writing $\boldsymbol{Y} = \cos(\varphi)\,\boldsymbol{\beta} + \sin(\varphi)\,\boldsymbol{\beta}_\perp$, one gets that $\varphi \mid U=u$ follows a von Mises distribution with density

$$\frac{1}{2\pi\mathcal{I}_0(\lambda(u))} e^{\lambda(u)\cos(\varphi-\alpha)}, \qquad \varphi \in [-\pi,\pi).$$

Moreover,

$$\boldsymbol{\omega}_2^\top \boldsymbol{X} = \rho U + \sqrt{1-\rho^2}\sqrt{1-U^2}\,\boldsymbol{\beta}^\top \boldsymbol{Y} = \rho U + \sqrt{1-\rho^2}\sqrt{1-U^2}\cos(\varphi).$$

Define

$$b(u) := \frac{a_2 - \rho u}{\sqrt{1-\rho^2}\sqrt{1-u^2}}.$$

Therefore, conditionally on $U=u$,

$$\{\boldsymbol{\omega}_2^\top \boldsymbol{X} \geq a_2\} = \{\cos(\varphi) \geq b(u)\}.$$

Hence, if $|m_1| < 1$ and $-1 < b(u) < 1$,

$$\mathsf{P}\big(\boldsymbol{\omega}_2^\top \boldsymbol{X} \geq a_2 \mid U=u\big) = \frac{1}{2\pi\mathcal{I}_0(\lambda(u))}\int_{-\arccos(b(u))}^{\arccos(b(u))} e^{\lambda(u)\cos(\varphi-\alpha)}\,\mathrm{d}\varphi. \tag{127}$$

If $|m_1| < 1$ and $b(u) \leq -1$, then $\mathsf{P}\left(\boldsymbol{\omega}_2^\top \boldsymbol{X} \geq a_2 \mid U=u\right) = 1$, whereas if $|m_1| < 1$ and $b(u) \geq 1$, then $\mathsf{P}\left(\boldsymbol{\omega}_2^\top \boldsymbol{X} \geq a_2 \mid U=u\right) = 0$.

If $|m_1| = 1$, then $\lambda(u) = 0$, so $\varphi \mid U=u$ is uniform on the circle. Therefore, if $-1 < b(u) < 1$,

$$\mathsf{P}\big(\boldsymbol{\omega}_2^\top \boldsymbol{X} \geq a_2 \mid U=u\big) = \frac{\arccos(b(u))}{\pi}, \tag{128}$$

whereas the conditional probability is again equal to 1 when $b(u) \leq -1$ and equal to 0 when $b(u) \geq 1$. Combining (127) and (128) with the density f_U given in (7) and with the definition of $\mathcal{C}$ in (10) yields (125).

Proof of (iii). Set

$$\rho := d_{\mathbb{H}^2}(\boldsymbol{\omega}_1, \boldsymbol{\omega}_2), \qquad \rho_\mu := d_{\mathbb{H}^2}(\boldsymbol{\omega}_1, \boldsymbol{\mu}), \qquad \lambda_H(r) := \kappa \sinh(r)\sinh(\rho_\mu).$$

For completeness, if $\rho = 0$, then $\boldsymbol{\omega}_1 = \boldsymbol{\omega}_2$, and the joint event is $\{R \leq \min(t_1, t_2)\}$. This boundary case is omitted from the statement.

Assume from now on that $\rho > 0$. Choose a hyperbolic transformation $\boldsymbol{A}$ such that

$$\boldsymbol{A}\boldsymbol{\omega}_1 = (1,0,0)^\top, \qquad \boldsymbol{A}\boldsymbol{\omega}_2 = (\cosh(\rho), \sinh(\rho), 0)^\top.$$

Write

$$\boldsymbol{A}\boldsymbol{X} = \big(\cosh(R), \sinh(R)\cos(\varphi), \sinh(R)\sin(\varphi)\big)^\top.$$

If $\rho_\mu > 0$, write also

$$\boldsymbol{A}\boldsymbol{\mu} = (\cosh(\rho_\mu), \sinh(\rho_\mu)\cos(\alpha_H), \sinh(\rho_\mu)\sin(\alpha_H))^\top.$$

Then, by the Minkowski inner product,

$$(\boldsymbol{A}\boldsymbol{\mu}, \boldsymbol{A}\boldsymbol{X}) = -\cosh(\rho_\mu)\cosh(R) + \sinh(\rho_\mu)\sinh(R)\cos(\varphi - \alpha_H).$$

Using the polar-coordinate representation from the proof of Proposition 2.1, it follows that the joint density of (R, φ) is proportional to

$$e^{\kappa(\boldsymbol{\mu}, \boldsymbol{\omega}_1)\cosh(R)} e^{\lambda_H(R)\cos(\varphi - \alpha_H)} \sinh(R),$$

and therefore the marginal density of R is f_R, while $\varphi \mid R = r$ follows a von Mises distribution with density

$$\frac{1}{2\pi \mathcal{I}_0(\lambda_H(r))} e^{\lambda_H(r)\cos(\varphi - \alpha_H)}, \qquad \varphi \in [-\pi, \pi).$$

If $\rho_\mu = 0$, then $\lambda_H(r) = 0$ and $\varphi \mid R = r$ is uniform on the circle.

Next, by the distance formula on $\mathbb{H}^2$,

$$\cosh\big(d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X})\big) = -(\boldsymbol{A}\boldsymbol{\omega}_2, \boldsymbol{A}\boldsymbol{X}) = \cosh(\rho)\cosh(R) - \sinh(\rho)\sinh(R)\cos(\varphi).$$

Define

$$b_H(r) := \frac{\cosh(\rho)\cosh(r) - \cosh(t_2)}{\sinh(\rho)\sinh(r)}.$$

Hence, conditionally on $R = r$,

$$\{d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X}) \leq t_2\} = \{\cos(\varphi) \geq b_H(r)\}.$$

Hence, if $\rho_\mu > 0$ and $-1 < b_H(r) < 1$,

$$\mathsf{P}\left(d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X}) \leq t_2 \mid R = r\right) = \frac{1}{2\pi \mathcal{I}_0(\lambda_H(r))} \int_{-\arccos(b_H(r))}^{\arccos(b_H(r))} e^{\lambda_H(r)\cos(\varphi - \alpha_H)}\, \mathrm{d}\varphi. \tag{129}$$

If $\rho_\mu > 0$ and $b_H(r) \leq -1$, then $\mathsf{P}\left(d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X}) \leq t_2 \mid R = r\right) = 1$, whereas if $\rho_\mu > 0$ and $b_H(r) \geq 1$, then $\mathsf{P}\left(d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X}) \leq t_2 \mid R = r\right) = 0$.

If $\rho_\mu = 0$, then $\lambda_H(r) = 0$, so $\varphi \mid R = r$ is uniform on the circle. Therefore, if $-1 < b_H(r) < 1$,

$$\mathsf{P}\left(d_{\mathbb{H}^2}(\boldsymbol{\omega}_2, \boldsymbol{X}) \leq t_2 \mid R = r\right) = \frac{\arccos(b_H(r))}{\pi}, \tag{130}$$

whereas the conditional probability is again equal to 1 when $b_H(r) \leq -1$ and equal to 0 when $b_H(r) \geq 1$. Combining (129) and (130) with the density f_R in (8) and with the definition of $\mathcal{C}$ in (10) yields (126). □

APPENDIX G: THE VARIANCE OF THE VMF DISTRIBUTION

Mardia and Jupp (1999) (Equation 10.3.16) show that the variance of $\boldsymbol{X} \sim \mathrm{vMF}(\boldsymbol{\mu}, \kappa)$ can be written as

$$\mathsf{Var}(\mathbf{X}) = \mathcal{I}_{(\kappa,\boldsymbol{\mu})} = \begin{pmatrix} A_p'(\kappa) & 0 \\ 0 & \dfrac{A_p(\kappa)}{\kappa}\left(\mathbf{I}_{q+1} - \boldsymbol{\mu}\boldsymbol{\mu}^T\right) \end{pmatrix}$$

Observe that, since $\boldsymbol{\mu} \in \mathbb{S}^q \subset \mathbb{R}^{q+1}$, this matrix has dimensionality $(q+2) \times (q+2)$. This is because the Fisher information matrix is expressed with respect to $(\kappa, \boldsymbol{\mu})$.

The expression for the variance shown in (123) can be derived from the properties of exponential families. For a $\mathrm{vMF}(\boldsymbol{\mu}, \kappa)$ distribution, this gives

$$\mathcal{I}_{\boldsymbol{\xi}} = \frac{\partial}{\partial \boldsymbol{\xi}^\top} A_q(\kappa)\boldsymbol{\mu} = \frac{\partial}{\partial \boldsymbol{\xi}^\top} A_q(\|\boldsymbol{\xi}\|)\frac{\boldsymbol{\xi}}{\|\boldsymbol{\xi}\|} = A_q'(\kappa)\boldsymbol{\mu}\boldsymbol{\mu}^\top + \frac{A_q(\kappa)}{\kappa}\left(\boldsymbol{I} - \boldsymbol{\mu}\boldsymbol{\mu}^\top\right).$$

Diego: Since I am not an expert in vMF distributions, I think it is a good practice to compare this result with Hillen et al. (2017) as a sanity check. (Hillen et al., 2017, Lemma 3.1) show that

$$\mathsf{Var}(\boldsymbol{X}) = \frac{1}{2}\left(1 - \frac{\mathcal{I}_2(k)}{\mathcal{I}_0(k)}\right)\boldsymbol{I}_2 + \left(\frac{\mathcal{I}_2(k)}{\mathcal{I}_0(k)} - \left(\frac{\mathcal{I}_1(k)}{\mathcal{I}_0(k)}\right)^2\right)\boldsymbol{\mu}\boldsymbol{\mu}^T. \tag{131}$$

It is easy to see that (131) equals (123) when $q = 1$, by observing that $\frac{\mathcal{I}_2(\kappa)}{\mathcal{I}_0(\kappa)} = 1 - \frac{2\mathcal{I}_2(\kappa)}{\kappa\mathcal{I}_0(\kappa)} = 1 - \frac{2}{\kappa}A_1(\kappa)$. This follows from the recurrence relation for modified Bessel functions of the first kind $\mathcal{I}_{\mu-1}(\kappa) - \mathcal{I}_{\mu+1}(\kappa) = \frac{2\mu}{\kappa}\mathcal{I}_\mu(\kappa)$ for $\mu \in \mathbb{Z}_{\geq 0}$.